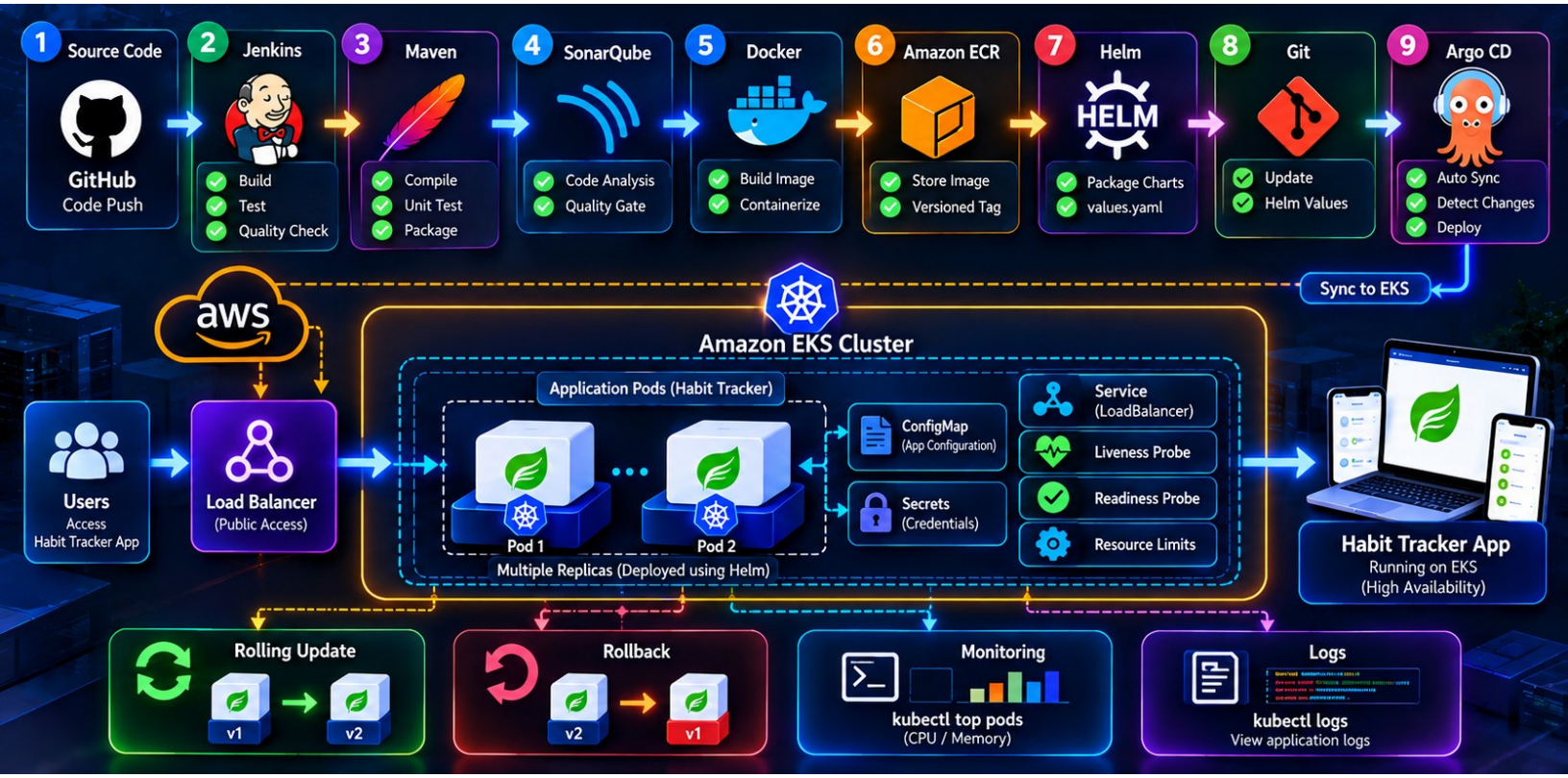
PROJECT - 5

Java Application CI/CD Using Jenkins, Maven, SonarQube, Helm & Argo CD



Task 1 : Java Application

Overview

The Habit Tracker Java application was cloned from the provided GitHub repository and successfully set up and verified locally as a Spring Boot application using Java 21 and Maven. The application was built, tested, packaged, and successfully executed both locally and inside a Docker container.

Application Structure

- Application: Spring Boot — Java 21
- Build Tool: Apache Maven 3.9.16
- Application Port: 8080
- Packaging: JAR
- Database: No external database; in-memory storage
- Configuration: src/main/resources/application.properties
- Unit Tests: src/test/java

Execution

The application was built using `mvn clean package`, generating `target/habit-tracker.jar`.

The application was tested using `mvn test`, and the packaged JAR was successfully executed.

A Docker image was created from the application JAR and the application was started inside a Docker container with port 8080 mapped to the host.

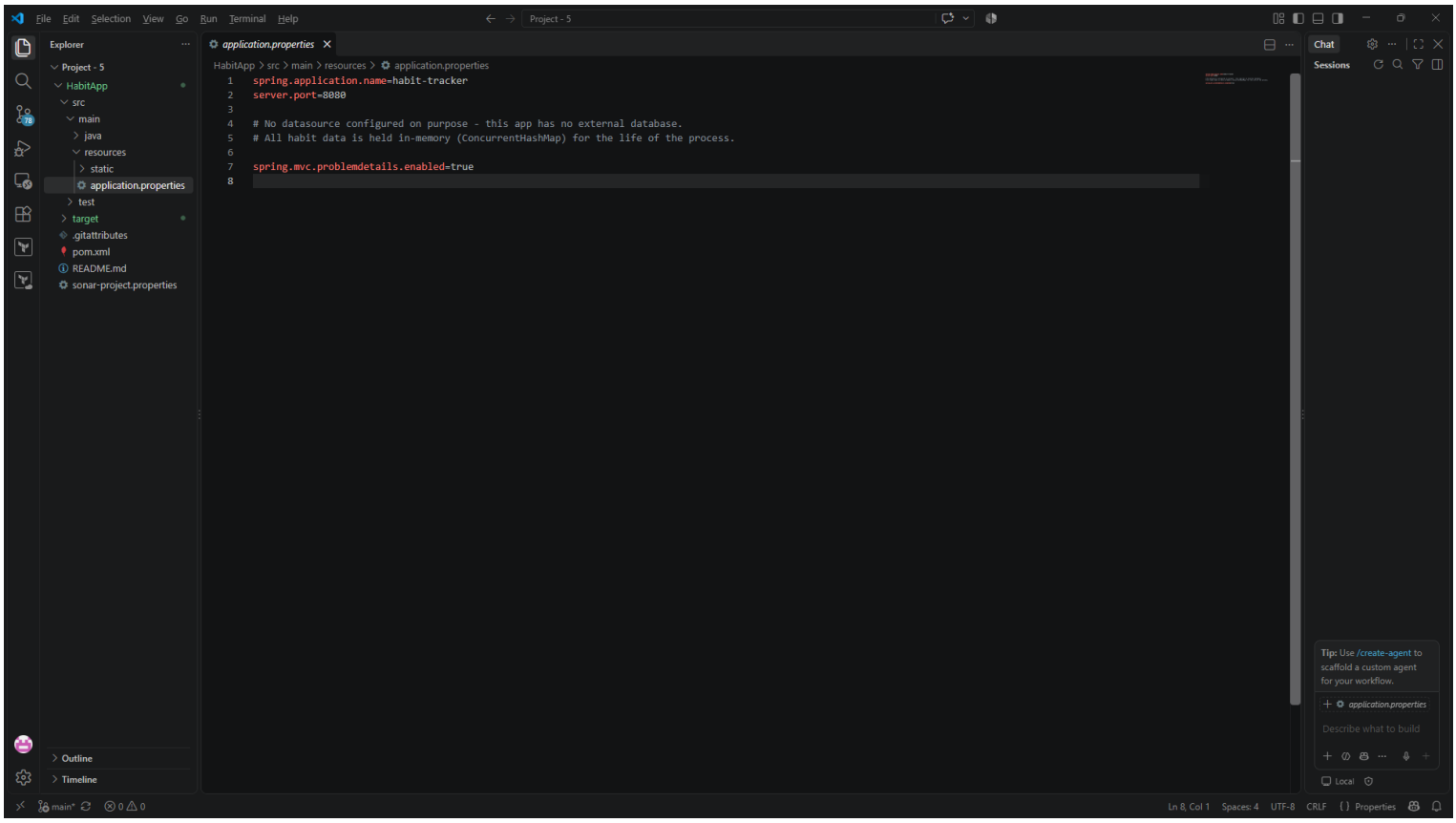
Verification

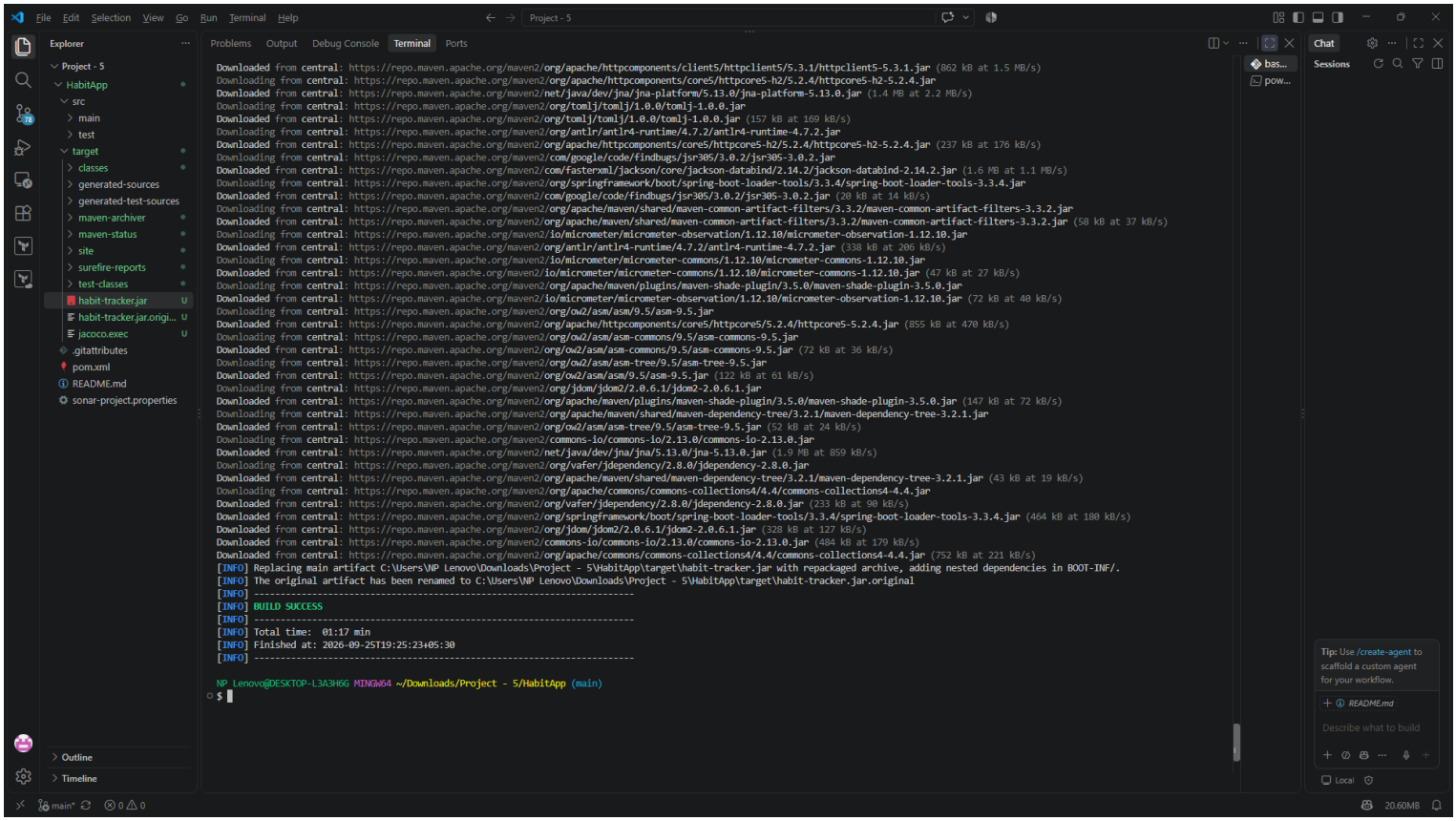
The application was successfully accessed through : http://localhost:8080
The application loaded successfully from the Docker container.
Unit test execution completed successfully:

Tests run: 1
Failures: 0
Errors: 0
Skipped: 0
BUILD SUCCESS
The Maven build also completed successfully and generated the application JAR.

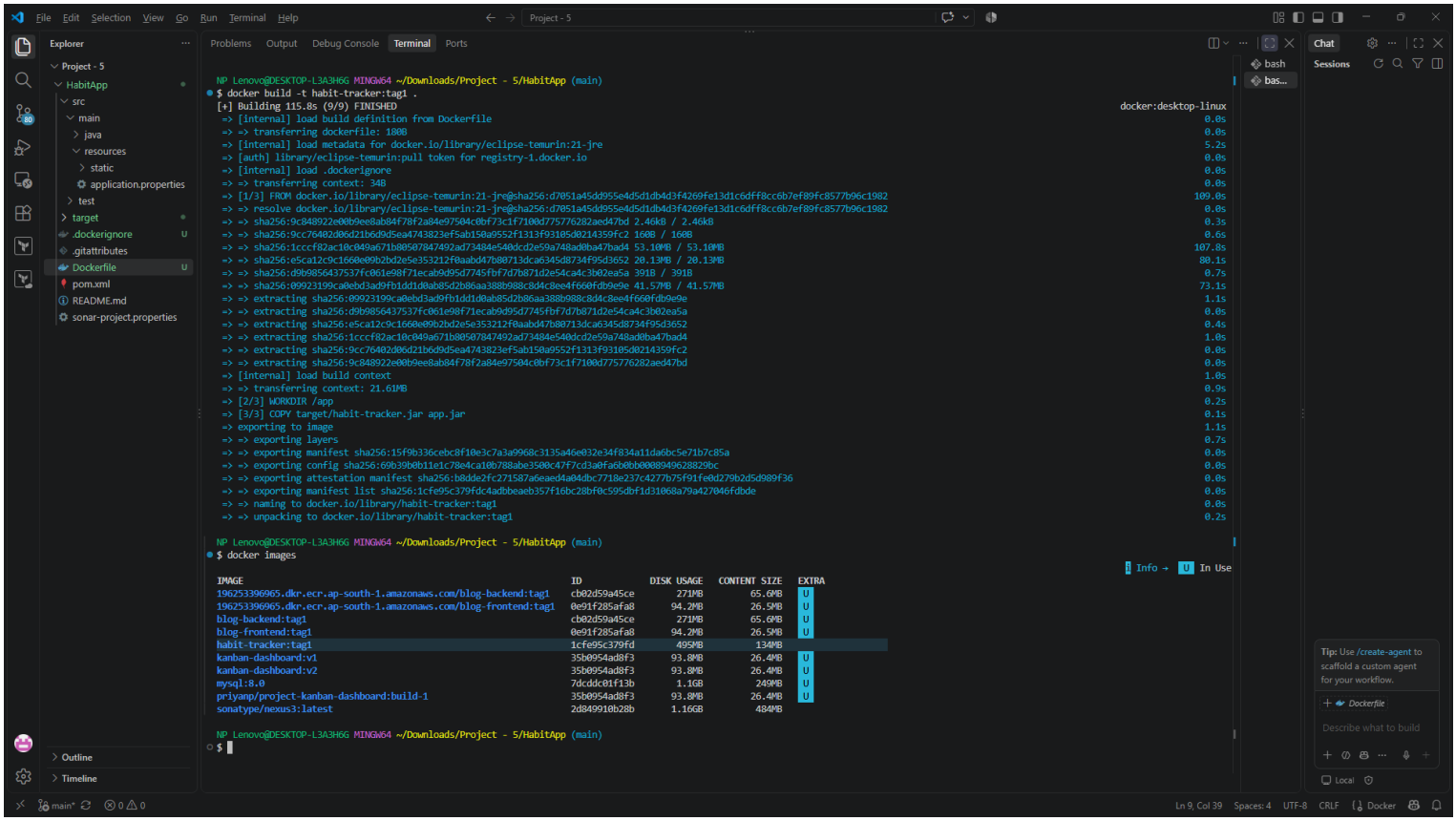
Result

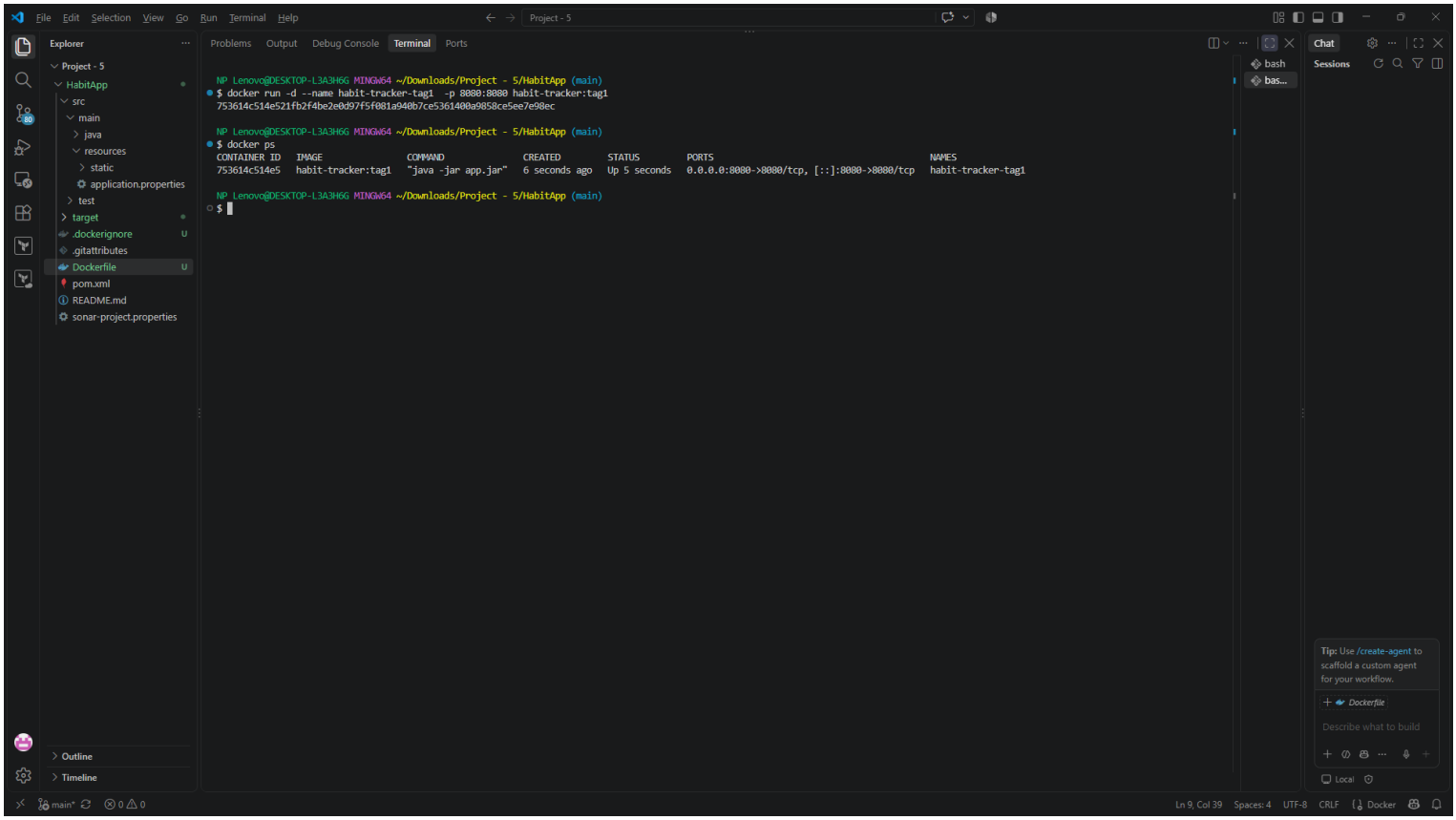
The Java application was successfully built, tested, packaged, and containerized. The application runs successfully in a Docker container and is accessible through port 8080.

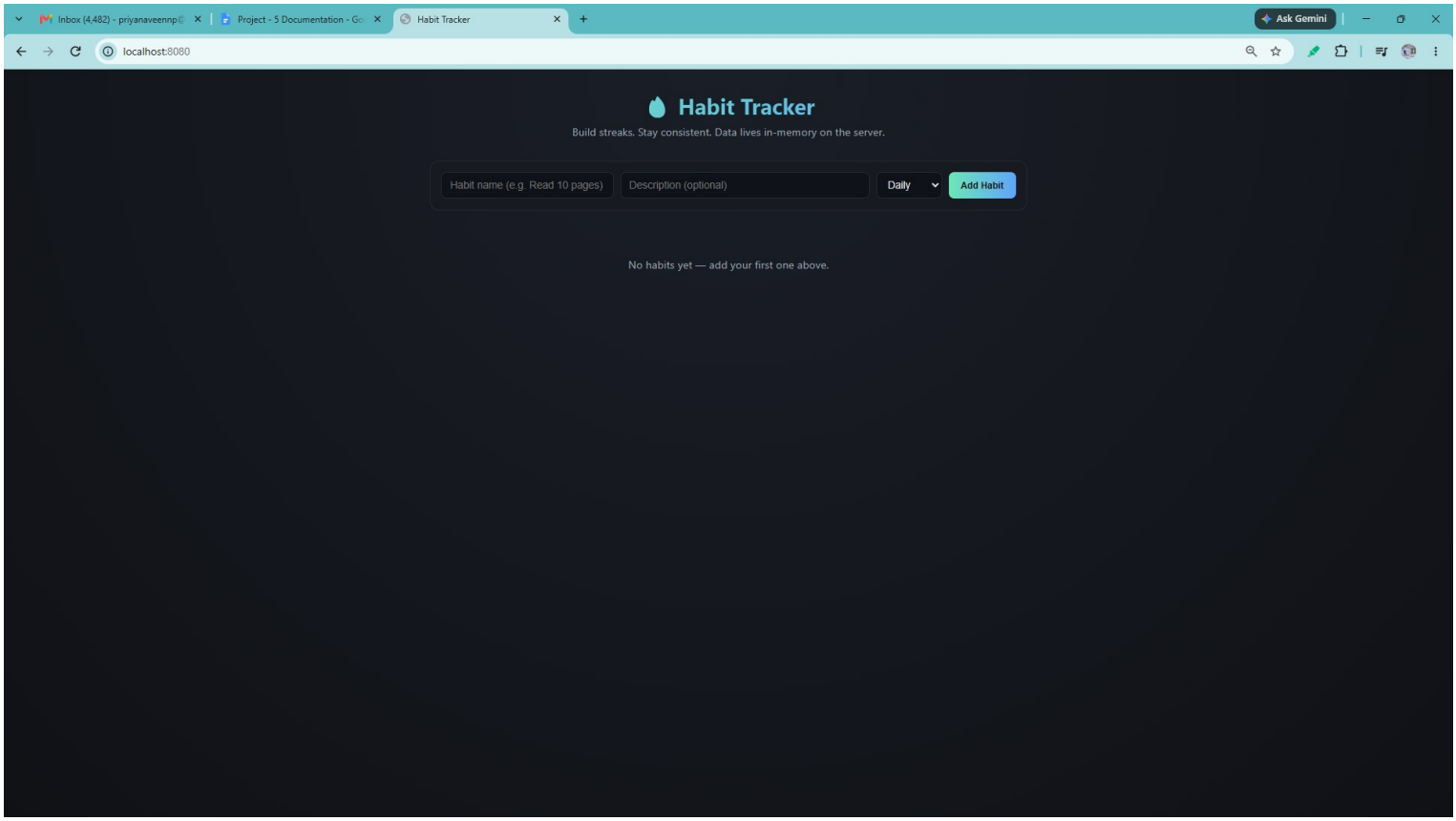



Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/httpcomponents/client5/httpclient5/5.3.1/httpclient5-5.3.1.jar (862 kB at 1.5 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/httpcomponents/core5/httpcore5-h2/5.2.4/httpcore5-h2-5.2.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/net/java/dev/jna/jna-platform/5.13.0/jna-platform-5.13.0.jar (1.4 MB at 2.2 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/tomlj/tomlj/1.0.0/tomlj-1.0.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/tomlj/tomlj/1.0.0/tomlj-1.0.0.jar (157 kB at 169 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/antlr/antlr4-runtime/4.7.2/antlr4-runtime-4.7.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/httpcomponents/core5/httpcore5-h2/5.2.4/httpcore5-h2-5.2.4.jar (237 kB at 176 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/com/google/code/findbugs/jsr305/3.0.2/jsr305-3.0.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/com/fasterxml/jackson/core/jackson-databind/2.14.2/jackson-databind-2.14.2.jar (1.6 MB at 1.1 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-loader-tools/3.3.4/spring-boot-loader-tools-3.3.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/com/google/code/findbugs/jsr305/3.0.2/jsr305-3.0.2.jar (20 kB at 14 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-common-artifact-filters/3.3.2/maven-common-artifact-filters-3.3.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-common-artifact-filters/3.3.2/maven-common-artifact-filters-3.3.2.jar (58 kB at 37 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/io/micrometer/micrometer-observation/1.12.10/micrometer-observation-1.12.10.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/antlr/antlr4-runtime/4.7.2/antlr4-runtime-4.7.2.jar (338 kB at 206 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/io/micrometer/micrometer-commons/1.12.10/micrometer-commons-1.12.10.jar
Downloaded from central: https://repo.maven.apache.org/maven2/io/micrometer/micrometer-commons/1.12.10/micrometer-commons-1.12.10.jar (47 kB at 27 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-shade-plugin/3.5.0/maven-shade-plugin-3.5.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/io/micrometer/micrometer-observation/1.12.10/micrometer-observation-1.12.10.jar (72 kB at 40 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.5/asm-9.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/httpcomponents/core5/httpcore5/5.2.4/httpcore5-5.2.4.jar (855 kB at 470 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-commons/9.5/asm-commons-9.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-commons/9.5/asm-commons-9.5.jar (72 kB at 36 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-tree/9.5/asm-tree-9.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.5/asm-9.5.jar (122 kB at 61 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/jdom/jdom2/2.0.6.1/jdom2-2.0.6.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-shade-plugin/3.5.0/maven-shade-plugin-3.5.0.jar (147 kB at 72 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-dependency-tree/3.2.1/maven-dependency-tree-3.2.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-tree/9.5/asm-tree-9.5.jar (52 kB at 24 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.13.0/commons-io-2.13.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/net/java/dev/jna/jna/5.13.0/jna-5.13.0.jar (1.9 MB at 859 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/vafer/jdependency/2.8.0/jdependency-2.8.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-dependency-tree/3.2.1/maven-dependency-tree-3.2.1.jar (43 kB at 19 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-collections4/4.4/commons-collections4-4.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/vafer/jdependency/2.8.0/jdependency-2.8.0.jar (233 kB at 90 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-loader-tools/3.3.4/spring-boot-loader-tools-3.3.4.jar (464 kB at 180 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/jdom/jdom2/2.0.6.1/jdom2-2.0.6.1.jar (328 kB at 127 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.13.0/commons-io-2.13.0.jar (484 kB at 179 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-collections4/4.4/commons-collections4-4.4.jar (752 kB at 221 kB/s)
[INFO] Replacing main artifact C:\Users\NP Lenovo\Downloads\Project - 5\HabitApp\target\habit-tracker.jar with repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to C:\Users\NP Lenovo\Downloads\Project - 5\HabitApp\target\habit-tracker.jar.original
[INFO] --
[INFO] BUILD SUCCESS
[INFO] --
[INFO] Total time: 01:17 min
[INFO] Finished at: 2026-09-25T19:25:23+05:30
[INFO] --
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$


NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$ mvn test
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ habit-tracker ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- surefire:3.2.5:test (default-test) @ habit-tracker ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO] T E S T S
[INFO] Running com.example.habittracker.HabitTrackerApplicationTests
19:39:15.528 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect default configuration classes for test class [com.example.habittracker.HabitTrackerApplicationTests]: HabitTrackerApplicationTests does not declare any static, non-private, non-final, nested classes annotated with @Configuration.
19:39:16.040 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found @SpringBootConfiguration com.example.habittracker.HabitTrackerApplication for test class com.example.habittracker.HabitTrackerApplicationTests
:: Spring Boot :: (v3.3.4)
2026-09-25T19:39:16.943+05:30 INFO 5364 --- [habit-tracker] [main] c.e.h.HabitTrackerApplicationTests : Starting HabitTrackerApplicationTests using Java 21.0.12.1 with PID 5364 (started by NP Lenovo in C:\Users\NP Lenovo\Downloads\Project - 5\HabitApp)
2026-09-25T19:39:16.952+05:30 INFO 5364 --- [habit-tracker] [main] c.e.h.HabitTrackerApplicationTests : No active profile set, falling back to 1 default profile: "default"
2026-09-25T19:39:18.766+05:30 INFO 5364 --- [habit-tracker] [main] o.s.b.a.w.s.WelcomePageHandlerMapping : Adding welcome page: class path resource [static/index.html]
2026-09-25T19:39:19.282+05:30 INFO 5364 --- [habit-tracker] [main] c.e.h.HabitTrackerApplicationTests : Started HabitTrackerApplicationTests in 2.844 seconds (process running for 5.935)
WARNING: A Java agent has been loaded dynamically (C:\Users\NP Lenovo\.m2\repository\net\bytebuddy\byte-buddy-agent\1.14.19\byte-buddy-agent-1.14.19.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 6.055 s -- in com.example.habittracker.HabitTrackerApplicationTests
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] --- jacoco:0.8.12:report (jacoco-report) @ habit-tracker ---
[INFO] Loading execution data file C:\Users\NP Lenovo\Downloads\Project - 5\HabitApp\target\jacoco.exec
[INFO] Analyzed bundle 'habit-tracker' with 10 classes
[INFO] BUILD SUCCESS
[INFO] Total time: 12.455 s
[INFO] Finished at: 2026-09-25T19:39:21+05:30
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$


File Edit Selection View Go Run Terminal Help
Project - 5
Explorer
Project - 5
HabitApp
src
main
java
resources
static
application.properties
test
target
.dockerignore
.gitattributes
Dockerfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$ docker build -t habit-tracker:tag1 .
[+] Building 115.8s (9/9) FINISHED docker:desktop-linux
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 180B 0.0s
=> [internal] load metadata for docker.io/library/eclipse-temurin:21-jre 5.2s
=> [auth] library/eclipse-temurin:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 34B 0.0s
=> [1/3] FROM docker.io/library/eclipse-temurin:21-jre@sha256:d7051a45dd955e4d5d1db4d3f4269fe13d1c6dff8cc6b7ef89fc8577b96c1982 109.0s
=> => resolve docker.io/library/eclipse-temurin:21-jre@sha256:d7051a45dd955e4d5d1db4d3f4269fe13d1c6dff8cc6b7ef89fc8577b96c1982 0.0s
=> => sha256:9c848922e00b9ee8ab84f78f2a84e97504c0bf73c1f7100d775776282aed47bd 2.46kB / 2.46kB 0.3s
=> => sha256:9cc76402d06d21b6d9d5ea4743823ef5ab150a9552f1313f93105d0214359fc2 160B / 160B 0.6s
=> => sha256:1cccf82ac10c049a671b80507847492ad73484e540dcd2e59a748ad0ba47bad4 53.10MB / 53.10MB 107.8s
=> => sha256:e5ca12c9c1660e09b2bd2e5e353212f0aabd47b80713dca6345d8734f95d3652 20.13MB / 20.13MB 80.1s
=> => sha256:d9b9856437537fc061e98f71ecab9d95d7745fbf7d7b871d2e54ca4c3b02ea5a 391B / 391B 0.7s
=> => sha256:09023199ca0ebd3ad9fb1dd1d0ab85d2b86aa388b988c8d4c8ee4f660fdb9e9e 41.57MB / 41.57MB 73.1s
=> => extracting sha256:09023199ca0ebd3ad9fb1dd1d0ab85d2b86aa388b988c8d4c8ee4f660fdb9e9e 1.1s
=> => extracting sha256:d9b9856437537fc061e98f71ecab9d95d7745fbf7d7b871d2e54ca4c3b02ea5a 0.0s
=> => extracting sha256:e5ca12c9c1660e09b2bd2e5e353212f0aabd47b80713dca6345d8734f95d3652 0.4s
=> => extracting sha256:1cccf82ac10c049a671b80507847492ad73484e540dcd2e59a748ad0ba47bad4 1.0s
=> => extracting sha256:9cc76402d06d21b6d9d5ea4743823ef5ab150a9552f1313f93105d0214359fc2 0.0s
=> => extracting sha256:9c848922e00b9ee8ab84f78f2a84e97504c0bf73c1f7100d775776282aed47bd 0.0s
=> [internal] load build context 1.0s
=> => transferring context: 21.61MB 0.9s
=> [2/3] WORKDIR /app 0.2s
=> [3/3] COPY target/habit-tracker.jar app.jar 0.1s
=> exporting to image 1.1s
=> => exporting layers 0.7s
=> => exporting manifest sha256:15f9b336cebc8f10e3c7a3a9968c3135a46e032e34f834a11da6bc5e71b7c85a 0.0s
=> => exporting config sha256:69b39b0b11e1c78e4ca10b788abe3500c47f7cd3a0fa6b0bb0008949628829bc 0.0s
=> => exporting attestation manifest sha256:b8dde2fc271587a6eaed4a04dbc7718e237c4277b75f91fe0d279b2d5d989f36 0.0s
=> => exporting manifest list sha256:1cfe95c379fdc4adbbeaeb357f16bc28bf0c595dbf1d31068a79a427046fdbde 0.0s
=> => naming to docker.io/library/habit-tracker:tag1 0.0s
=> => unpacking to docker.io/library/habit-tracker:tag1 0.2s
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$ docker images
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
196253396965.dkr.ecr.ap-south-1.amazonaws.com/blog-backend:tag1 cb02d59a45ce 271MB 65.6MB U
196253396965.dkr.ecr.ap-south-1.amazonaws.com/blog-frontend:tag1 0e91f285afa8 94.2MB 26.5MB U
blog-backend:tag1 cb02d59a45ce 271MB 65.6MB U
blog-frontend:tag1 0e91f285afa8 94.2MB 26.5MB U
habit-tracker:tag1 1cfe95c379fd 495MB 134MB
kanban-dashboard:v1 35b0954ad8f3 93.8MB 26.4MB U
kanban-dashboard:v2 35b0954ad8f3 93.8MB 26.4MB U
mysql:8.0 7dcddc01f13b 1.1GB 249MB U
priyanp/project-kanban-dashboard:build-1 35b0954ad8f3 93.8MB 26.4MB U
sonatype/nexus3:latest 2d849910b28b 1.16GB 484MB
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$
Outline
Timeline
Chat
Sessions
bash
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Dockerfile
Describe what to build
Local
main*
Ln 9, Col 39 Spaces: 4 UTF-8 CRLF Docker


File Edit Selection View Go Run Terminal Help
Project - 5
Explorer
Project - 5
HabitApp
src
main
java
resources
static
application.properties
test
target
.dockerignore
.gitattributes
Dockerfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$ docker run -d --name habit-tracker-tag1 -p 8080:8080 habit-tracker:tag1
753614c514e521fb2f4be2e0d97f5f081a940b7ce536140a9858ce5ee7e98ec
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
753614c514e5 habit-tracker:tag1 "java -jar app.jar" 6 seconds ago Up 5 seconds 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp habit-tracker-tag1
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$
Outline
Timeline
Chat
Sessions
bash
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Dockerfile
Describe what to build
Local
main*
Ln 9, Col 39 Spaces: 4 UTF-8 CRLF Docker


Inbox (4,482) - priyanaveennp@
Project - 5 Documentation - Go
Habit Tracker
Ask Gemini
localhost:8080
Habit Tracker
Build streaks. Stay consistent. Data lives in-memory on the server.
Habit name (e.g. Read 10 pages)
Description (optional)
Daily
Add Habit
No habits yet — add your first one above.

Task 2 : Maven Build & Test

Overview

Maven was integrated with Jenkins to automate the Java application's dependency management, compilation, unit testing, and packaging.

Execution

Jenkins checked out the Habit Tracker application from the GitHub repository.

Maven executed : mvn clean package

This performed dependency resolution, compilation, unit testing, and application packaging.

The application was successfully compiled using Java 21 and packaged as : target/habit-tracker.jar

A separate Maven test stage executed : mvn test

Verification

Unit tests completed successfully:

Tests run: 1
Failures: 0
Errors: 0
Skipped: 0

Maven successfully created the application JAR and reported : BUILD SUCCESS

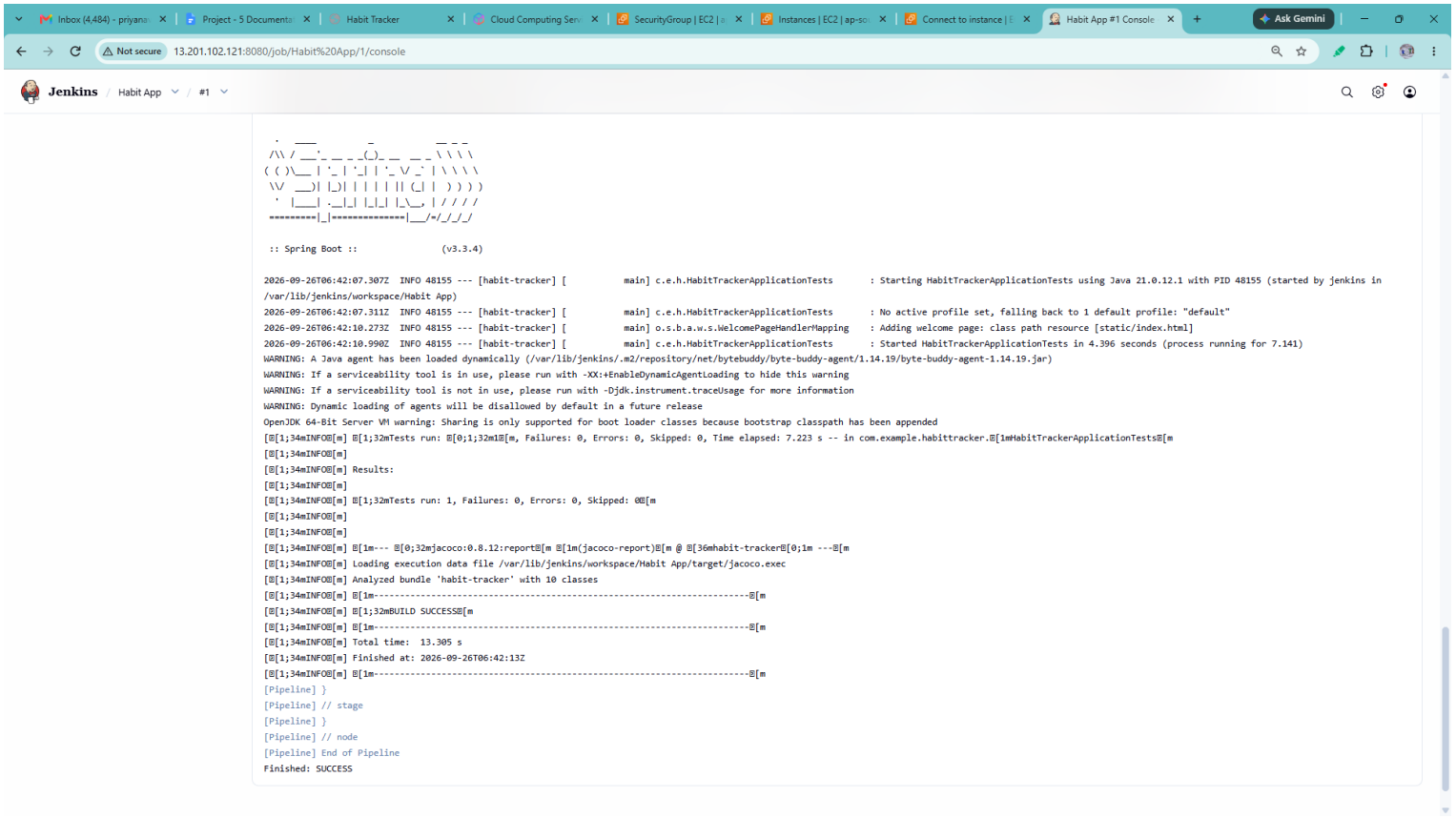
Jenkins completed the entire pipeline successfully:

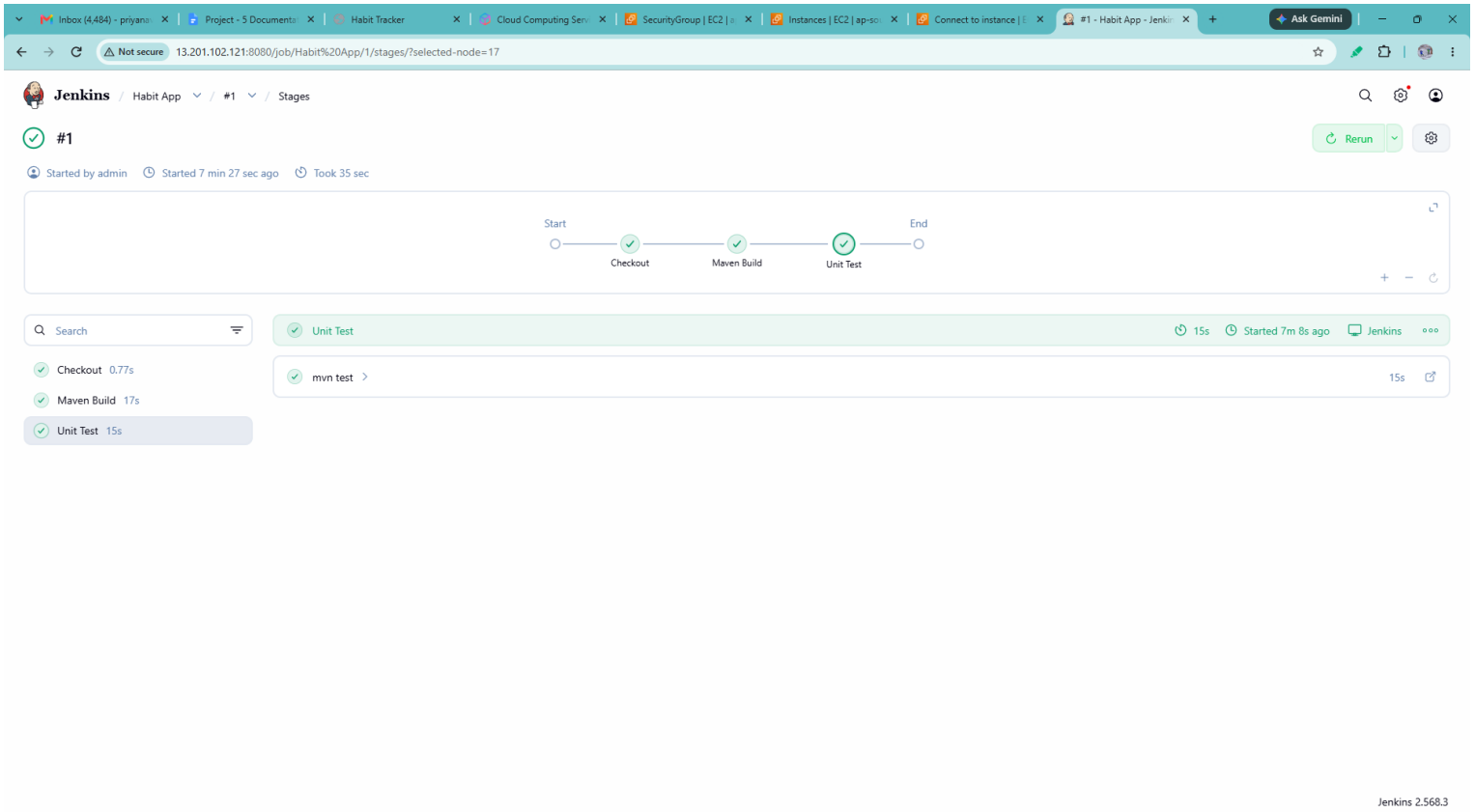
Finished: SUCCESS

The Jenkins console output provides evidence of the Maven build, compilation, unit test results, and application packaging.

Result

The Java application successfully passed the Maven build and unit testing process through Jenkins.


Not secure 13.201.102.121:8080/job/Habit%20App/1/console
Jenkins / Habit App / #1
:: Spring Boot :: (v3.3.4)
2026-09-26T06:42:07.307Z INFO 48155 --- [habit-tracker] [main] c.e.h.HabitTrackerApplicationTests : Starting HabitTrackerApplicationTests using Java 21.0.12.1 with PID 48155 (started by jenkins in /var/lib/jenkins/workspace/Habit App)
2026-09-26T06:42:07.311Z INFO 48155 --- [habit-tracker] [main] c.e.h.HabitTrackerApplicationTests : No active profile set, falling back to 1 default profile: "default"
2026-09-26T06:42:10.273Z INFO 48155 --- [habit-tracker] [main] o.s.b.a.w.s.WelcomePageHandlerMapping : Adding welcome page: class path resource [static/index.html]
2026-09-26T06:42:10.990Z INFO 48155 --- [habit-tracker] [main] c.e.h.HabitTrackerApplicationTests : Started HabitTrackerApplicationTests in 4.396 seconds (process running for 7.141)
WARNING: A Java agent has been loaded dynamically (/var/lib/jenkins/.m2/repository/net/bytebuddy/byte-buddy-agent/1.14.19/byte-buddy-agent-1.14.19.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 7.223 s -- in com.example.habittracker.HabitTrackerApplicationTests
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] --- jacoco:0.8.12:report (jacoco-report) @ habit-tracker ---
[INFO] Loading execution data file /var/lib/jenkins/workspace/Habit App/target/jacoco.exec
[INFO] Analyzed bundle 'habit-tracker' with 10 classes
[INFO] BUILD SUCCESS
[INFO] Total time: 13.305 s
[INFO] Finished at: 2026-09-26T06:42:13Z
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS


Not secure 13.201.102.121:8080/job/Habit%20App/1/stages/?selected-node=17
Jenkins / Habit App / #1 / Stages
#1
Rerun
Started by admin
Started 7 min 27 sec ago
Took 35 sec
Start
Checkout
Maven Build
Unit Test
End
Search
Checkout 0.77s
Maven Build 17s
Unit Test 15s
Unit Test 15s Started 7m 8s ago Jenkins
mvn test 15s
Jenkins 2.568.3

Task 3 : SonarQube

Overview

SonarQube was integrated with Jenkins to perform automated source-code analysis and Quality Gate validation for the Habit Tracker application.

Execution

Jenkins executed the Maven build and unit tests, followed by SonarQube source-code analysis.

The SonarQube analysis evaluated the application source code and returned the Quality Gate result to Jenkins.

Jenkins waited for the Quality Gate result before completing the pipeline.

Verification

The Jenkins pipeline completed successfully with the following stages:

- Checkout
- Maven Build
- Unit Test
- SonarQube Analysis
- Quality Gate

The SonarQube project Habit Tracker was successfully analyzed.

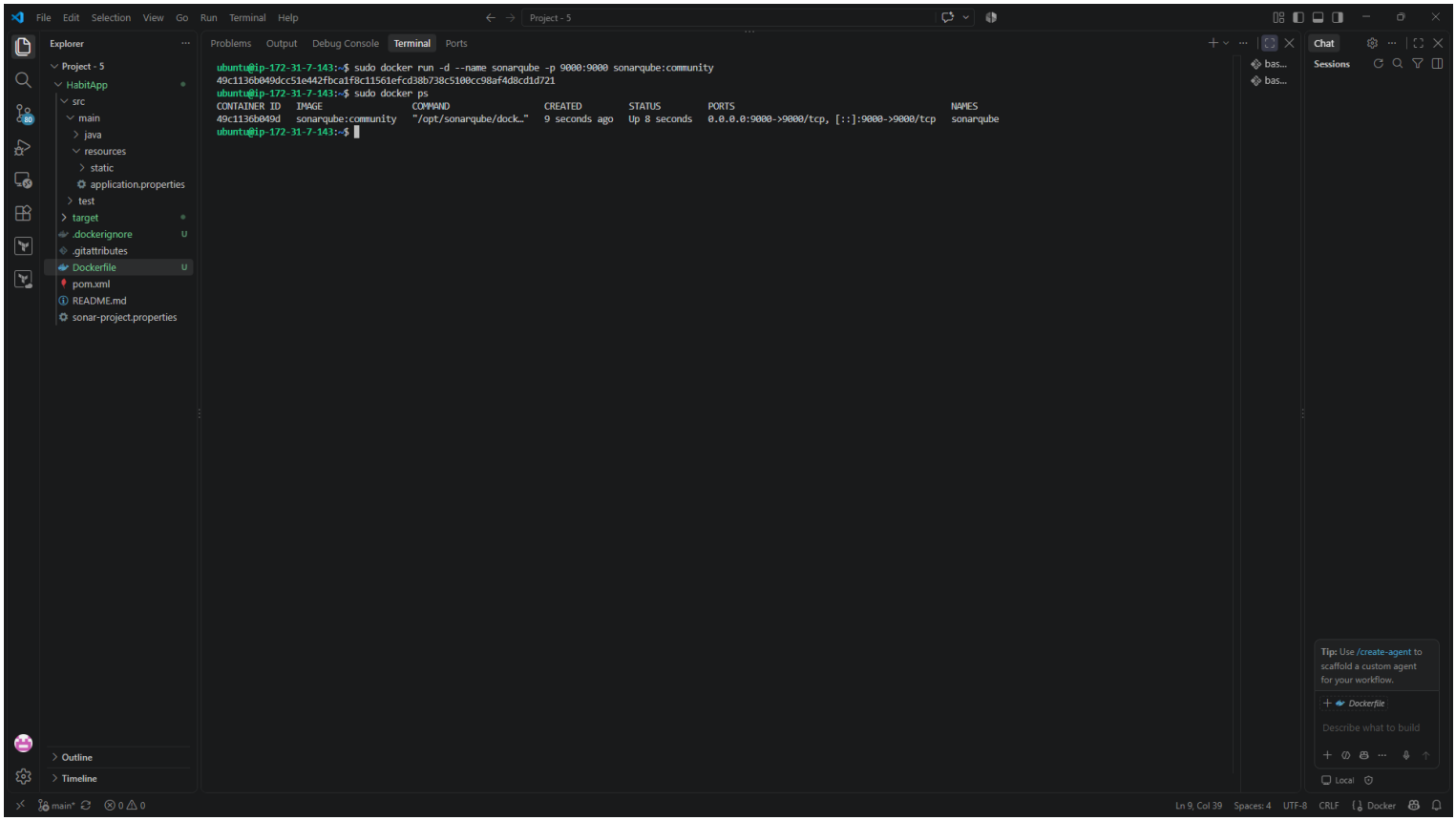
The SonarQube dashboard showed:

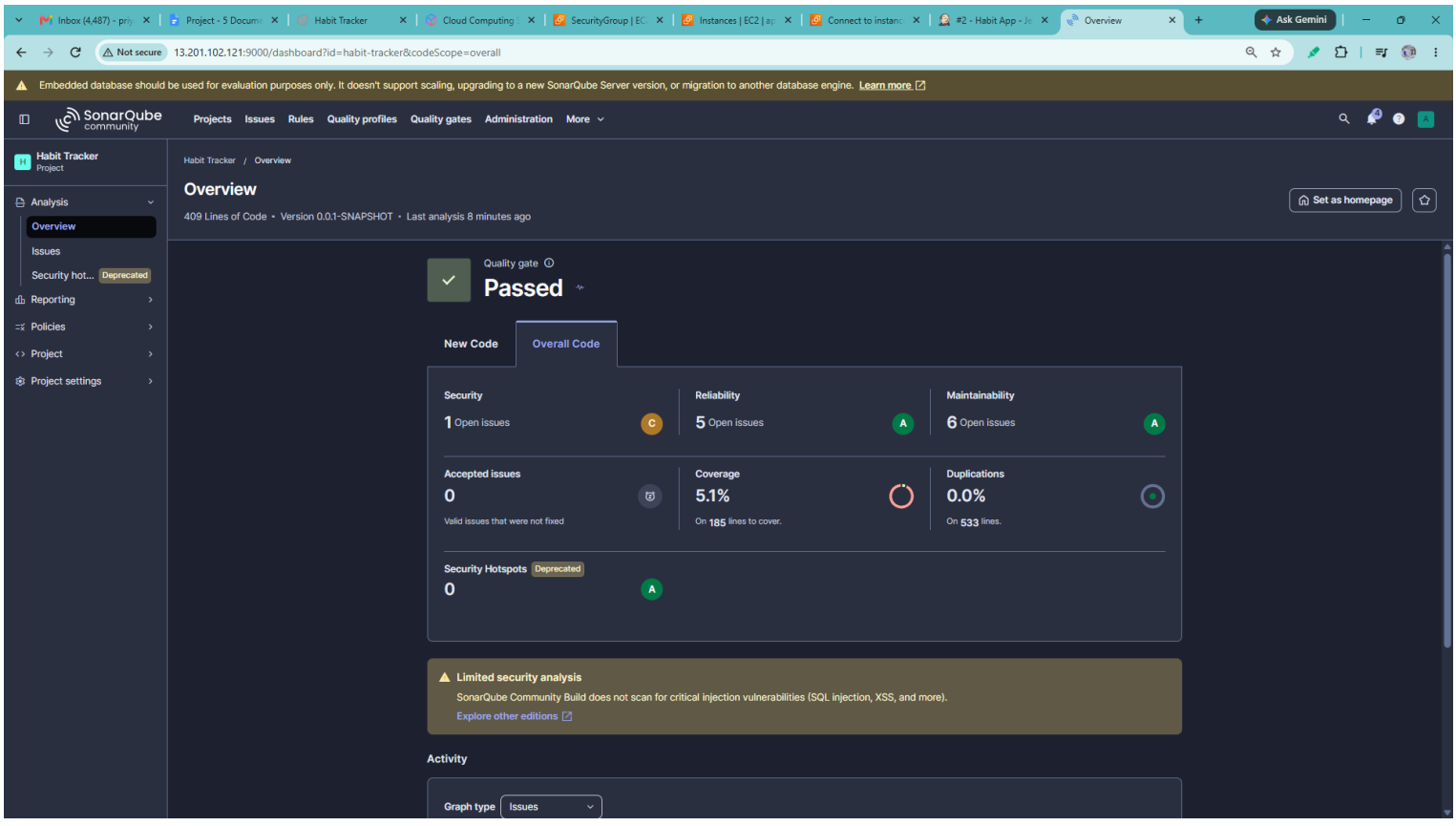
Quality Gate: Passed

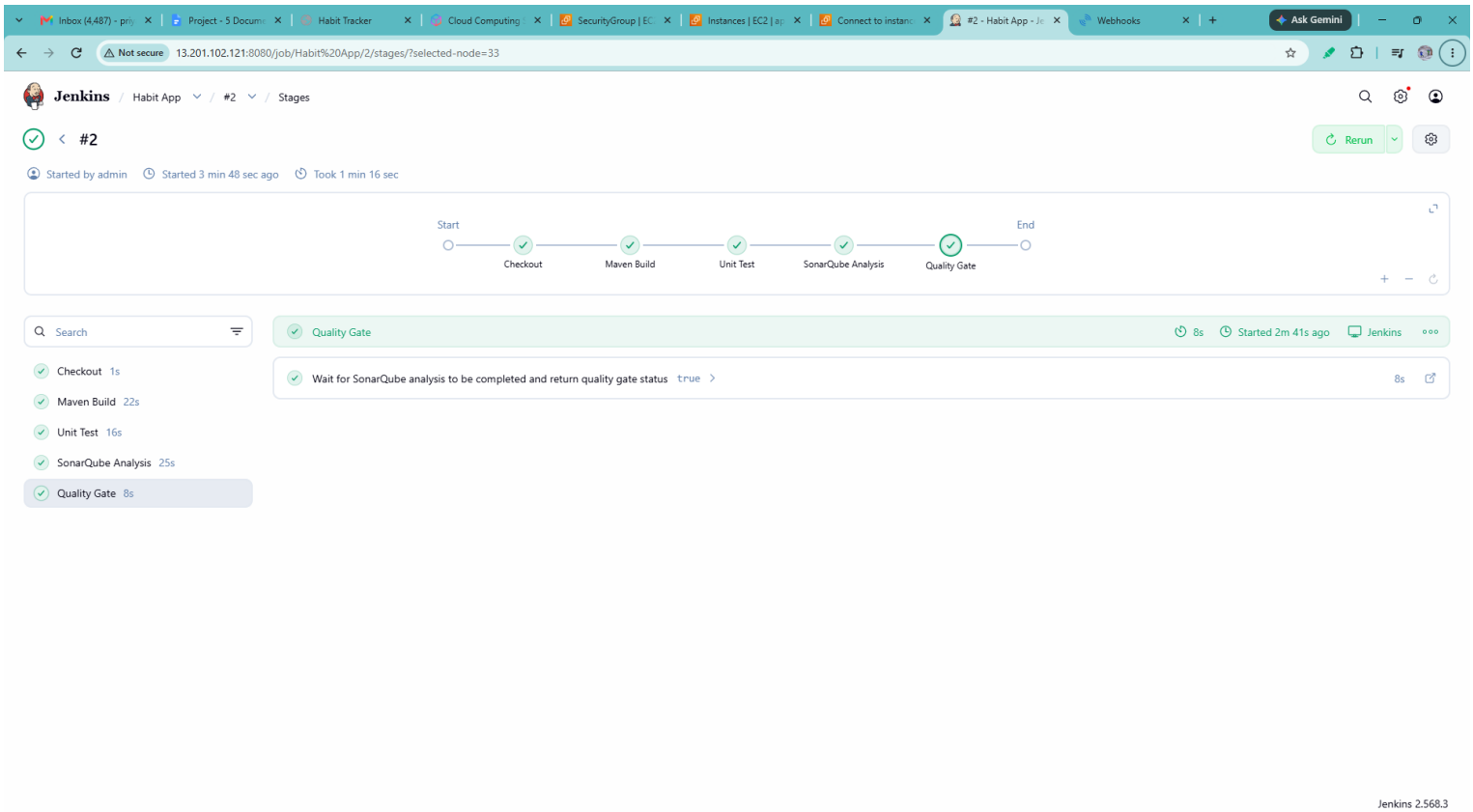
Result

SonarQube source-code analysis and Quality Gate validation were successfully integrated into the Jenkins pipeline.

The application passed the SonarQube Quality Gate before the pipeline completed.


File Edit Selection View Go Run Terminal Help
Project - 5
Explorer
Project - 5
HabitApp
src
main
java
resources
static
application.properties
test
target
.dockerignore
.gitattributes
Dockerfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
ubuntu@ip-172-31-7-143:~$ sudo docker run -d --name sonarqube -p 9000:9000 sonarqube:community
49c1136b049dcc51e442fbca1f8c11561efcd38b738c5100cc98af4d8cd1d721
ubuntu@ip-172-31-7-143:~$ sudo docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
49c1136b049d sonarqube:community "/opt/sonarqube/dock…" 9 seconds ago Up 8 seconds 0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp sonarqube
ubuntu@ip-172-31-7-143:~$
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Dockerfile
Describe what to build
Local
Outline
Timeline
Ln 9, Col 39 Spaces: 4 UTF-8 CRLF Docker


Not secure 13.201.102.121:9000/dashboard?id=habit-tracker&codeScope=overall
Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. Learn more
SonarQube community
Projects Issues Rules Quality profiles Quality gates Administration More
Habit Tracker
Project
Analysis
Overview
Issues
Security hot... Deprecated
Reporting
Policies
Project
Project settings
Habit Tracker / Overview
Overview
409 Lines of Code • Version 0.0.1-SNAPSHOT • Last analysis 8 minutes ago
Set as homepage
Quality gate
Passed
New Code
Overall Code
Security
1 Open issues
Reliability
5 Open issues
Maintainability
6 Open issues
Accepted issues
0
Valid issues that were not fixed
Coverage
5.1%
On 185 lines to cover.
Duplications
0.0%
On 533 lines.
Security Hotspots Deprecated
0
Limited security analysis
SonarQube Community Build does not scan for critical injection vulnerabilities (SQL injection, XSS, and more).
Explore other editions
Activity
Graph type Issues



Jenkins 2.568.3

Task 4 : Docker

Overview

The Habit Tracker Java application was containerized using Docker with a production-oriented Docker configuration.

A multi-stage Docker build was used with Java 21, a separate runtime stage, non-root execution, and a .dockerignore file.

Application Structure

- Application: Habit Tracker
- Java: 21
- Docker Image: habit-tracker:production
- Application Port: 8080
- Runtime User: appuser

Execution

A multi-stage Dockerfile was created with Maven and Java 21 for the build stage and Java 21 JRE for the runtime stage.

The application was packaged into the Docker image and configured to run as the non-root user appuser.

The production Docker image was successfully built and the container was started with port 8080 mapped to the host.

Verification

The Docker image was successfully created as habit-tracker:production.

The production container was running with port 8080 mapped correctly.

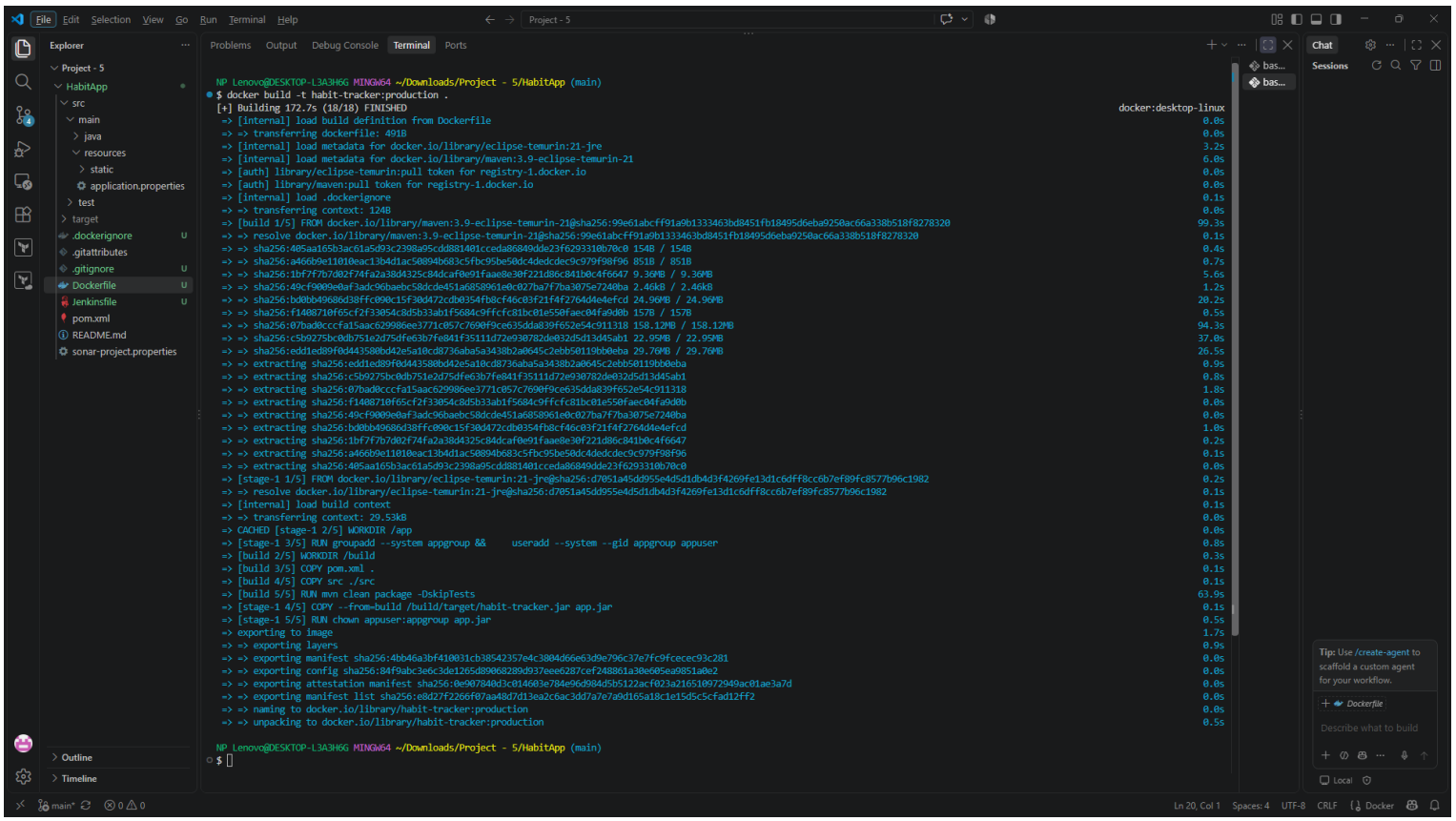
Container logs confirmed that the application started successfully using Java 21 and was running as appuser.

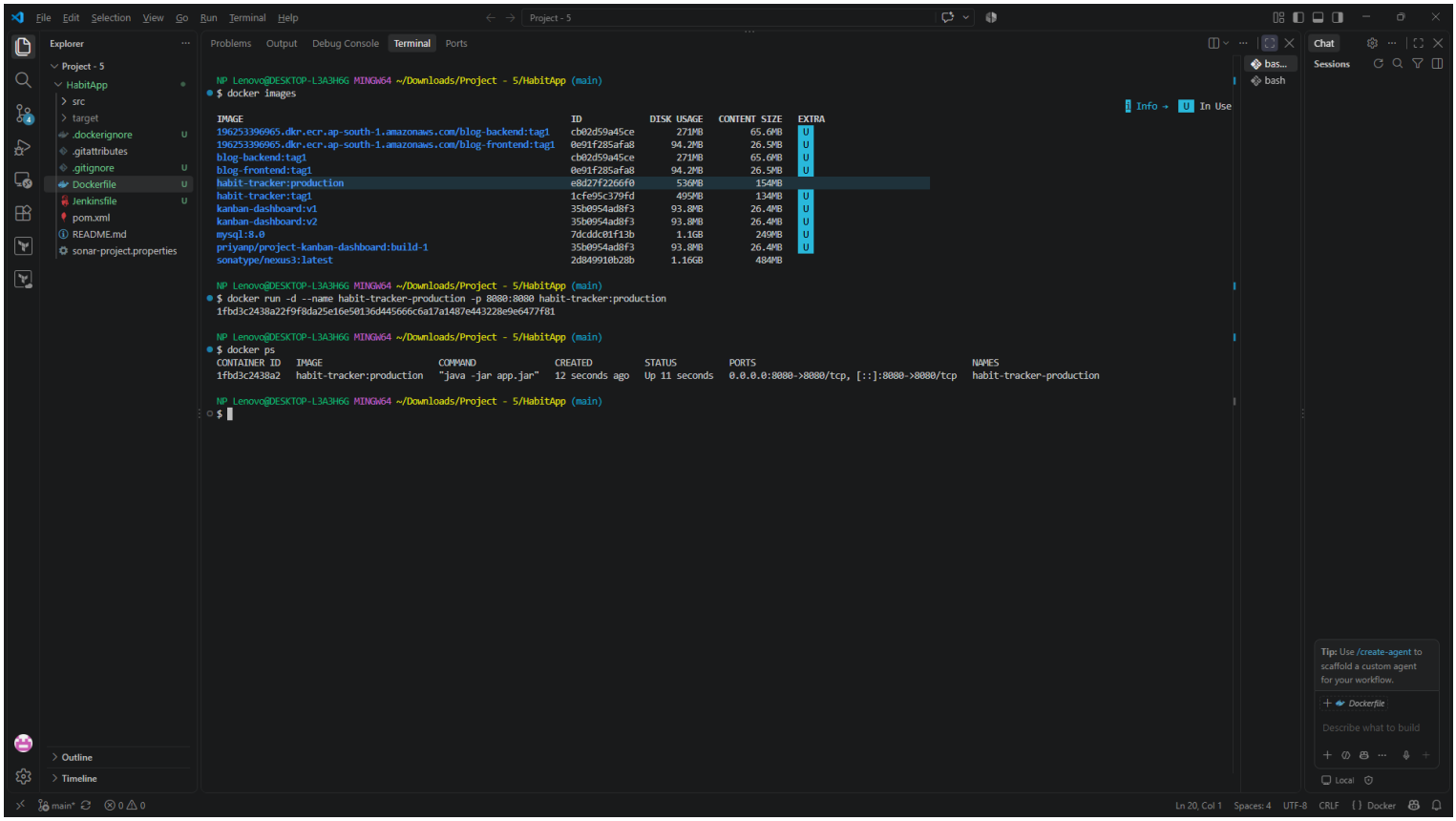
The Habit Tracker application was successfully accessed through http://localhost:8080

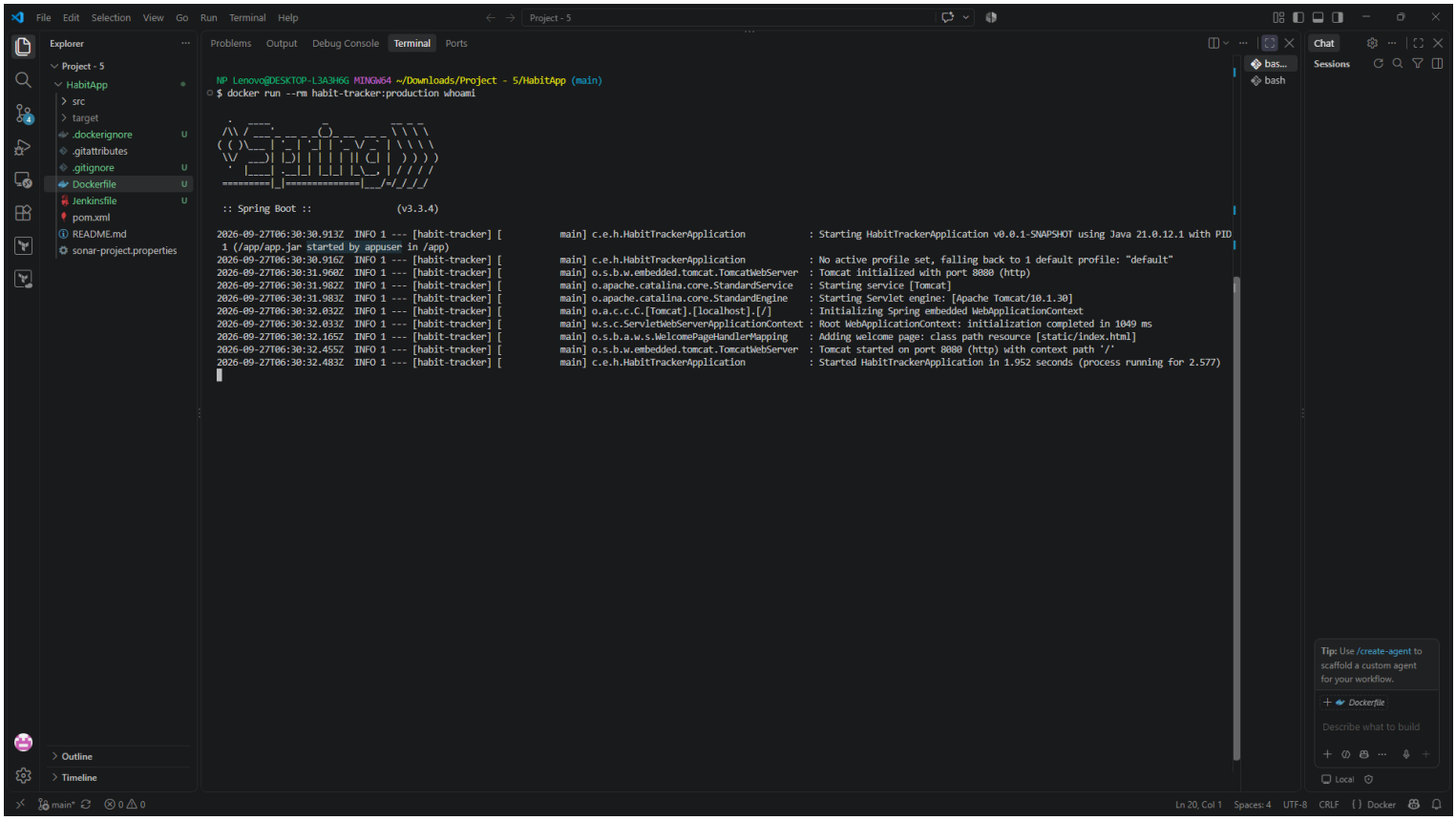
Docker image, container, logs, and application screenshots were captured as evidence.

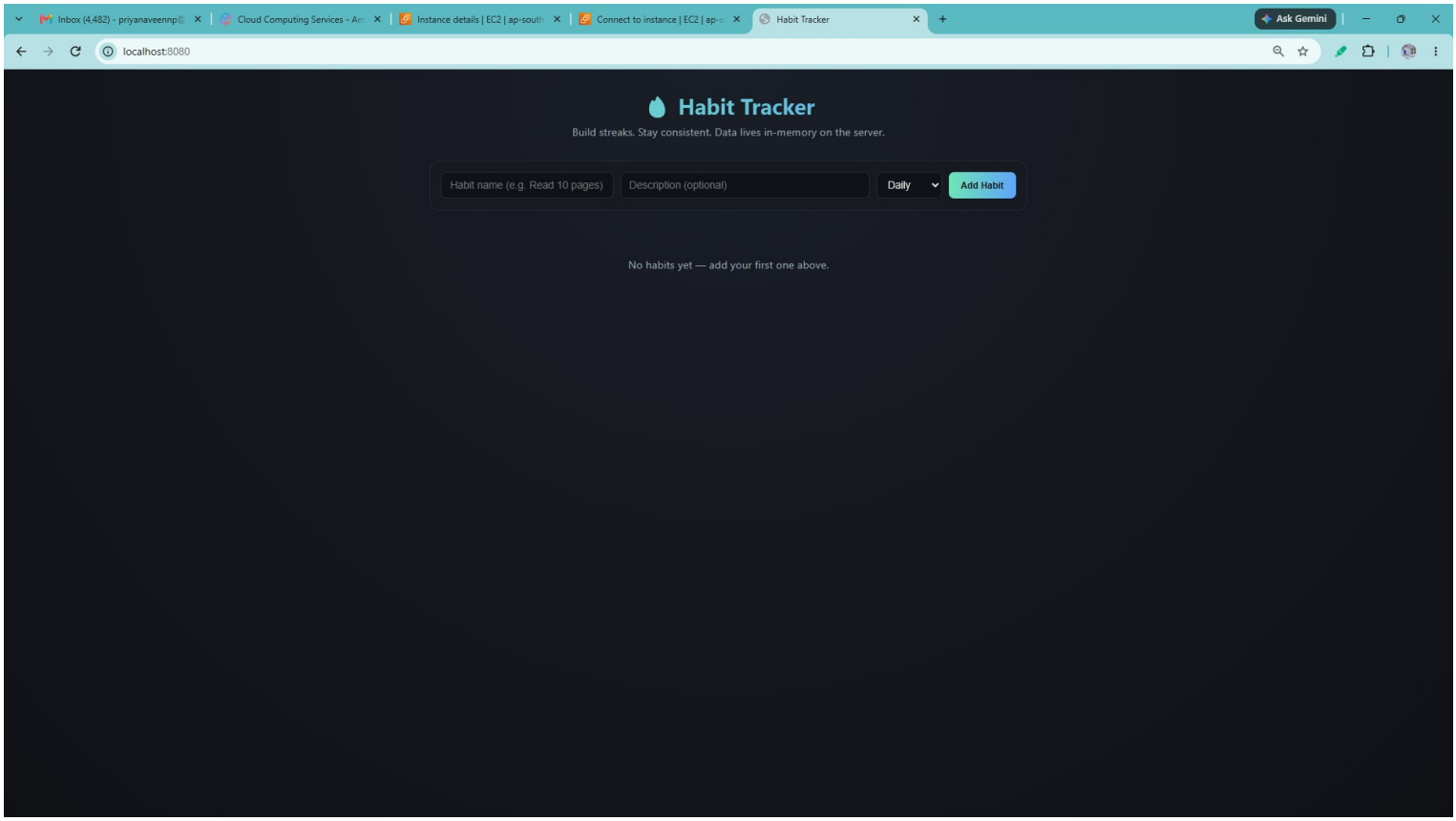
Result

The Habit Tracker application was successfully containerized using a production-oriented Docker configuration. The image built successfully, the application ran successfully inside the container, and the application was accessible on port 8080.




File Edit Selection View Go Run Terminal Help
Project - 5
Explorer
Project - 5
HabitApp
src
target
.dockerignore
.gitattributes
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$ docker images
Info U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
196253396965.dkr.ecr.ap-south-1.amazonaws.com/blog-backend:tag1 cb02d59a45ce 271MB 65.6MB U
196253396965.dkr.ecr.ap-south-1.amazonaws.com/blog-frontend:tag1 0e91f285afa8 94.2MB 26.5MB U
blog-backend:tag1 cb02d59a45ce 271MB 65.6MB U
blog-frontend:tag1 0e91f285afa8 94.2MB 26.5MB U
habit-tracker:production e8d27f2266f0 536MB 154MB
habit-tracker:tag1 1cfe95c379fd 495MB 134MB U
kanban-dashboard:v1 35b0954ad8f3 93.8MB 26.4MB U
kanban-dashboard:v2 35b0954ad8f3 93.8MB 26.4MB U
mysql:8.0 7dcddc01f13b 1.1GB 249MB U
priyanp/project-kanban-dashboard:build-1 35b0954ad8f3 93.8MB 26.4MB U
sonatype/nexus3:latest 2d849910b28b 1.16GB 484MB
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$ docker run -d --name habit-tracker-production -p 8080:8080 habit-tracker:production
1fbd3c2438a22f9f8da25e16e50136d445666c6a17a1487e443228e9e6477f81
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
1fbd3c2438a2 habit-tracker:production "java -jar app.jar" 12 seconds ago Up 11 seconds 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp habit-tracker-production
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$
Chat
Sessions
bas...
bash
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Dockerfile
Describe what to build
Local
Outline
Timeline
main*
Ln 20, Col 1 Spaces: 4 UTF-8 CRLF Docker


File Edit Selection View Go Run Terminal Help
Project - 5
Explorer
Project - 5
HabitApp
src
target
.dockerignore
.gitattributes
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$ docker run --rm habit-tracker:production whoami
:: Spring Boot :: (v3.3.4)
2026-09-27T06:30:30.913Z INFO 1 --- [habit-tracker] [main] c.e.h.HabitTrackerApplication : Starting HabitTrackerApplication v0.0.1-SNAPSHOT using Java 21.0.12.1 with PID 1 (/app/app.jar started by appuser in /app)
2026-09-27T06:30:30.916Z INFO 1 --- [habit-tracker] [main] c.e.h.HabitTrackerApplication : No active profile set, falling back to 1 default profile: "default"
2026-09-27T06:30:31.960Z INFO 1 --- [habit-tracker] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-09-27T06:30:31.982Z INFO 1 --- [habit-tracker] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-09-27T06:30:31.983Z INFO 1 --- [habit-tracker] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.30]
2026-09-27T06:30:32.032Z INFO 1 --- [habit-tracker] [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-09-27T06:30:32.033Z INFO 1 --- [habit-tracker] [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 1049 ms
2026-09-27T06:30:32.165Z INFO 1 --- [habit-tracker] [main] o.s.b.a.w.s.WelcomePageHandlerMapping : Adding welcome page: class path resource [static/index.html]
2026-09-27T06:30:32.455Z INFO 1 --- [habit-tracker] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-09-27T06:30:32.483Z INFO 1 --- [habit-tracker] [main] c.e.h.HabitTrackerApplication : Started HabitTrackerApplication in 1.952 seconds (process running for 2.577)
Chat
Sessions
bas...
bash
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Dockerfile
Describe what to build
Local
Outline
Timeline
main*
Ln 20, Col 1 Spaces: 4 UTF-8 CRLF Docker


Inbox (4,482) - priyanaveennp@
Cloud Computing Services - An
Instance details | EC2 | ap-south
Connect to instance | EC2 | ap-s
Habit Tracker
Ask Gemini
localhost:8080
Habit Tracker
Build streaks. Stay consistent. Data lives in-memory on the server.
Habit name (e.g. Read 10 pages)
Description (optional)
Daily
Add Habit
No habits yet — add your first one above.

Task 5 : Container Registry

Overview

The Habit Tracker Docker image was pushed to Amazon Elastic Container Registry (ECR) for centralized container image storage.

A Git commit SHA was used as the image version to provide traceability between the source code version and the container image.

Application Structure

- Container Registry: Amazon ECR
- Repository: habit-tracker
- Region: ap-south-1
- Image Version: 03117ad
- Registry URI: 196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker

Execution

The production Docker image was tagged using the Git commit SHA 03117ad and pushed to the Amazon ECR habit-tracker repository.

The image was successfully uploaded and became available in the ECR repository.

Verification

The ECR repository was verified successfully.

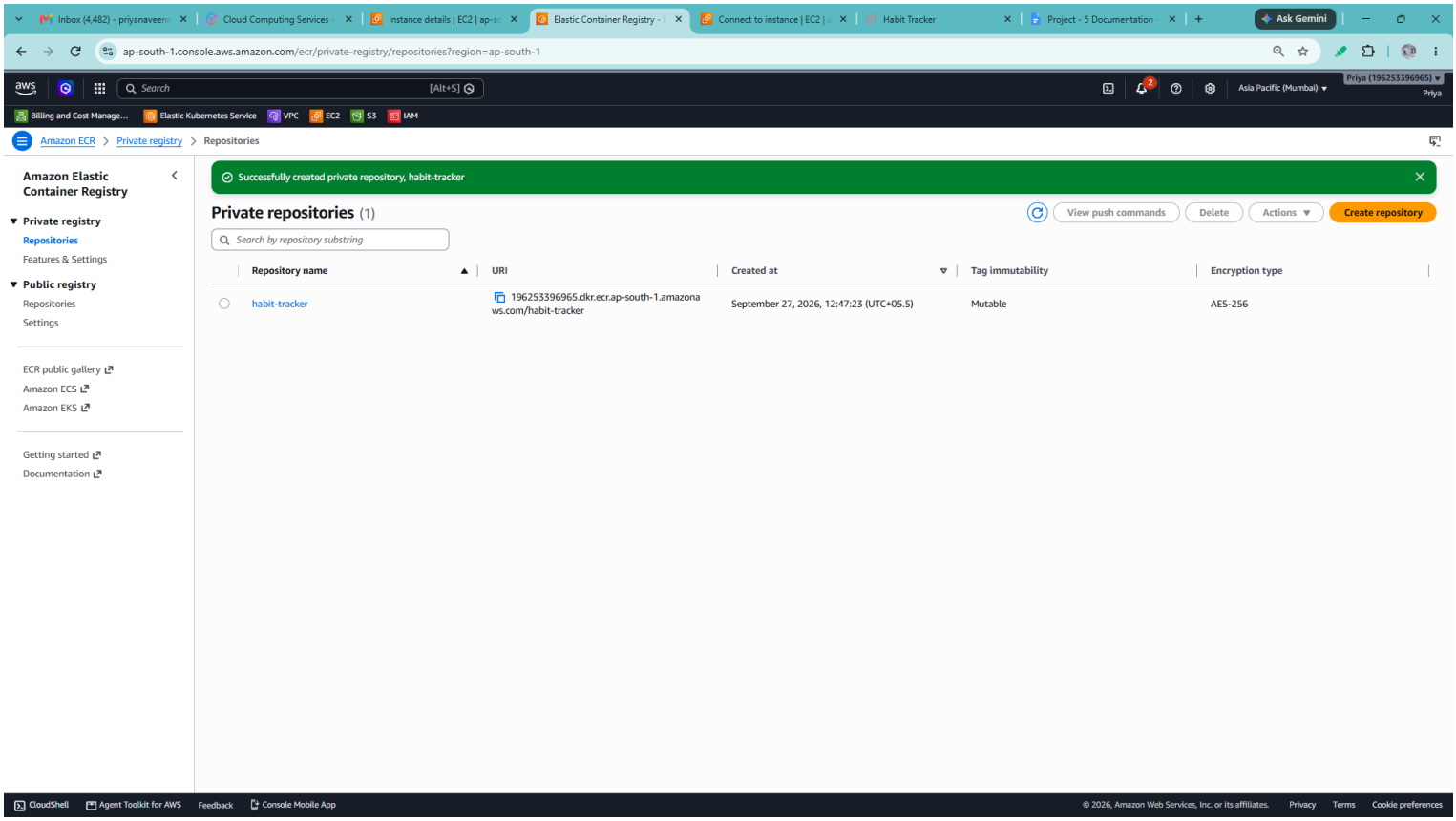
The image was available with the tag : 03117ad

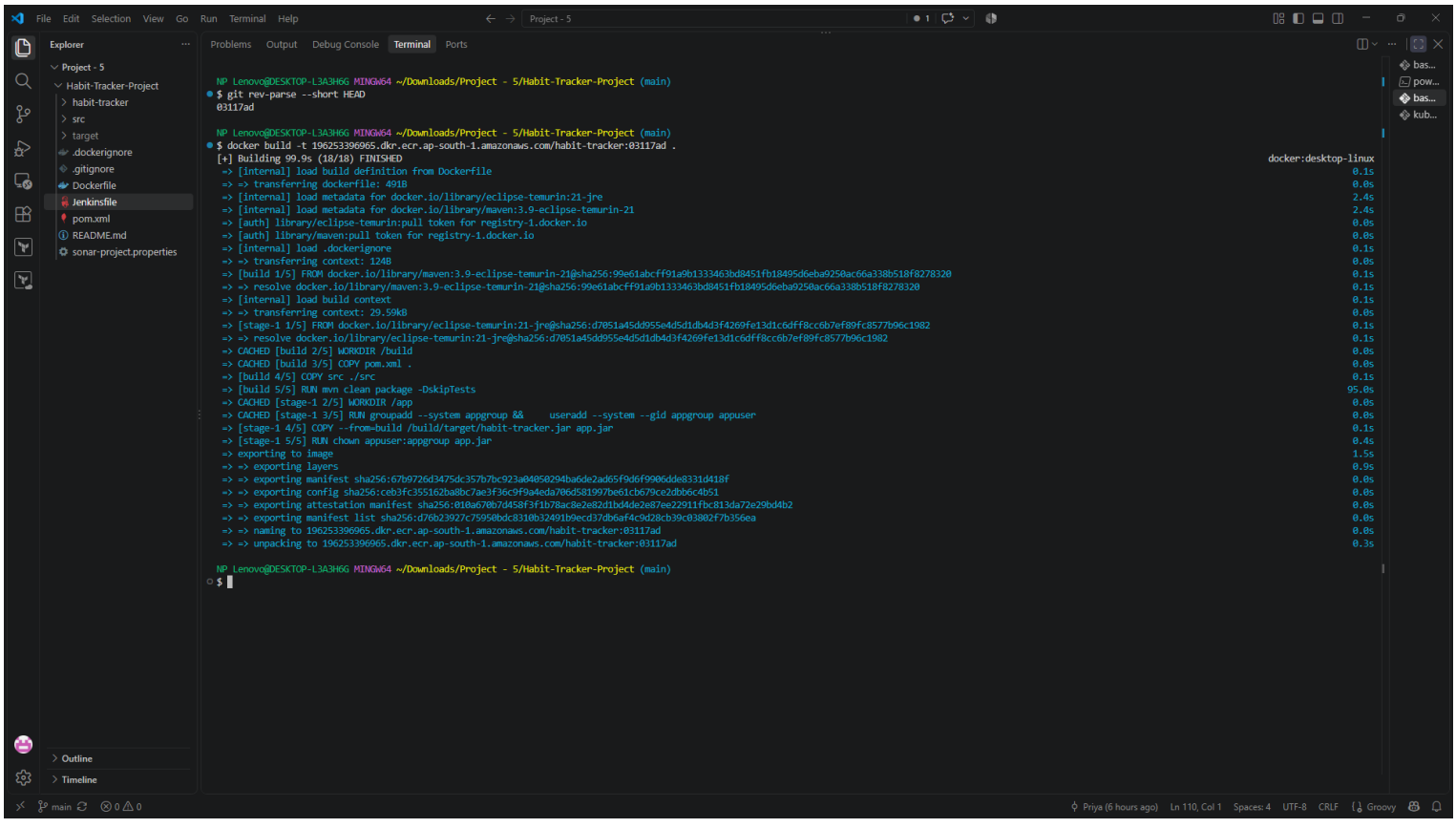
The Git SHA-based image provides a traceable version that can be used by the Kubernetes deployment in the later deployment stages.

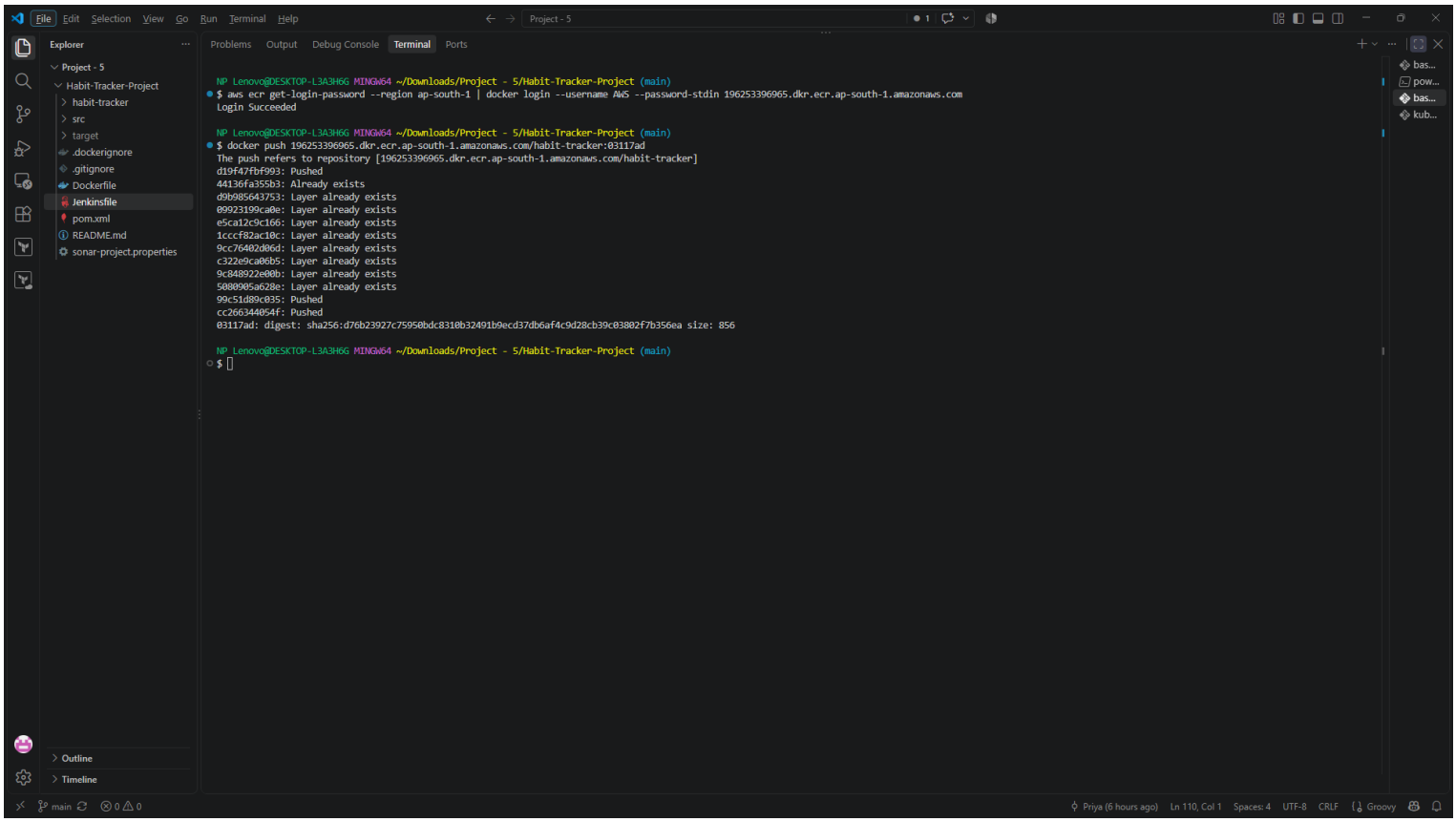
ECR repository and image screenshots were captured as evidence.

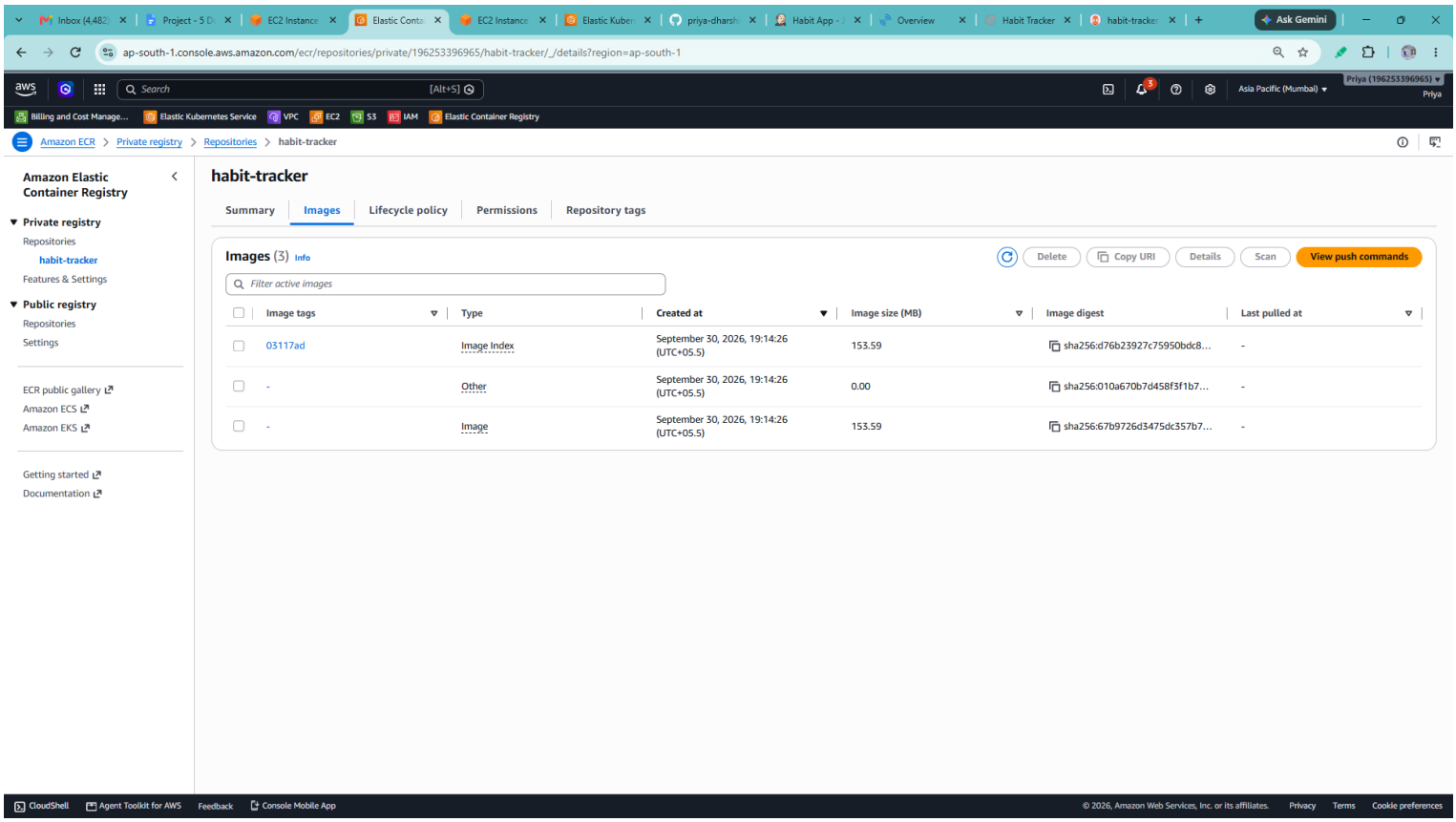
Result

The production Docker image was successfully pushed to Amazon ECR using the Git commit SHA 03117ad as the image version.


Amazon Elastic Container Registry
Private registry
Repositories
Features & Settings
Public registry
Repositories
Settings
ECR public gallery
Amazon ECS
Amazon EKS
Getting started
Documentation
Successfully created private repository, habit-tracker
Private repositories (1)
View push commands
Delete
Actions
Create repository
Search by repository substring
Repository name
URI
Created at
Tag immutability
Encryption type
habit-tracker
196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker
September 27, 2026, 12:47:23 (UTC+05.5)
Mutable
AES-256


Explorer
Project - 5
Habit-Tracker-Project
habit-tracker
src
target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems
Output
Debug Console
Terminal
Ports
$ git rev-parse --short HEAD
03117ad
$ docker build -t 196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker:03117ad .
[+] Building 99.9s (18/18) FINISHED
docker:desktop-linux
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 491B
=> [internal] load metadata for docker.io/library/eclipse-temurin:21-jre
=> [internal] load metadata for docker.io/library/maven:3.9-eclipse-temurin-21
=> [auth] library/eclipse-temurin:pull token for registry-1.docker.io
=> [auth] library/maven:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 124B
=> [build 1/5] FROM docker.io/library/maven:3.9-eclipse-temurin-21@sha256:99e61abcff91a9b1333463bd8451fb18495d6eba9250ac66a338b518f8278320
=> => resolve docker.io/library/maven:3.9-eclipse-temurin-21@sha256:99e61abcff91a9b1333463bd8451fb18495d6eba9250ac66a338b518f8278320
=> [internal] load build context
=> => transferring context: 29.59kB
=> [stage-1 1/5] FROM docker.io/library/eclipse-temurin:21-jre@sha256:d7051a45dd955e4d5d1db4d3f4269fe13d1c6dff8cc6b7ef89fc8577b96c1982
=> => resolve docker.io/library/eclipse-temurin:21-jre@sha256:d7051a45dd955e4d5d1db4d3f4269fe13d1c6dff8cc6b7ef89fc8577b96c1982
=> CACHED [build 2/5] WORKDIR /build
=> CACHED [build 3/5] COPY pom.xml .
=> [build 4/5] COPY src ./src
=> [build 5/5] RUN mvn clean package -DskipTests
=> CACHED [stage-1 2/5] WORKDIR /app
=> CACHED [stage-1 3/5] RUN groupadd --system appgroup && useradd --system --gid appgroup appuser
=> [stage-1 4/5] COPY --from=build /build/target/habit-tracker.jar app.jar
=> [stage-1 5/5] RUN chown appuser:appgroup app.jar
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:67b9726d3475dc357b7bc923a04050294ba6de2ad65f9d6f9906dde8331d418f
=> => exporting config sha256:ceb3fc355162ba8bc7ae3f36c9f9a4eda706d581997be61cb679ce2dbb6c4b51
=> => exporting attestation manifest sha256:010a670b7d458f3f1b78ac8e2e82d1bd4de2e87ee22911fbc813da72e29bd4b2
=> => exporting manifest list sha256:d76b23927c75950bdc8310b32491b9ecd37db6af4c9d28cb39c03802f7b356ea
=> => naming to 196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker:03117ad
=> => unpacking to 196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker:03117ad
$
Outline
Timeline


File Edit Selection View Go Run Terminal Help
Project - 5
Explorer
Project - 5
Habit-Tracker-Project
habit-tracker
src
target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin 196253396965.dkr.ecr.ap-south-1.amazonaws.com
Login Succeeded
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ docker push 196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker:03117ad
The push refers to repository [196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker]
d19f47fbf993: Pushed
44136fa355b3: Already exists
d9b985643753: Layer already exists
09923199ca0e: Layer already exists
e5ca12c9c166: Layer already exists
1cccf82ac10c: Layer already exists
9cc76402d06d: Layer already exists
c322e9ca06b5: Layer already exists
9c848922e00b: Layer already exists
5080905a628e: Layer already exists
99c51d89c035: Pushed
cc266344054f: Pushed
03117ad: digest: sha256:d76b23927c75950bdc8310b32491b9ecd37db6af4c9d28cb39c03802f7b356ea size: 856
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Outline
Timeline
main
Priya (6 hours ago) Ln 110, Col 1 Spaces: 4 UTF-8 CRLF Groovy


ap-south-1.console.aws.amazon.com/ecr/repositories/private/196253396965/habit-tracker/_/details?region=ap-south-1
Search [Alt+S]
Asia Pacific (Mumbai)
Priya (196253396965)
Billing and Cost Manage... Elastic Kubernetes Service VPC EC2 S3 IAM Elastic Container Registry
Amazon ECR > Private registry > Repositories > habit-tracker
Amazon Elastic Container Registry
Private registry
Repositories
habit-tracker
Features & Settings
Public registry
Repositories
Settings
ECR public gallery
Amazon ECS
Amazon EKS
Getting started
Documentation
habit-tracker
Summary Images Lifecycle policy Permissions Repository tags
Images (3) Info
Delete Copy URI Details Scan View push commands
Filter active images
Image tags Type Created at Image size (MB) Image digest Last pulled at
03117ad Image Index September 30, 2026, 19:14:26 (UTC+05.5) 153.59 sha256:d76b23927c75950bdc8... -
- Other September 30, 2026, 19:14:26 (UTC+05.5) 0.00 sha256:010a670b7d458f3f1b7... -
- Image September 30, 2026, 19:14:26 (UTC+05.5) 153.59 sha256:67b9726d3475dc357b7... -
CloudShell Agent Toolkit for AWS Feedback Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Task 6 : Helm

Overview

A reusable Helm chart was created to deploy the Habit Tracker Spring Boot application to Kubernetes. The chart manages the Deployment, Service, ConfigMap, optional Secret configuration, resource settings, and health probes using configurable values in values.yaml.

Application Structure

- Helm Chart: habit-tracker
- Kubernetes: Amazon EKS
- Container Registry: Amazon ECR
- Image Version: 03117ad
- Replicas: 2
- Service Type: LoadBalancer
- Application Port: 8080

Execution

The Helm chart was configured using values.yaml and Helm templates for:

- Deployment
- Service
- ConfigMap
- Optional Secret configuration
- Resource requests and limits
- Liveness probe and Readiness probe

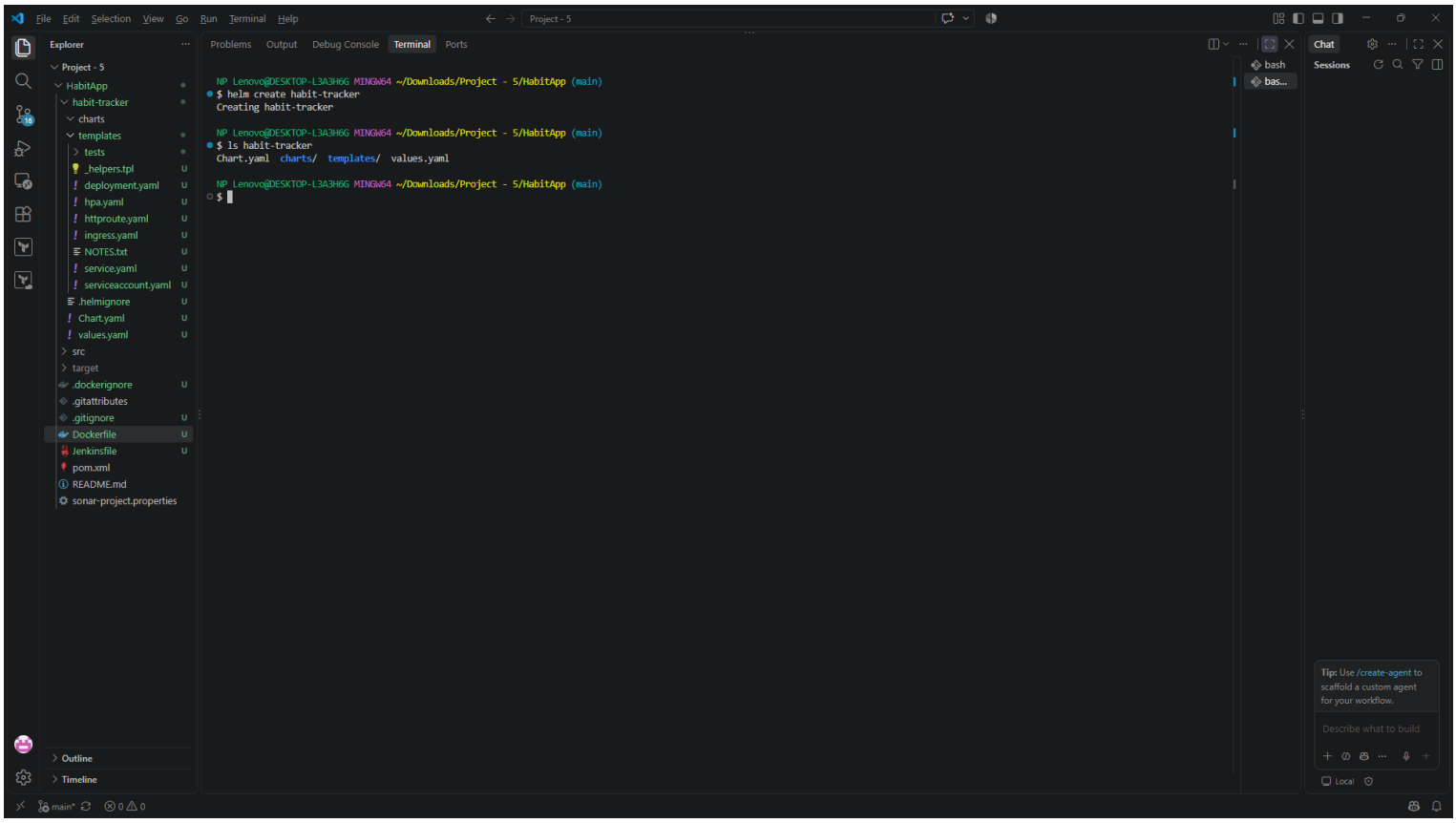
The Helm chart was validated successfully using helm lint. The result was 1 chart(s) linted, 0 chart(s) failed.

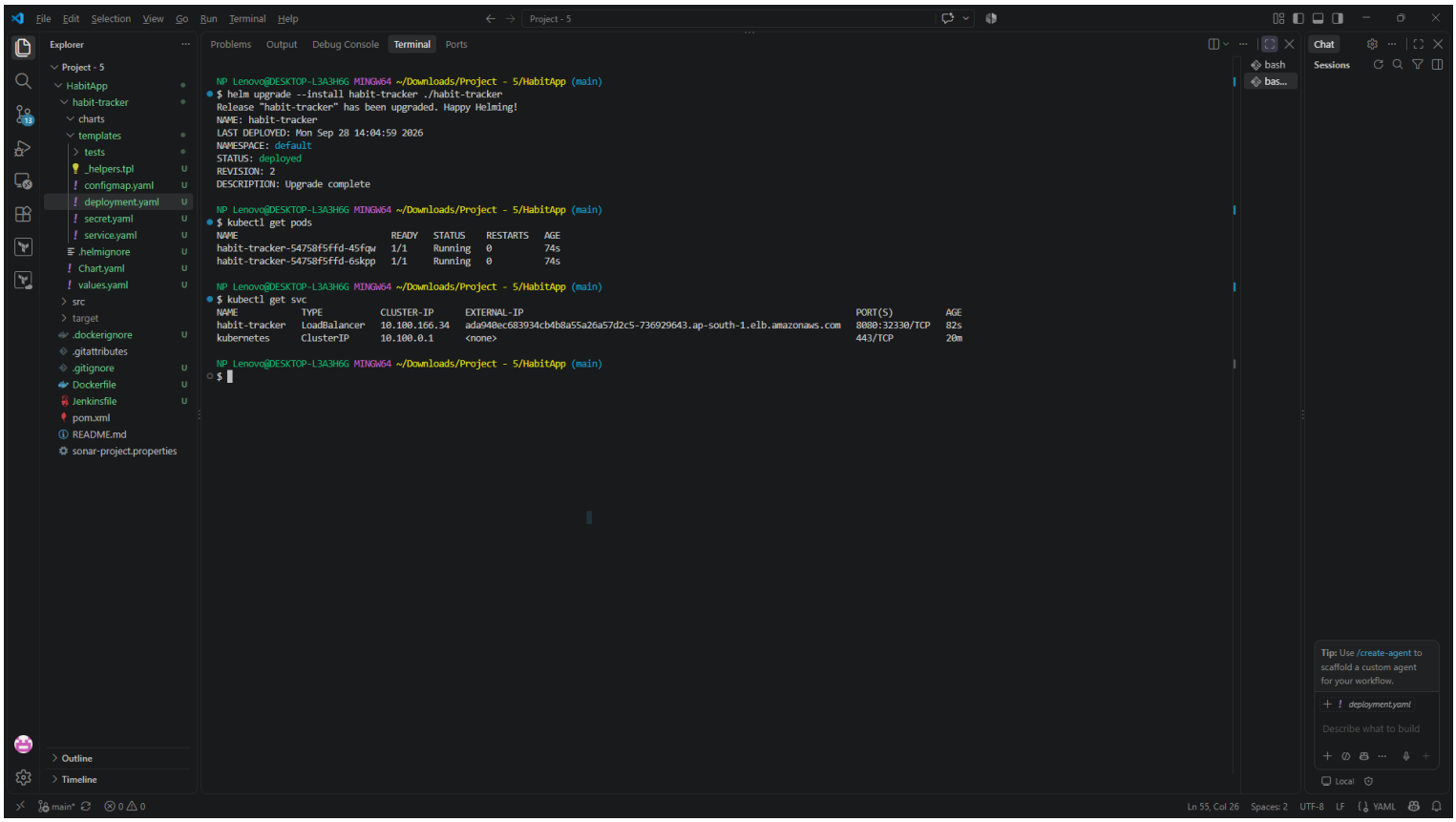
Verification

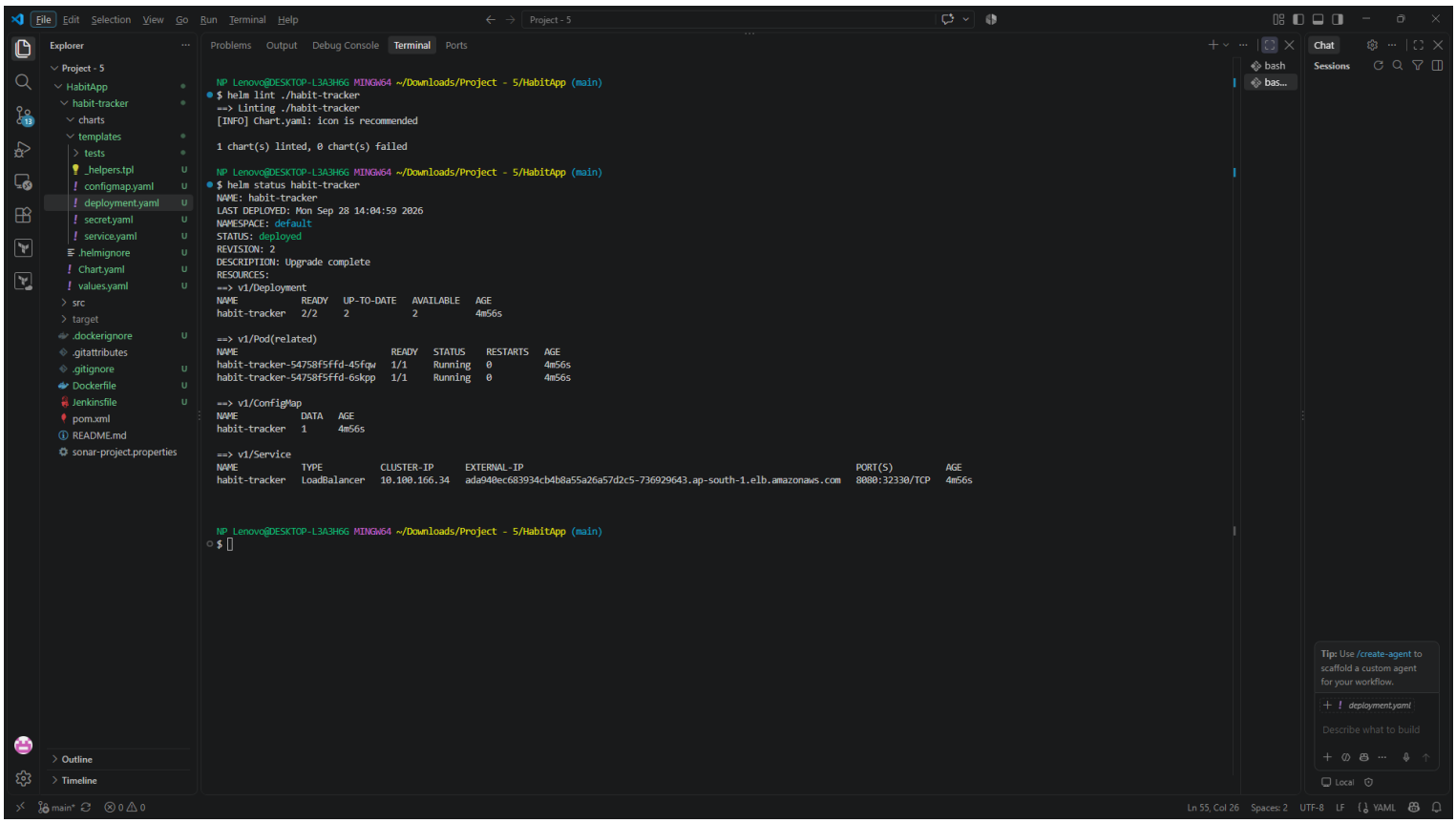
- Helm chart validation completed successfully.
- Deployment: 2/2 replicas available
- Pods: 2/2 Running
- ConfigMap: Created
- Service: LoadBalancer
- Application: Successfully accessible through the LoadBalancer

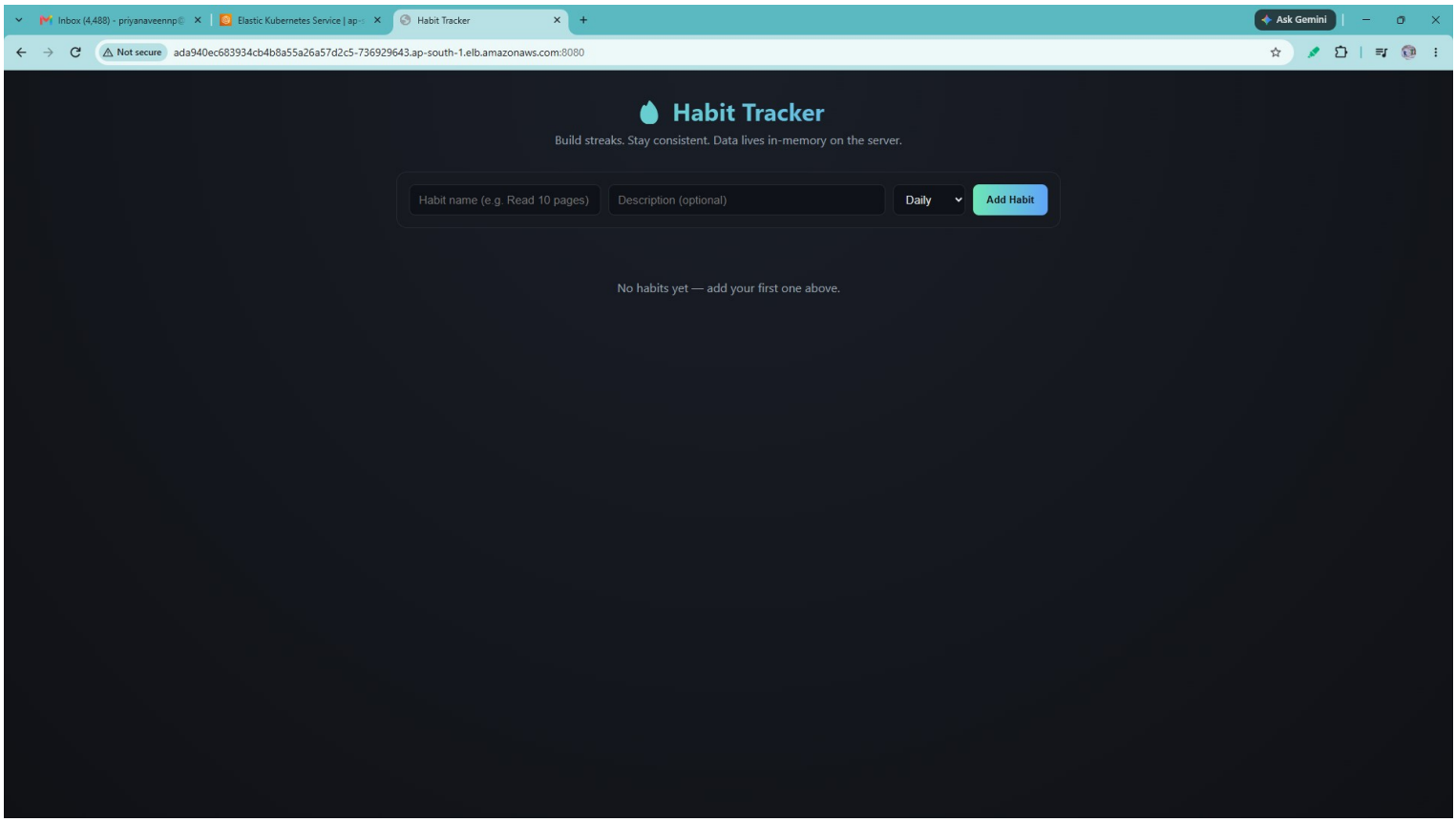
Result

The Habit Tracker application was successfully configured and deployed using a reusable Helm chart with configurable deployment values and the required Kubernetes resources.




File Edit Selection View Go Run Terminal Help
Project - 5
Explorer
Project - 5
HabitApp
habit-tracker
charts
templates
tests
_helpers.tpl
configmap.yaml
deployment.yaml
secret.yaml
service.yaml
.helmignore
Chart.yaml
values.yaml
src
target
.dockerignore
.gitattributes
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Outline
Timeline
Problems Output Debug Console Terminal Ports
bash
bas...
Chat
Sessions
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$ helm upgrade --install habit-tracker ./habit-tracker
Release "habit-tracker" has been upgraded. Happy Helming!
NAME: habit-tracker
LAST DEPLOYED: Mon Sep 28 14:04:59 2026
NAMESPACE: default
STATUS: deployed
REVISION: 2
DESCRIPTION: Upgrade complete
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$ kubectl get pods
NAME READY STATUS RESTARTS AGE
habit-tracker-54758f5ffd-45fqw 1/1 Running 0 74s
habit-tracker-54758f5ffd-6skpp 1/1 Running 0 74s
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$ kubectl get svc
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
habit-tracker LoadBalancer 10.100.166.34 ada940ec683934cb4b8a55a26a57d2c5-736929643.ap-south-1.elb.amazonaws.com 8080:32330/TCP 82s
kubernetes ClusterIP 10.100.0.1 <none> 443/TCP 20m
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$
Tip: Use /create-agent to scaffold a custom agent for your workflow.
deployment.yaml
Describe what to build
Local
main*
Ln 55, Col 26 Spaces: 2 UTF-8 LF YAML


File Edit Selection View Go Run Terminal Help
Project - 5
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Project - 5
HabitApp
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service.yaml
.helmignore
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values.yaml
src
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.dockerignore
.gitattributes
.gitignore
Dockerfile
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bash
bas...
Chat
Sessions
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$ helm lint ./habit-tracker
==> Linting ./habit-tracker
[INFO] Chart.yaml: icon is recommended
1 chart(s) linted, 0 chart(s) failed
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$ helm status habit-tracker
NAME: habit-tracker
LAST DEPLOYED: Mon Sep 28 14:04:59 2026
NAMESPACE: default
STATUS: deployed
REVISION: 2
DESCRIPTION: Upgrade complete
RESOURCES:
==> v1/Deployment
NAME READY UP-TO-DATE AVAILABLE AGE
habit-tracker 2/2 2 2 4m56s
==> v1/Pod(related)
NAME READY STATUS RESTARTS AGE
habit-tracker-54758f5ffd-45fqw 1/1 Running 0 4m56s
habit-tracker-54758f5ffd-6skpp 1/1 Running 0 4m56s
==> v1/ConfigMap
NAME DATA AGE
habit-tracker 1 4m56s
==> v1/Service
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
habit-tracker LoadBalancer 10.100.166.34 ada940ec683934cb4b8a55a26a57d2c5-736929643.ap-south-1.elb.amazonaws.com 8080:32330/TCP 4m56s
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/HabitApp (main)
$
Tip: Use /create-agent to scaffold a custom agent for your workflow.
deployment.yaml
Describe what to build
Local
main*
Ln 55, Col 26 Spaces: 2 UTF-8 LF YAML


Inbox (4,488) - priyanaveennp@
Elastic Kubernetes Service | ap
Habit Tracker
Ask Gemini
Not secure
ada940ec683934cb4b8a55a26a57d2c5-736929643.ap-south-1.elb.amazonaws.com:8080
Habit Tracker
Build streaks. Stay consistent. Data lives in-memory on the server.
Habit name (e.g. Read 10 pages)
Description (optional)
Daily
Add Habit
No habits yet — add your first one above.

Task 7 : Kubernetes

Overview

The Habit Tracker application was deployed to Amazon EKS using Helm with the required Kubernetes configuration for availability, health checks, resource management, application configuration, and rolling updates.

Application Structure

- Kubernetes: Amazon EKS
- Deployment: habit-tracker
- Replicas: 2
- Service: LoadBalancer
- Application Port: 8080

Execution

The Kubernetes Deployment was configured with multiple replicas, resource requests and limits, readiness and liveness probes, application configuration, and a RollingUpdate strategy.

The deployment was updated using a rolling restart to verify that Kubernetes could replace the application pods without stopping the application.

Verification

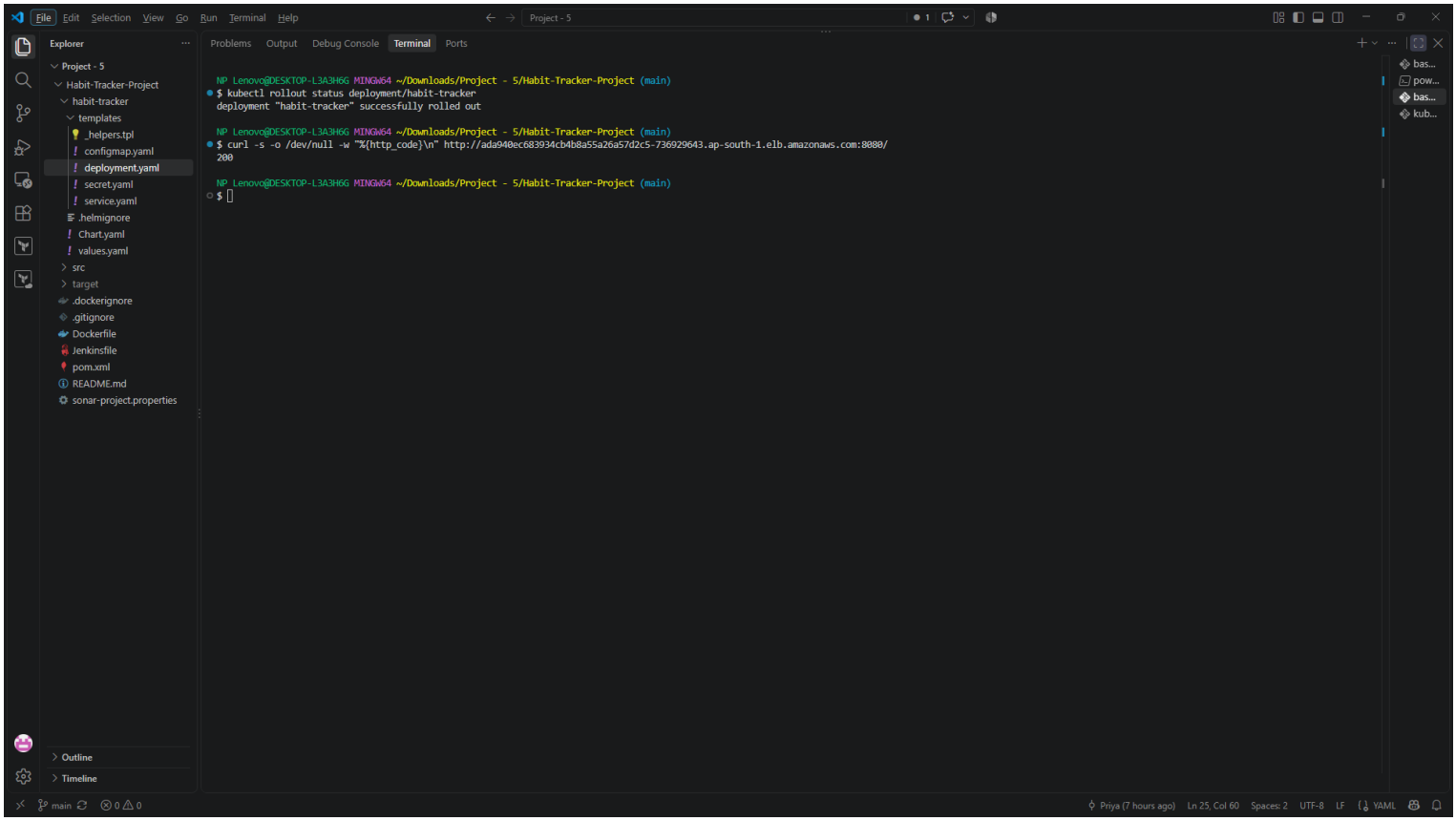
The deployment strategy was verified as : RollingUpdate

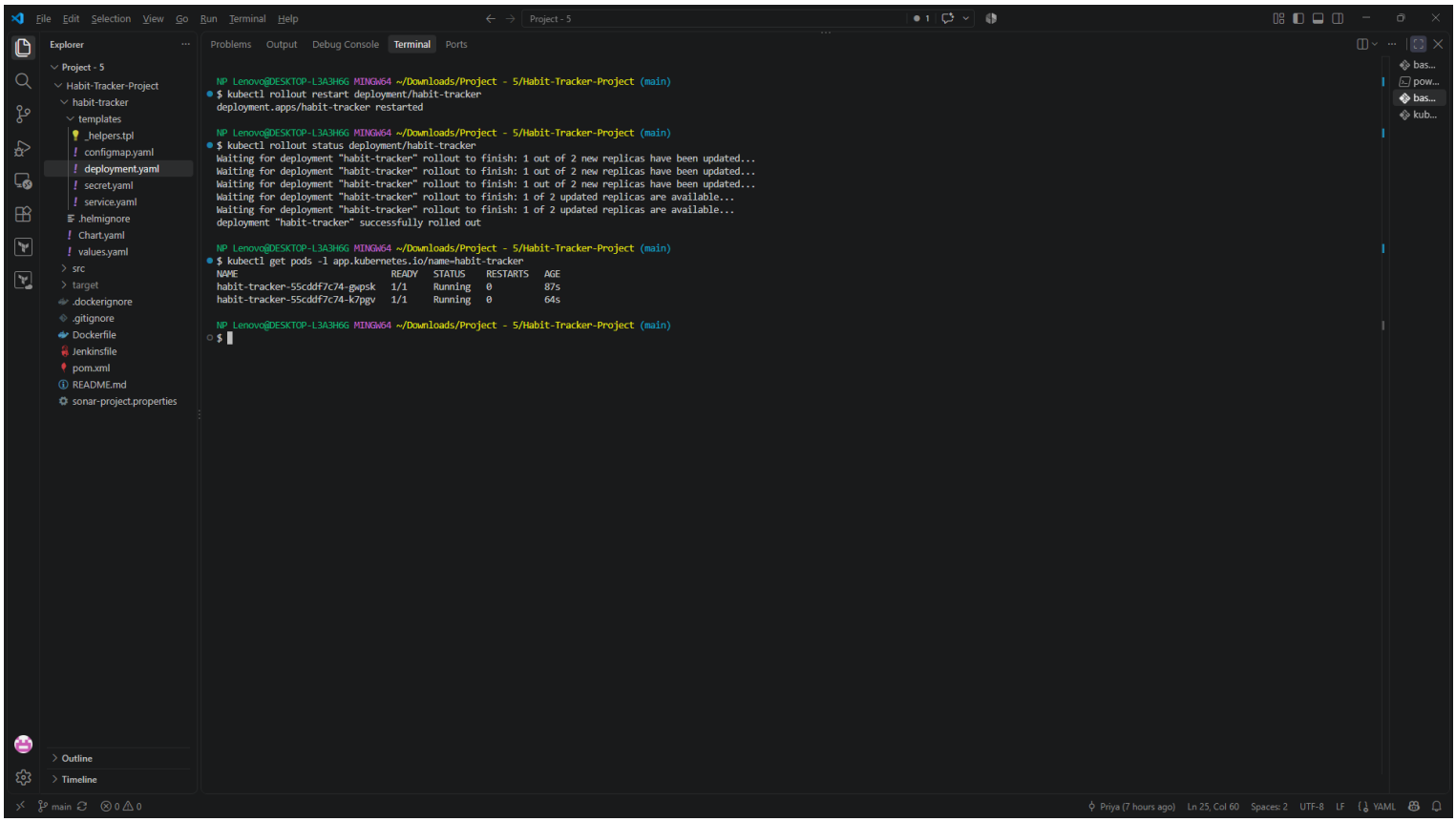
The rolling update completed successfully.

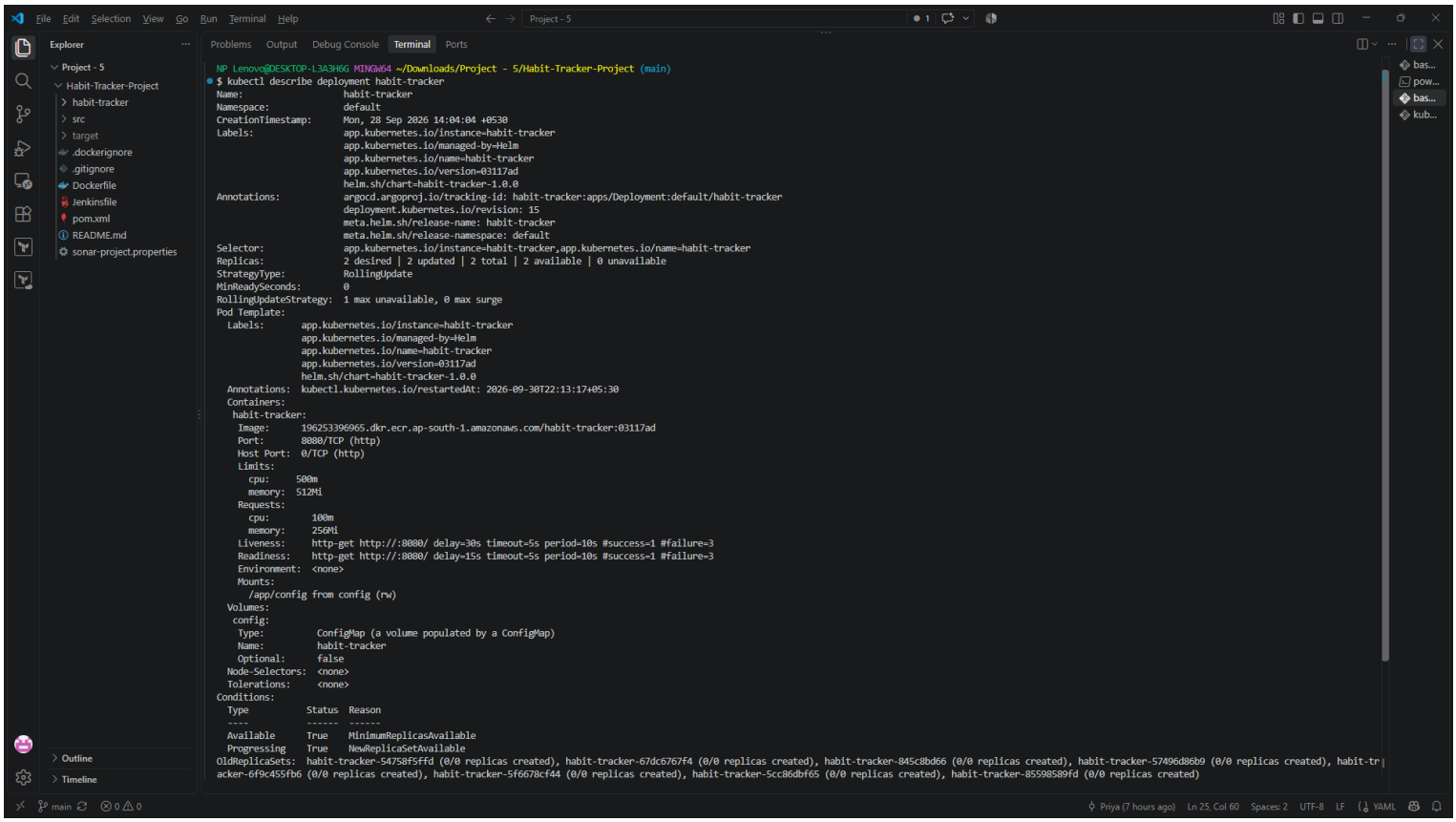
- Replicas: 2/2
- Pods: 2/2 Running
- Restarts: 0
- Application: HTTP 200 responses verified after rolling update

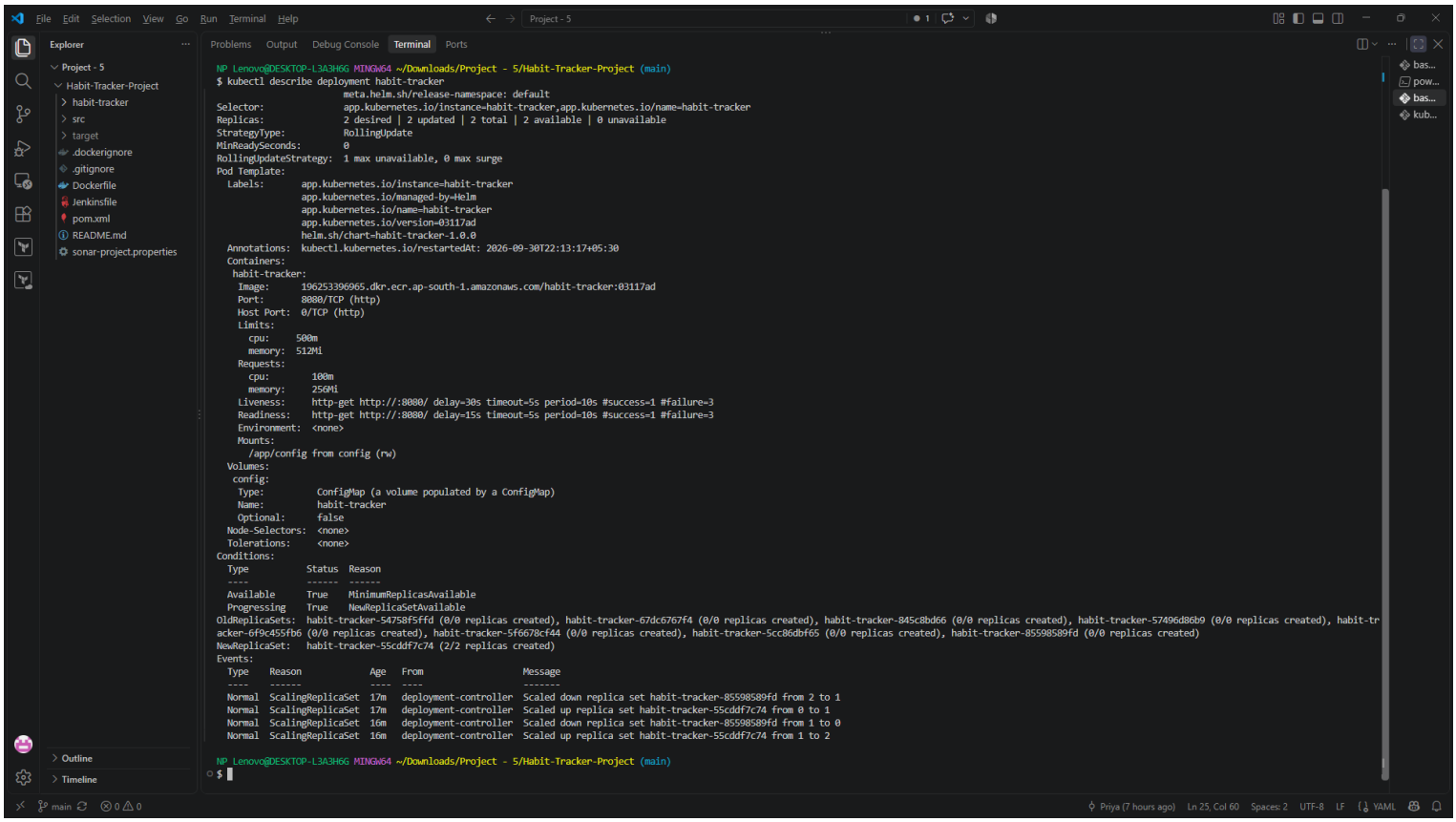
Result

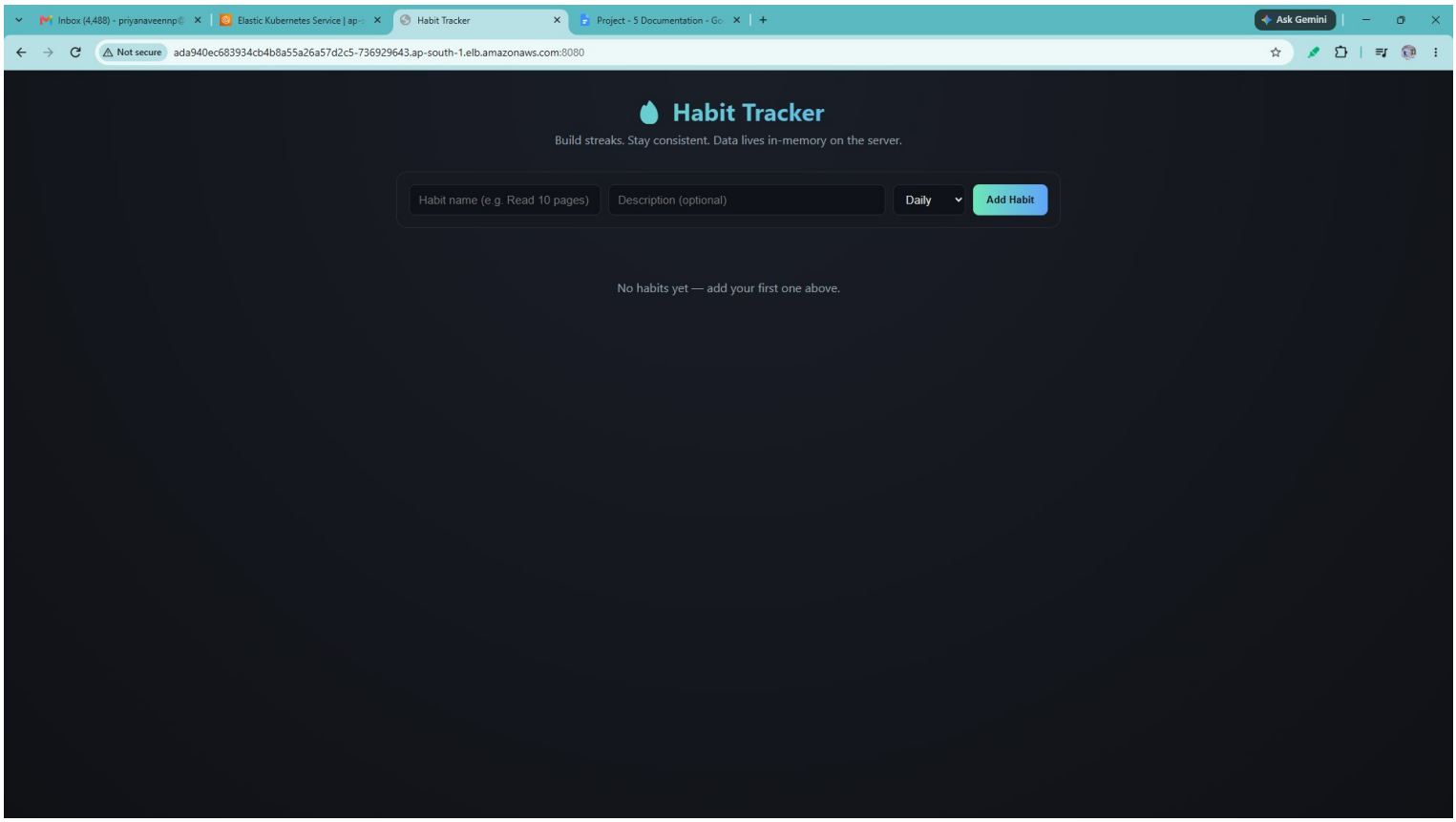
The Habit Tracker application was successfully deployed and verified on Amazon EKS. The rolling update completed successfully while the application remained available.


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Project - 5
Habit-Tracker-Project
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.helmignore
Chart.yaml
values.yaml
src
target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl rollout status deployment/habit-tracker
deployment "habit-tracker" successfully rolled out
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ curl -s -o /dev/null -w "%{http_code}\n" http://ada940ec683934cb4b8a55a26a57d2c5-736929643.ap-south-1.elb.amazonaws.com:8080/
200
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Outline
Timeline
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Priya (7 hours ago) Ln 25, Col 60 Spaces: 2 UTF-8 LF YAML


File Edit Selection View Go Run Terminal Help
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Problems Output Debug Console Terminal Ports
Project - 5
Habit-Tracker-Project
habit-tracker
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secret.yaml
service.yaml
.helmignore
Chart.yaml
values.yaml
src
target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl rollout restart deployment/habit-tracker
deployment.apps/habit-tracker restarted
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl rollout status deployment/habit-tracker
Waiting for deployment "habit-tracker" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "habit-tracker" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "habit-tracker" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "habit-tracker" rollout to finish: 1 of 2 updated replicas are available...
Waiting for deployment "habit-tracker" rollout to finish: 1 of 2 updated replicas are available...
deployment "habit-tracker" successfully rolled out
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get pods -l app.kubernetes.io/name=habit-tracker
NAME READY STATUS RESTARTS AGE
habit-tracker-55cddf7c74-gwpsk 1/1 Running 0 87s
habit-tracker-55cddf7c74-k7pgv 1/1 Running 0 64s
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Outline
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Priya (7 hours ago) Ln 25, Col 60 Spaces: 2 UTF-8 LF YAML


File Edit Selection View Go Run Terminal Help
Project - 5
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Habit-Tracker-Project
habit-tracker
src
target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl describe deployment habit-tracker
Name: habit-tracker
Namespace: default
CreationTimestamp: Mon, 28 Sep 2026 14:04:04 +0530
Labels: app.kubernetes.io/instance=habit-tracker
app.kubernetes.io/managed-by=Helm
app.kubernetes.io/name=habit-tracker
app.kubernetes.io/version=03117ad
helm.sh/chart=habit-tracker-1.0.0
Annotations: argocd.argoproj.io/tracking-id: habit-tracker:apps/Deployment:default/habit-tracker
deployment.kubernetes.io/revision: 15
meta.helm.sh/release-name: habit-tracker
meta.helm.sh/release-namespace: default
Selector: app.kubernetes.io/instance=habit-tracker,app.kubernetes.io/name=habit-tracker
Replicas: 2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType: RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 1 max unavailable, 0 max surge
Pod Template:
Labels: app.kubernetes.io/instance=habit-tracker
app.kubernetes.io/managed-by=Helm
app.kubernetes.io/name=habit-tracker
app.kubernetes.io/version=03117ad
helm.sh/chart=habit-tracker-1.0.0
Annotations: kubectl.kubernetes.io/restartedAt: 2026-09-30T22:13:17+05:30
Containers:
habit-tracker:
Image: 196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker:03117ad
Port: 8080/TCP (http)
Host Port: 0/TCP (http)
Limits:
cpu: 500m
memory: 512Mi
Requests:
cpu: 100m
memory: 256Mi
Liveness: http-get http://:8080/ delay=30s timeout=5s period=10s #success=1 #failure=3
Readiness: http-get http://:8080/ delay=15s timeout=5s period=10s #success=1 #failure=3
Environment: <none>
Mounts:
/app/config from config (rw)
Volumes:
config:
Type: ConfigMap (a volume populated by a ConfigMap)
Name: habit-tracker
Optional: false
Node-Selectors: <none>
Tolerations: <none>
Conditions:
Type Status Reason
Available True MinimumReplicasAvailable
Progressing True NewReplicaSetAvailable
OldReplicaSets: habit-tracker-54758f5ffd (0/0 replicas created), habit-tracker-67dc6767f4 (0/0 replicas created), habit-tracker-845c8bd66 (0/0 replicas created), habit-tracker-57496d86b9 (0/0 replicas created), habit-tracker-6f9c455fb6 (0/0 replicas created), habit-tracker-5f6678cf44 (0/0 replicas created), habit-tracker-5cc86dbf65 (0/0 replicas created), habit-tracker-85598589fd (0/0 replicas created)
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Timeline
main
Priya (7 hours ago) Ln 25, Col 60 Spaces: 2 UTF-8 LF YAML


File Edit Selection View Go Run Terminal Help
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Explorer
Project - 5
Habit-Tracker-Project
habit-tracker
src
target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl describe deployment habit-tracker
meta.helm.sh/release-namespace: default
Selector: app.kubernetes.io/instance=habit-tracker,app.kubernetes.io/name=habit-tracker
Replicas: 2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType: RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 1 max unavailable, 0 max surge
Pod Template:
Labels: app.kubernetes.io/instance=habit-tracker
app.kubernetes.io/managed-by=Helm
app.kubernetes.io/name=habit-tracker
app.kubernetes.io/version=03117ad
helm.sh/chart=habit-tracker-1.0.0
Annotations: kubectl.kubernetes.io/restartedAt: 2026-09-30T22:13:17+05:30
Containers:
habit-tracker:
Image: 196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker:03117ad
Port: 8080/TCP (http)
Host Port: 0/TCP (http)
Limits:
cpu: 500m
memory: 512Mi
Requests:
cpu: 100m
memory: 256Mi
Liveness: http-get http://:8080/ delay=30s timeout=5s period=10s #success=1 #failure=3
Readiness: http-get http://:8080/ delay=15s timeout=5s period=10s #success=1 #failure=3
Environment: <none>
Mounts:
/app/config from config (rw)
Volumes:
config:
Type: ConfigMap (a volume populated by a ConfigMap)
Name: habit-tracker
Optional: false
Node-Selectors: <none>
Tolerations: <none>
Conditions:
Type Status Reason
Available True MinimumReplicasAvailable
Progressing True NewReplicaSetAvailable
OldReplicaSets: habit-tracker-54758f5ffd (0/0 replicas created), habit-tracker-67dc6767f4 (0/0 replicas created), habit-tracker-845c8bd66 (0/0 replicas created), habit-tracker-57496d86b9 (0/0 replicas created), habit-tracker-6f9c455fb6 (0/0 replicas created), habit-tracker-5f6678cf44 (0/0 replicas created), habit-tracker-5cc86dbf65 (0/0 replicas created), habit-tracker-85598589fd (0/0 replicas created)
NewReplicaSet: habit-tracker-55cddf7c74 (2/2 replicas created)
Events:
Type Reason Age From Message
Normal ScalingReplicaSet 17m deployment-controller Scaled down replica set habit-tracker-85598589fd from 2 to 1
Normal ScalingReplicaSet 17m deployment-controller Scaled up replica set habit-tracker-55cddf7c74 from 0 to 1
Normal ScalingReplicaSet 16m deployment-controller Scaled down replica set habit-tracker-85598589fd from 1 to 0
Normal ScalingReplicaSet 16m deployment-controller Scaled up replica set habit-tracker-55cddf7c74 from 1 to 2
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Outline
Timeline
main
Priya (7 hours ago) Ln 25, Col 60 Spaces: 2 UTF-8 LF YAML


Inbox (4,488) - priyanaveennp@
Elastic Kubernetes Service | ap
Habit Tracker
Project - 5 Documentation - Go
Ask Gemini
Not secure
ada940ec683934cb4b8a55a26a57d2c5-736929643.ap-south-1.elb.amazonaws.com:8080
Habit Tracker
Build streaks. Stay consistent. Data lives in-memory on the server.
Habit name (e.g. Read 10 pages)
Description (optional)
Daily
Add Habit
No habits yet — add your first one above.

Task 8 : Argo CD

Overview

Argo CD was configured for GitOps-based deployment of the Habit Tracker application. The Habit-Tracker-Project GitHub repository is used to store the Helm deployment configuration, which Argo CD monitors and synchronizes with Amazon EKS.

Application Structure

- GitOps Repository: Habit-Tracker-Project
- Deployment Tool: Argo CD
- Kubernetes: Amazon EKS
- Application: habit-tracker
- Deployment Configuration: Helm Chart
- Sync Policy: Automatic

Execution

- Verified Argo CD application habit-tracker.
- Changed replicas from 2 → 3 and pushed commit 1574d2e.
- Third pod remained Pending because of the available vCPU capacity.
- Restored replicas from 3 → 2 and pushed commit 4779952.
- Argo CD synchronized the restoration successfully.

Verification

Argo CD detected the GitHub configuration changes.

Final Git revision: 4779952

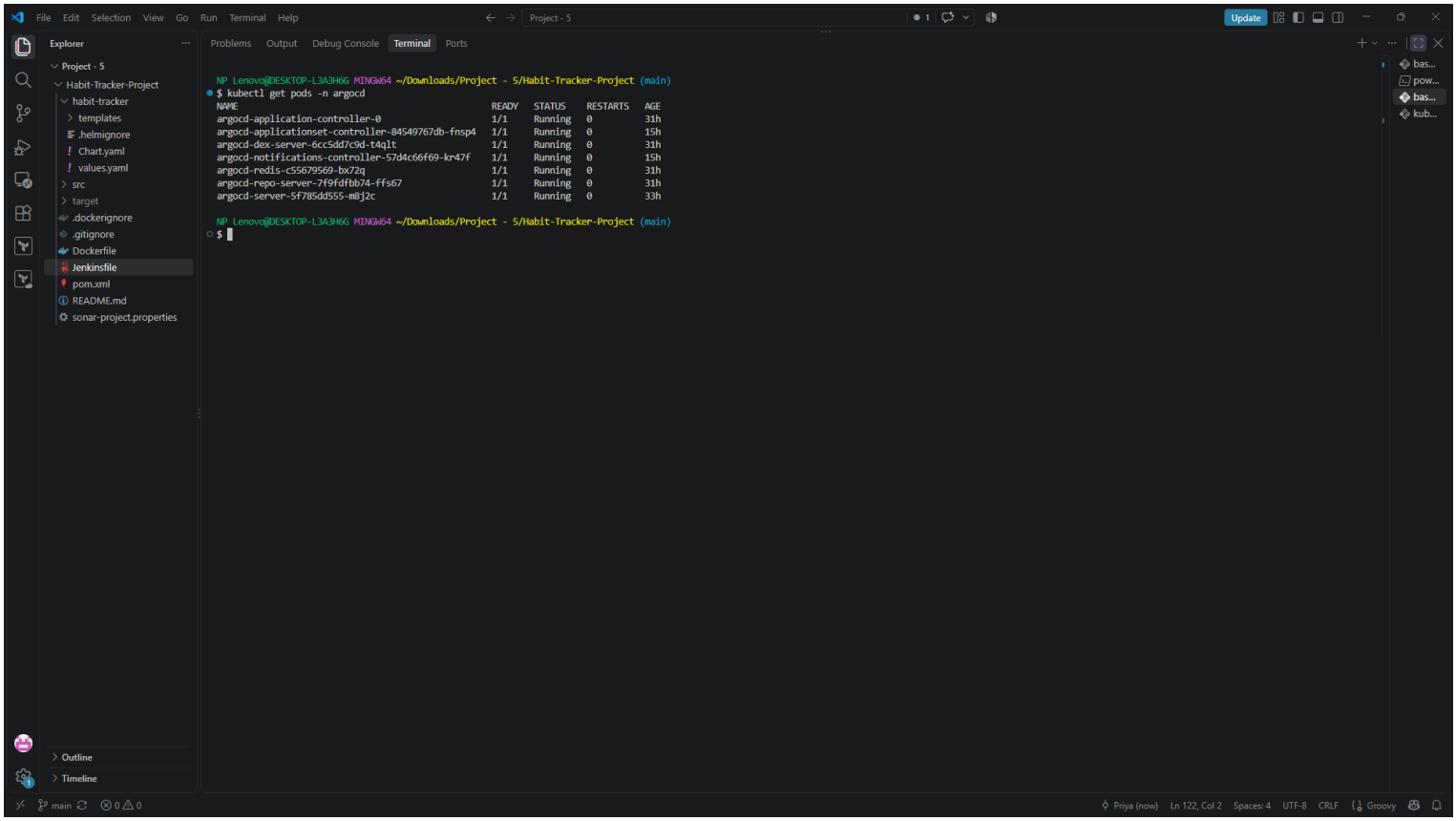
After synchronization:

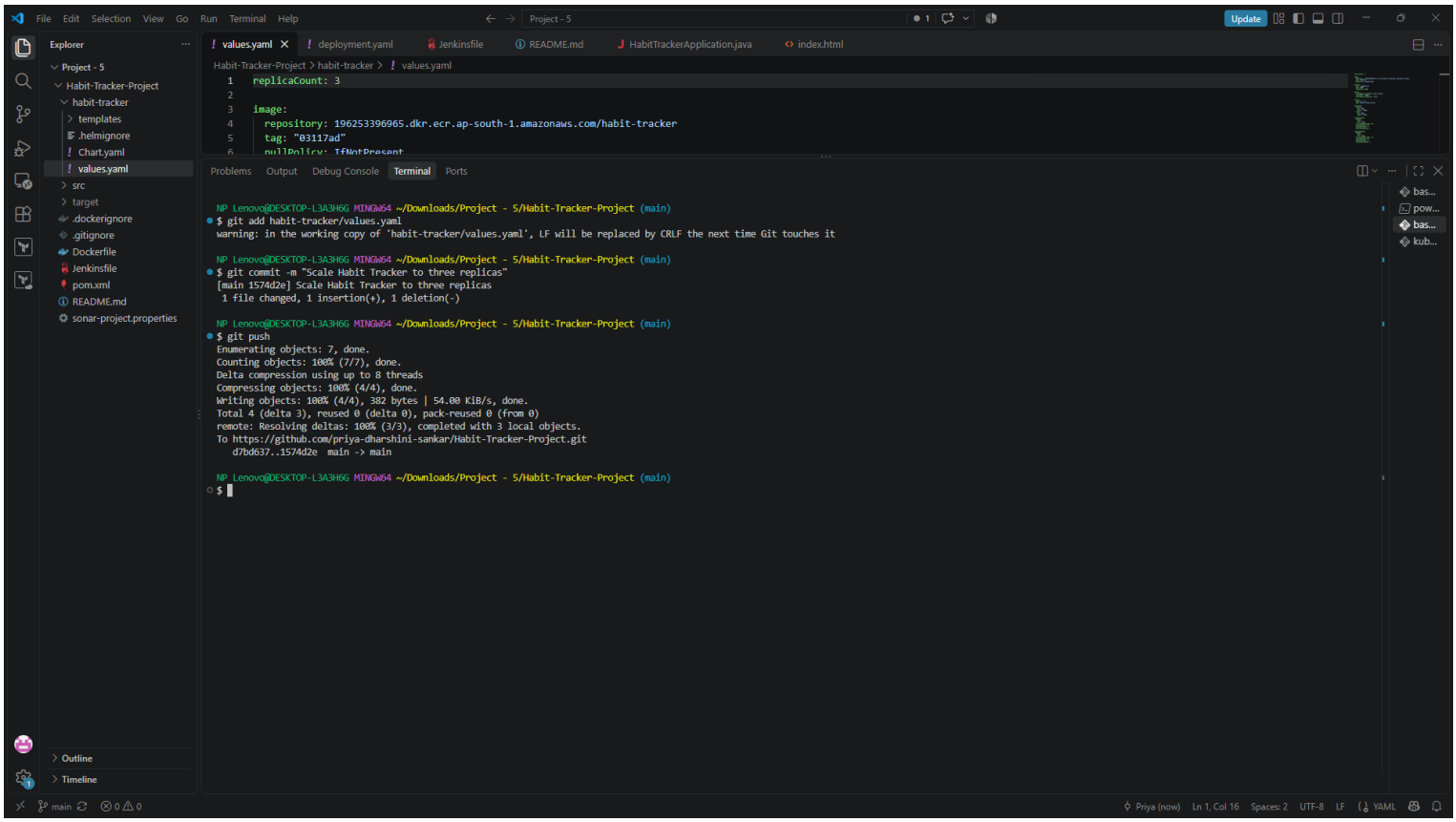
- Sync Status: Synced
- Application Health: Healthy
- OutOfSync: 0
- Progressing: 0
- Replicas: 2
- Pods: 2/2 Running

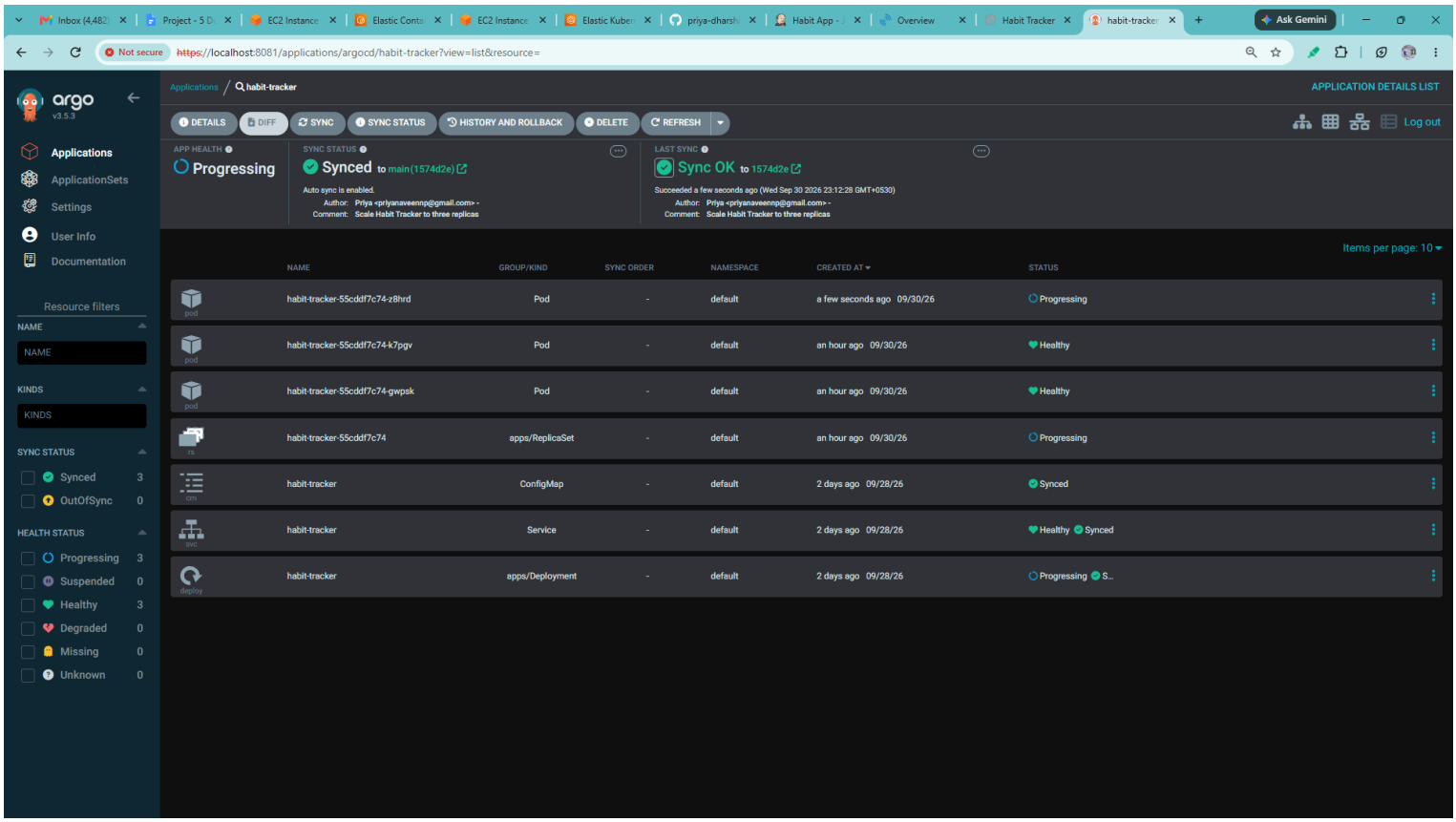
The application was successfully restored and remained healthy.

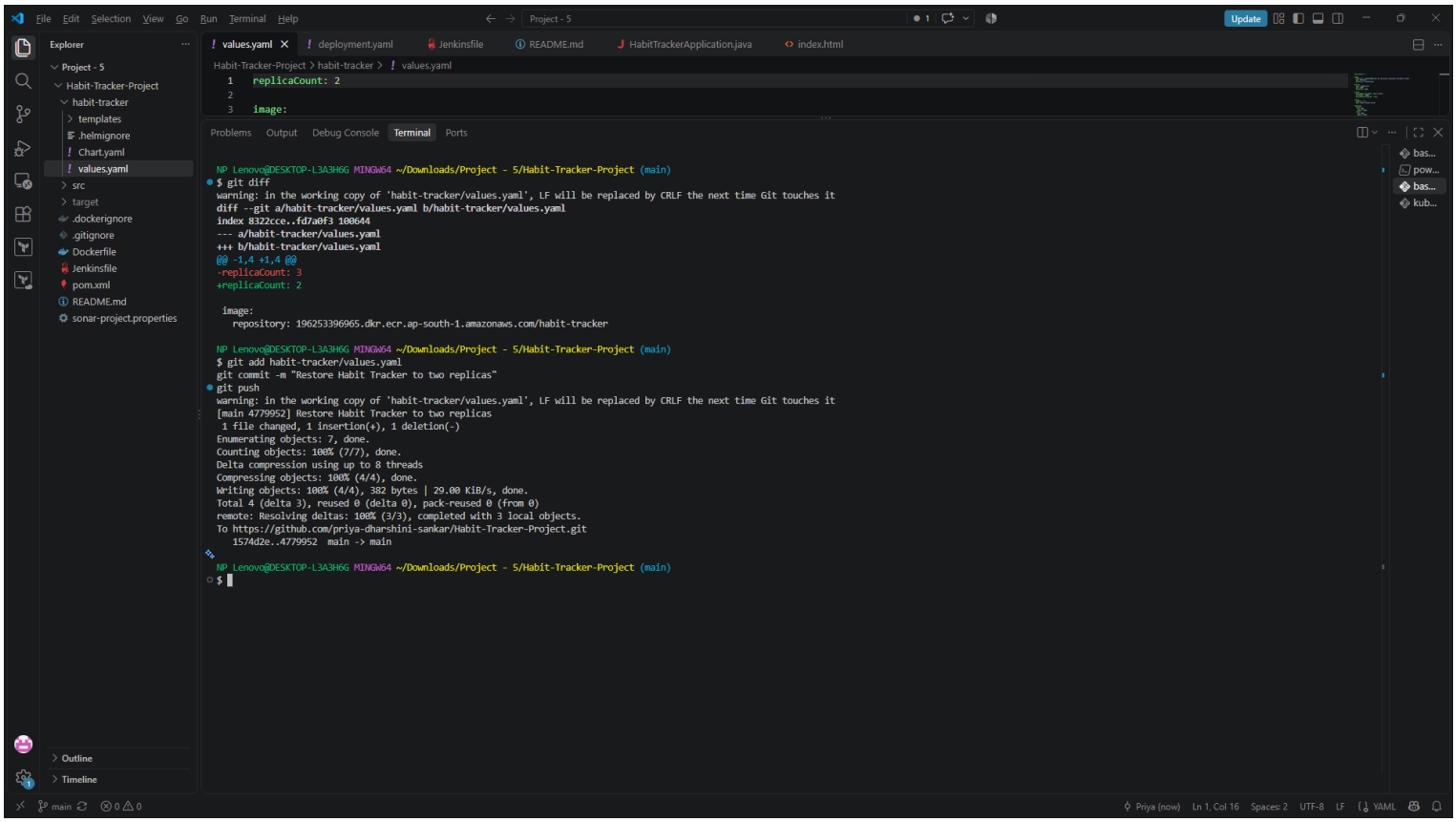
Result

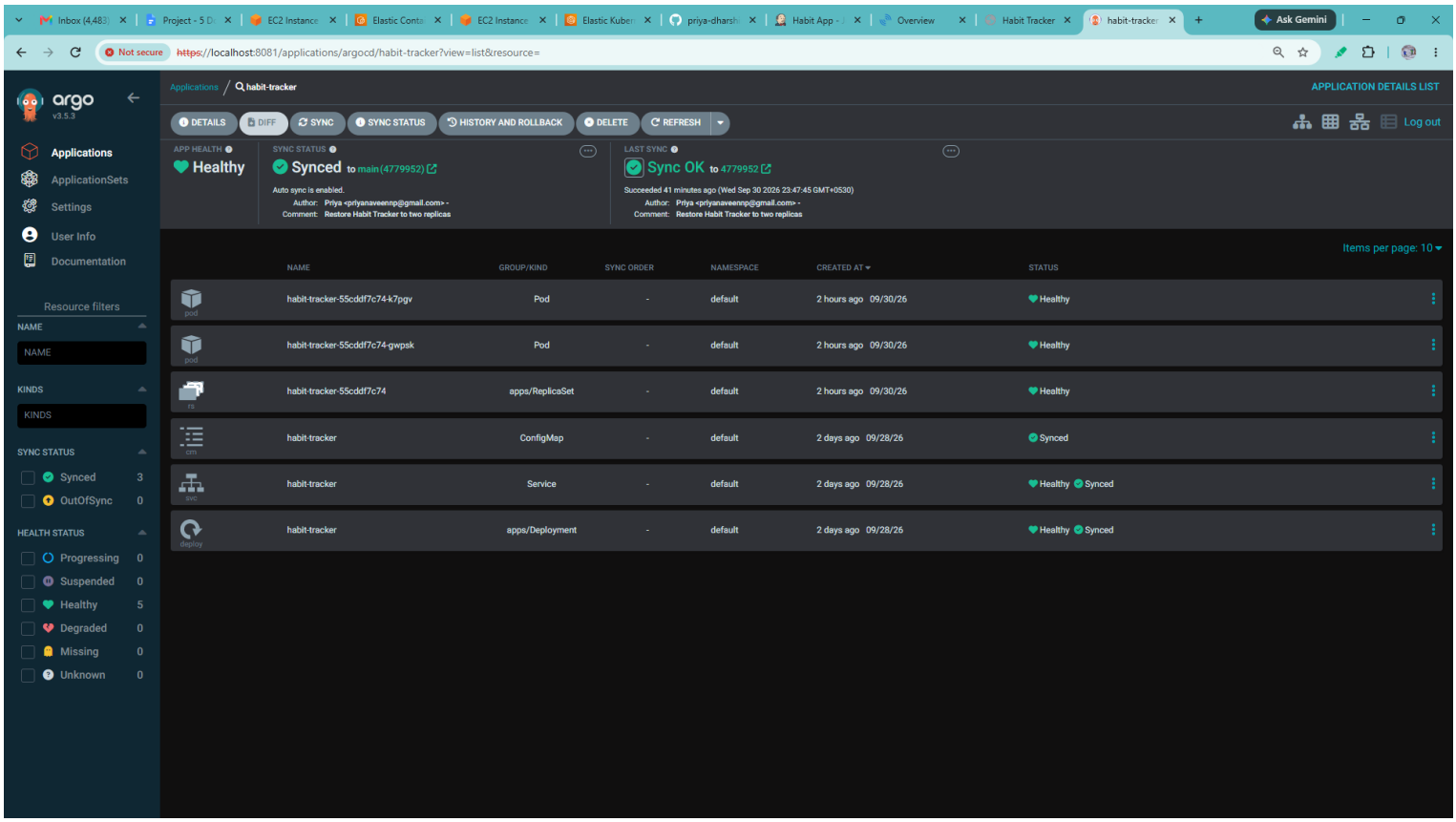
Argo CD successfully implemented GitOps-based deployment for the Habit Tracker application. The replica configuration changes were detected and synchronized from GitHub to Amazon EKS, and the application was safely restored to 2 running replicas.

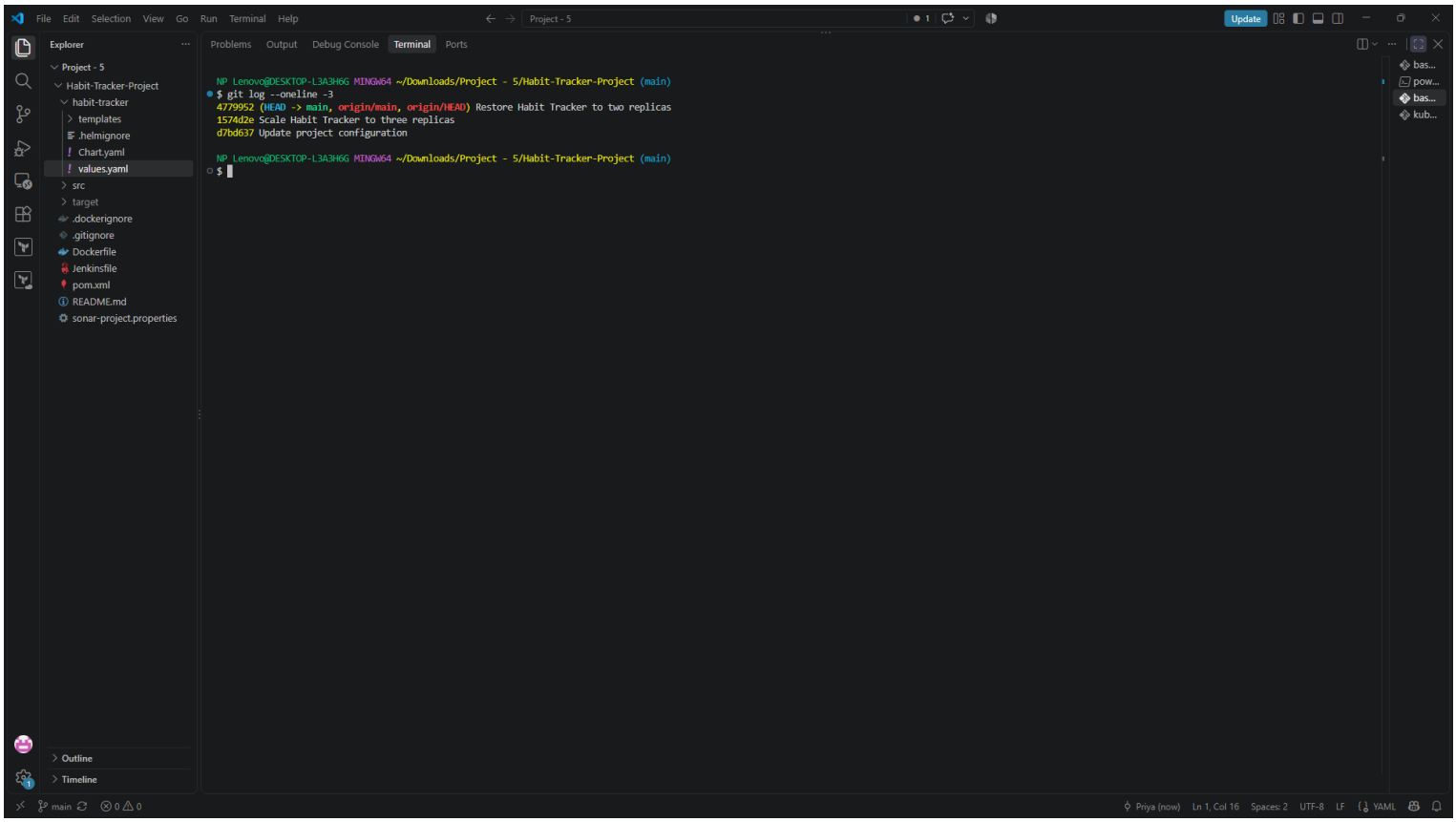



values.yaml
deployment.yaml
Jenkinsfile
README.md
HabitTrackerApplication.java
index.html
Habit-Tracker-Project > habit-tracker > values.yaml
replicaCount: 3
image:
repository: 196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker
tag: "03117ad"
Problems
Output
Debug Console
Terminal
Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ git add habit-tracker/values.yaml
warning: in the working copy of 'habit-tracker/values.yaml', LF will be replaced by CRLF the next time Git touches it
$ git commit -m "Scale Habit Tracker to three replicas"
[main 1574d2e] Scale Habit Tracker to three replicas
1 file changed, 1 insertion(+), 1 deletion(-)
$ git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 382 bytes | 54.00 KiB/s, done.
Total 4 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/priya-dharshini-sankar/Habit-Tracker-Project.git
d7bd637..1574d2e main -> main


https://localhost:8081/applications/argocd/habit-tracker?view=list&resource=
argo
v3.5.3
Applications
ApplicationSets
Settings
User Info
Documentation
Resource filters
Applications / habit-tracker
APPLICATION DETAILS LIST
DETAILS
DIFF
SYNC
SYNC STATUS
HISTORY AND ROLLBACK
DELETE
REFRESH
Log out
APP HEALTH
Progressing
SYNC STATUS
Synced to main(1574d2e)
Auto sync is enabled.
Author: Priya <priyanaveennp@gmail.com> -
Comment: Scale Habit Tracker to three replicas
LAST SYNC
Sync OK to 1574d2e
Succeeded a few seconds ago (Wed Sep 30 2026 23:12:28 GMT+0530)
Author: Priya <priyanaveennp@gmail.com> -
Comment: Scale Habit Tracker to three replicas
Items per page: 10
NAME
GROUP/KIND
SYNC ORDER
NAMESPACE
CREATED AT
STATUS
habit-tracker-55cddf7c74-z8hrd
Pod
default
a few seconds ago 09/30/26
Progressing
habit-tracker-55cddf7c74-k7pgv
Pod
default
an hour ago 09/30/26
Healthy
habit-tracker-55cddf7c74-gwpsk
Pod
default
an hour ago 09/30/26
Healthy
habit-tracker-55cddf7c74
apps/ReplicaSet
default
an hour ago 09/30/26
Progressing
habit-tracker
ConfigMap
default
2 days ago 09/28/26
Synced
habit-tracker
Service
default
2 days ago 09/28/26
Healthy Synced
habit-tracker
apps/Deployment
default
2 days ago 09/28/26
Progressing
NAME
KINDS
SYNC STATUS
Synced 3
OutOfSync 0
HEALTH STATUS
Progressing 3
Suspended 0
Healthy 3
Degraded 0
Missing 0
Unknown 0


values.yaml
deployment.yaml
Jenkinsfile
README.md
HabitTrackerApplication.java
index.html
Habit-Tracker-Project > habit-tracker > values.yaml
1 replicaCount: 2
2
3 image:
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ git diff
warning: in the working copy of 'habit-tracker/values.yaml', LF will be replaced by CRLF the next time Git touches it
diff --git a/habit-tracker/values.yaml b/habit-tracker/values.yaml
index 8322cce..fd7a0f3 100644
--- a/habit-tracker/values.yaml
+++ b/habit-tracker/values.yaml
@@ -1,4 +1,4 @@
-replicaCount: 3
+replicaCount: 2
image:
repository: 196253396965.dkr.ap-south-1.amazonaws.com/habit-tracker
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ git add habit-tracker/values.yaml
git commit -m "Restore Habit Tracker to two replicas"
git push
warning: in the working copy of 'habit-tracker/values.yaml', LF will be replaced by CRLF the next time Git touches it
[main 4779952] Restore Habit Tracker to two replicas
1 file changed, 1 insertion(+), 1 deletion(-)
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 382 bytes | 29.00 KiB/s, done.
Total 4 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/priya-dharshini-sankar/Habit-Tracker-Project.git
1574d2e..4779952 main -> main
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$


Not secure https://localhost:8081/applications/argocd/habit-tracker?view=list&resource=
argo v3.5.3
Applications
ApplicationSets
Settings
User Info
Documentation
Resource filters
NAME
KINDS
SYNC STATUS
Synced 3
OutOfSync 0
HEALTH STATUS
Progressing 0
Suspended 0
Healthy 5
Degraded 0
Missing 0
Unknown 0
Applications / habit-tracker
APPLICATION DETAILS LIST
DETAILS DIFF SYNC SYNC STATUS HISTORY AND ROLLBACK DELETE REFRESH
Log out
APP HEALTH Healthy
SYNC STATUS Synced to main (4779952)
Auto sync is enabled.
Author: Priya <priyanaveennp@gmail.com>
Comment: Restore Habit Tracker to two replicas
LAST SYNC Sync OK to 4779952
Succeeded 41 minutes ago (Wed Sep 30 2026 23:47:45 GMT+0530)
Author: Priya <priyanaveennp@gmail.com>
Comment: Restore Habit Tracker to two replicas
Items per page: 10
NAME GROUP/KIND SYNC ORDER NAMESPACE CREATED AT STATUS
habit-tracker-55cddf7c74-k7pgv Pod - default 2 hours ago 09/30/26 Healthy
habit-tracker-55cddf7c74-gwpsk Pod - default 2 hours ago 09/30/26 Healthy
habit-tracker-55cddf7c74 apps/ReplicaSet - default 2 hours ago 09/30/26 Healthy
habit-tracker ConfigMap - default 2 days ago 09/28/26 Synced
habit-tracker Service - default 2 days ago 09/28/26 Healthy Synced
habit-tracker apps/Deployment - default 2 days ago 09/28/26 Healthy Synced


File Edit Selection View Go Run Terminal Help
Project - 5
Update
Explorer
Project - 5
Habit-Tracker-Project
habit-tracker
templates
.helmignore
Chart.yaml
values.yaml
src
target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Outline
Timeline
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ git log --oneline -3
4779952 (HEAD -> main, origin/main, origin/HEAD) Restore Habit Tracker to two replicas
1574d2e Scale Habit Tracker to three replicas
d7bd637 Update project configuration
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
bas...
pow...
bas...
kub...
main
Priya (now) Ln 1, Col 16 Spaces: 2 UTF-8 LF YAML

Task 9 : Jenkins CI Pipeline

Overview

Created a Declarative Jenkins Pipeline to automate the CI process for the Habit Tracker Java application.

Execution

The pipeline was configured to:

- Checkout the source code from GitHub
- Build and test the application using Maven
- Perform SonarQube analysis and Quality Gate validation
- Build and push the Docker image to Amazon ECR
- Generate the image tag from the Git commit SHA
- Update the image tag in `habit-tracker/values.yaml`
- Commit and push the deployment configuration to GitHub

The Docker image was versioned using the Git commit SHA.

The `values.yaml` image tag was automatically updated with the new image version.

Verification

Jenkins Build completed successfully.

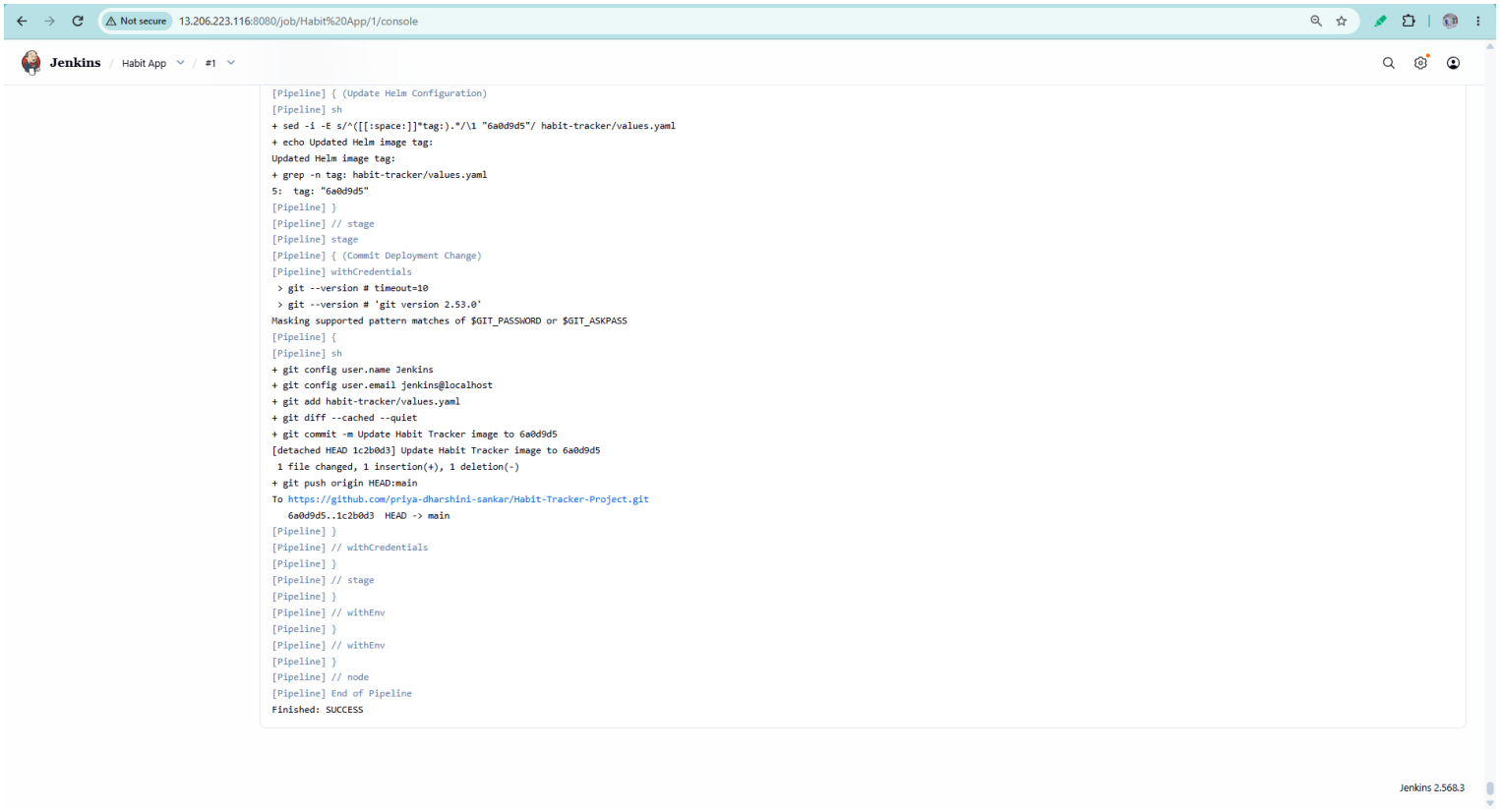
`Image version : 6a0d9d5`

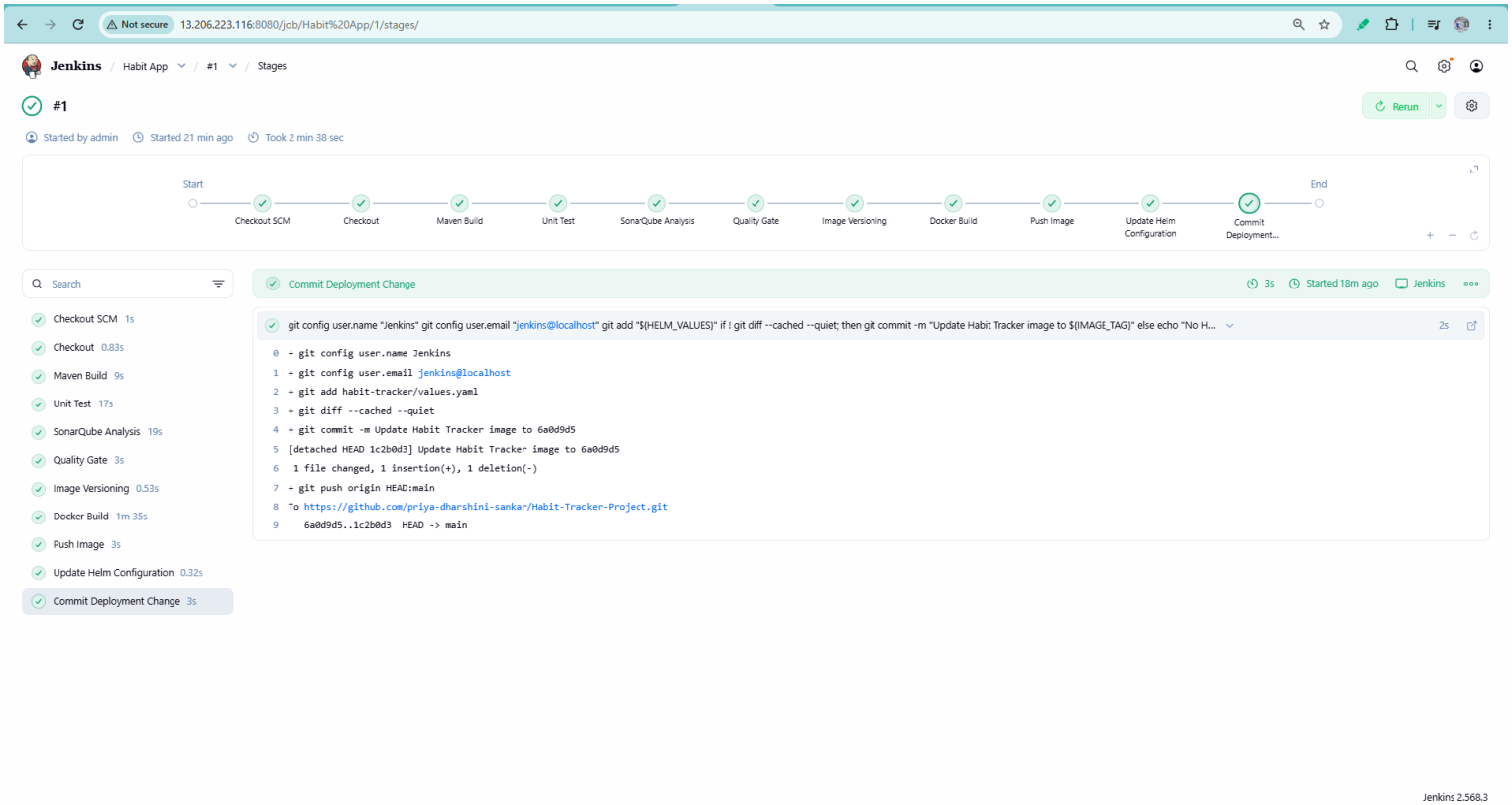
Updated Helm image tag => tag: "6a0d9d5"

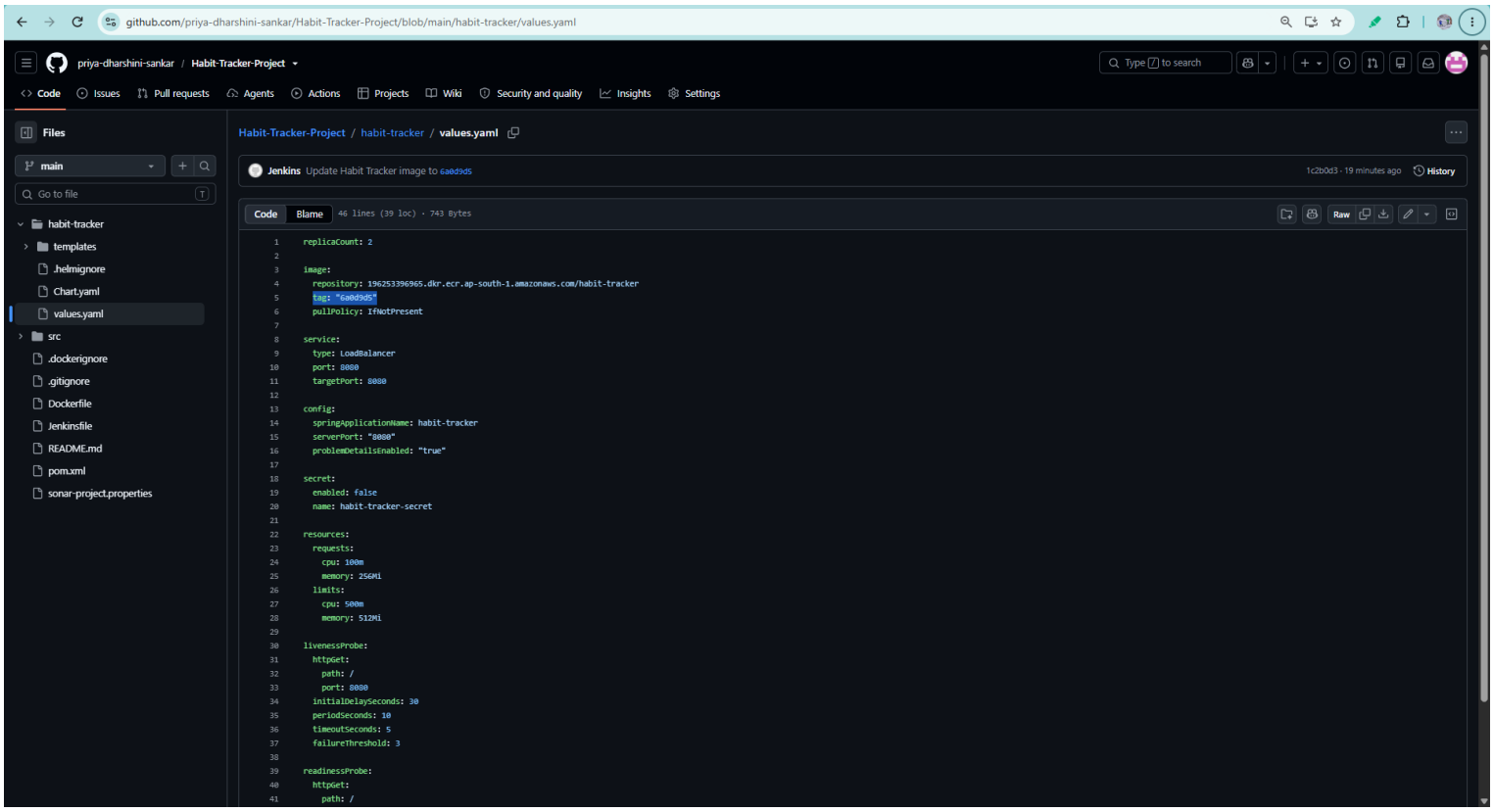
The updated deployment configuration was successfully committed and pushed to GitHub.

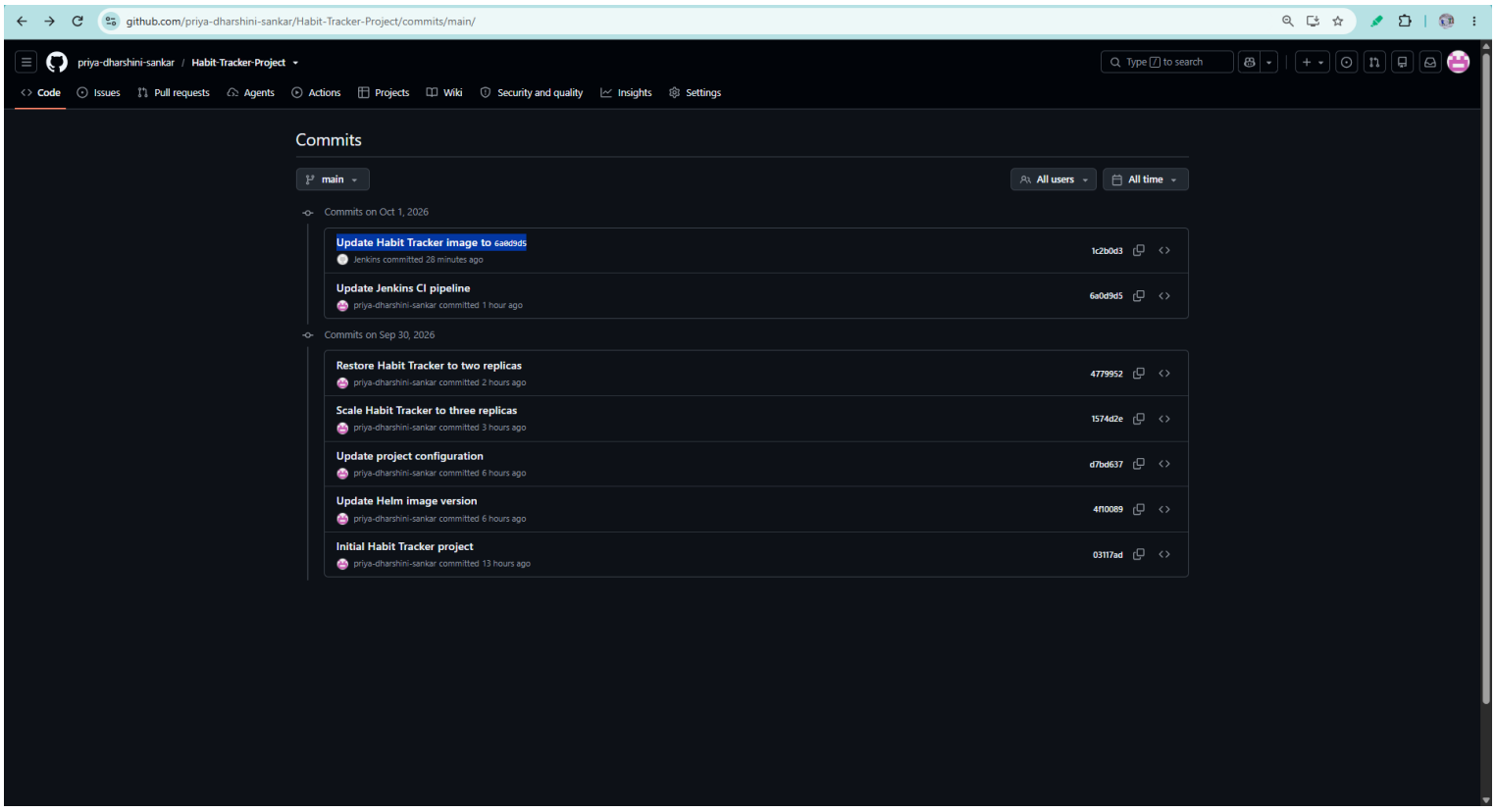
Result

Jenkins now performs the complete CI process and updates the deployment configuration with the new application image version.


Not secure 13.206.223.116:8080/job/Habit%20App/1/console
Jenkins / Habit App / #1
[Pipeline] { (Update Helm Configuration)
[Pipeline] sh
+ sed -i -E s/^([[:space:]]*tag:).*/\1 "6a0d9d5"/ habit-tracker/values.yaml
+ echo Updated Helm image tag:
Updated Helm image tag:
+ grep -n tag: habit-tracker/values.yaml
5: tag: "6a0d9d5"
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Commit Deployment Change)
[Pipeline] withCredentials
 > git --version # timeout=10
 > git --version # 'git version 2.53.0'
Masking supported pattern matches of $GIT_PASSWORD or $GIT_ASKPASS
[Pipeline] {
[Pipeline] sh
+ git config user.name Jenkins
+ git config user.email jenkins@localhost
+ git add habit-tracker/values.yaml
+ git diff --cached --quiet
+ git commit -m Update Habit Tracker image to 6a0d9d5
[detached HEAD 1c2b0d3] Update Habit Tracker image to 6a0d9d5
 1 file changed, 1 insertion(+), 1 deletion(-)
+ git push origin HEAD:main
To https://github.com/priya-dharshini-sankar/Habit-Tracker-Project.git
   6a0d9d5..1c2b0d3 HEAD -> main
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Jenkins 2.568.3


Not secure 13.206.223.116:8080/job/Habit%20App/1/stages/
Jenkins / Habit App / #1 / Stages
#1
Rerun
Started by admin
Started 21 min ago
Took 2 min 38 sec
Start
Checkout SCM
Checkout
Maven Build
Unit Test
SonarQube Analysis
Quality Gate
Image Versioning
Docker Build
Push Image
Update Helm Configuration
Commit Deployment...
End
Search
Checkout SCM 1s
Checkout 0.83s
Maven Build 9s
Unit Test 17s
SonarQube Analysis 19s
Quality Gate 3s
Image Versioning 0.53s
Docker Build 1m 35s
Push Image 3s
Update Helm Configuration 0.32s
Commit Deployment Change 3s
Commit Deployment Change
3s
Started 18m ago
Jenkins
git config user.name "Jenkins" git config user.email "jenkins@localhost" git add "${HELM_VALUES}" if ! git diff --cached --quiet; then git commit -m "Update Habit Tracker image to ${IMAGE_TAG}" else echo "No H...
2s
0 + git config user.name Jenkins
1 + git config user.email jenkins@localhost
2 + git add habit-tracker/values.yaml
3 + git diff --cached --quiet
4 + git commit -m Update Habit Tracker image to 6a0d9d5
5 [detached HEAD 1c2b0d3] Update Habit Tracker image to 6a0d9d5
6 1 file changed, 1 insertion(+), 1 deletion(-)
7 + git push origin HEAD:main
8 To https://github.com/priya-dharshini-sankar/Habit-Tracker-Project.git
9   6a0d9d5..1c2b0d3 HEAD -> main
Jenkins 2.568.3


github.com/priya-dharshini-sankar/Habit-Tracker-Project/blob/main/habit-tracker/values.yaml
priya-dharshini-sankar / Habit-Tracker-Project
Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings
Files
main
Go to file
habit-tracker
templates
.helmignore
Chart.yaml
values.yaml
src
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
README.md
pom.xml
sonar-project.properties
Habit-Tracker-Project / habit-tracker / values.yaml
Jenkins Update Habit Tracker image to 6a0d9d5
1c2b0d3 · 19 minutes ago
History
Code Blame 46 lines (39 loc) · 743 Bytes
Raw
replicaCount: 2
image:
repository: 196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker
tag: "6a0d9d5"
pullPolicy: IfNotPresent
service:
type: LoadBalancer
port: 8080
targetPort: 8080
config:
springApplicationName: habit-tracker
serverPort: "8080"
problemDetailsEnabled: "true"
secret:
enabled: false
name: habit-tracker-secret
resources:
requests:
cpu: 100m
memory: 256Mi
limits:
cpu: 500m
memory: 512Mi
livenessProbe:
httpGet:
path: /
port: 8080
initialDelaySeconds: 30
periodSeconds: 10
timeoutSeconds: 5
failureThreshold: 3
readinessProbe:
httpGet:
path: /


github.com/priya-dharshini-sankar/Habit-Tracker-Project/commits/main/
priya-dharshini-sankar / Habit-Tracker-Project
Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings
Commits
main
All users
All time
Commits on Oct 1, 2026
Update Habit Tracker image to 6a0d9d5
Jenkins committed 28 minutes ago
1c2b0d3
Update Jenkins CI pipeline
priya-dharshini-sankar committed 1 hour ago
6a0d9d5
Commits on Sep 30, 2026
Restore Habit Tracker to two replicas
priya-dharshini-sankar committed 2 hours ago
4779952
Scale Habit Tracker to three replicas
priya-dharshini-sankar committed 3 hours ago
1574d2e
Update project configuration
priya-dharshini-sankar committed 6 hours ago
d7bd637
Update Helm image version
priya-dharshini-sankar committed 6 hours ago
4f10089
Initial Habit Tracker project
priya-dharshini-sankar committed 13 hours ago
03117ad

Task 10 : Jenkins → Argo CD GitOps

Overview

Configured Jenkins for Continuous Integration and Argo CD for Continuous Deployment using the GitOps approach.

Jenkins builds, tests, analyzes, containerizes, and versions the application, then updates the Helm deployment configuration in GitHub. Argo CD monitors the Git repository and automatically synchronizes the desired state to Kubernetes.

Execution

The CI/CD flow was configured as:

GitHub → Jenkins → Maven Build/Test → SonarQube → Docker Build → Amazon ECR → Update Helm Values → Git → Argo CD → Kubernetes

Jenkins updates the Helm image tag and commits the deployment configuration to GitHub.

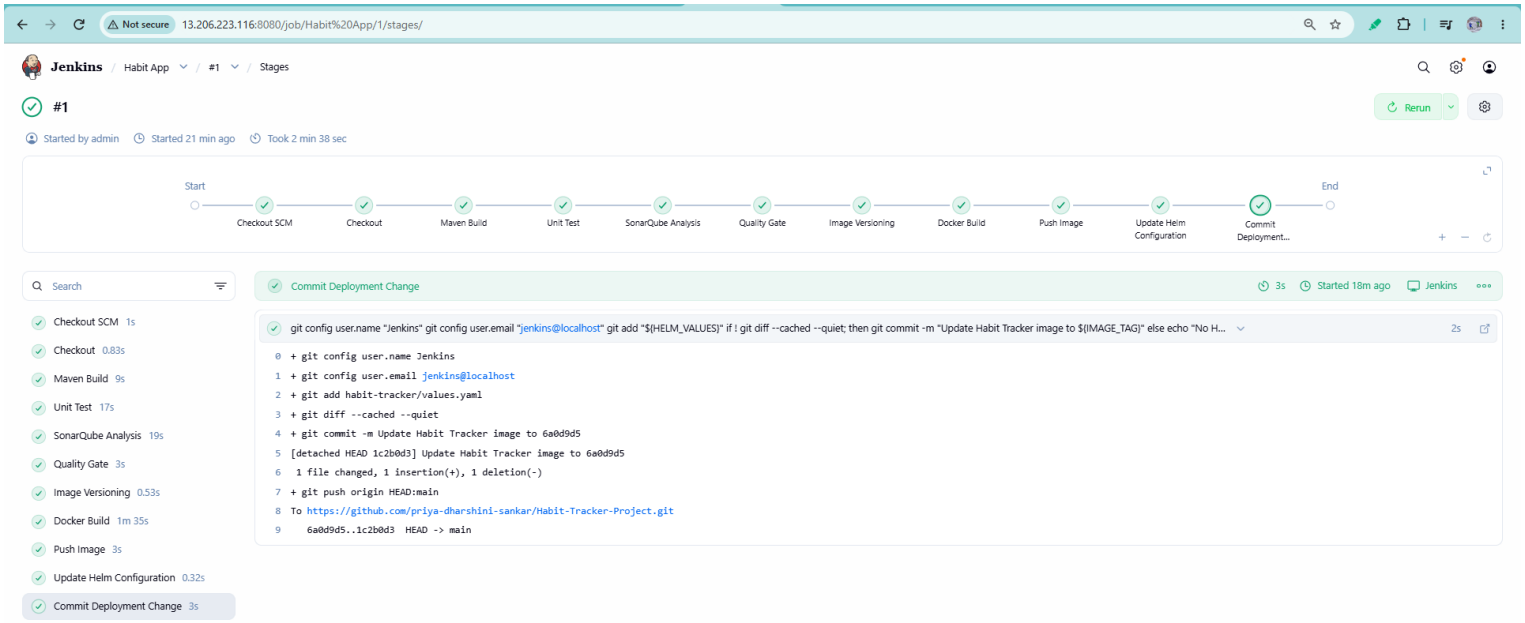
Jenkins does not directly deploy the application using kubectl.

Verification

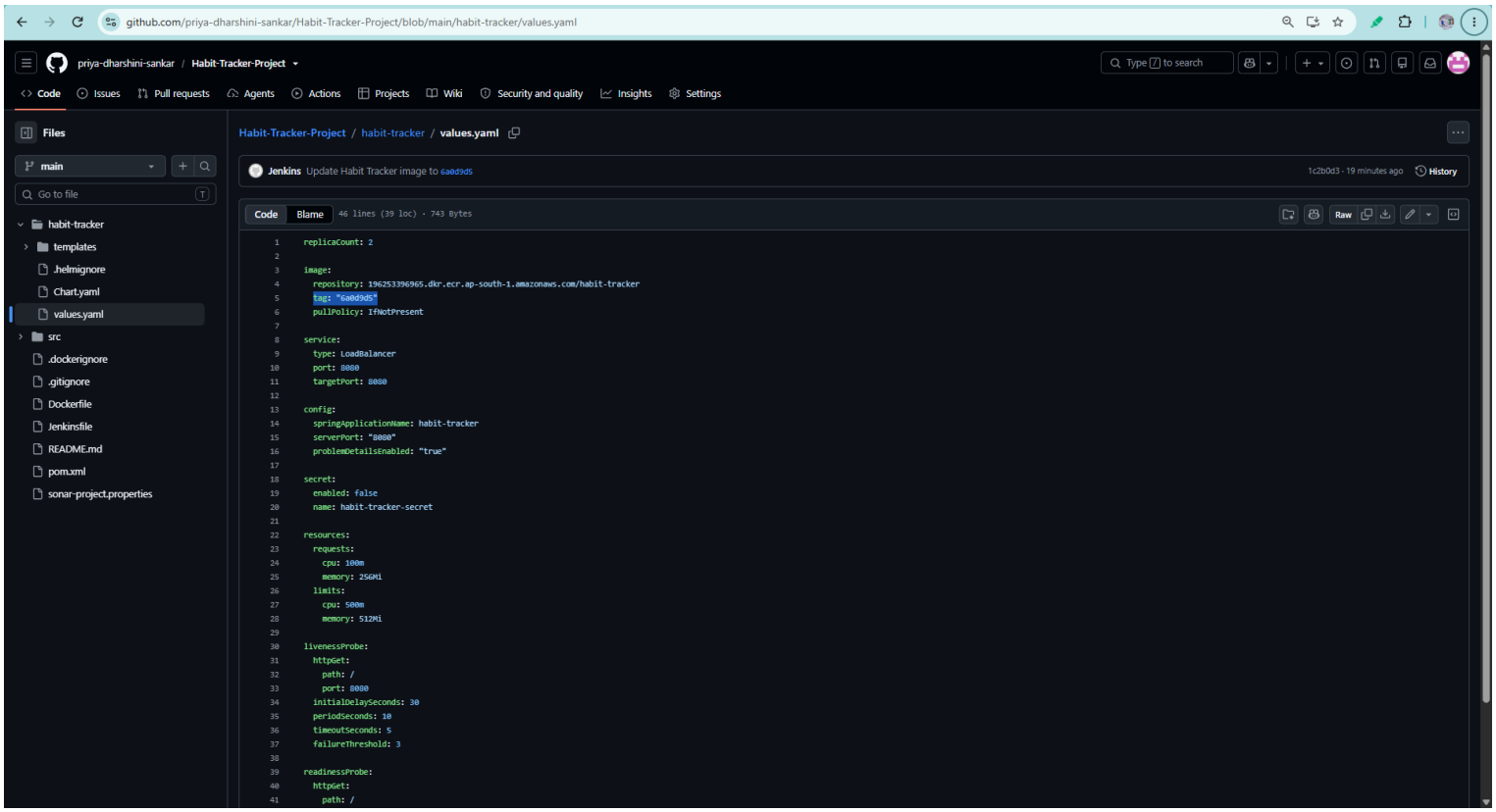
- Jenkins pipeline completed successfully.
- Image version: 6a0d9d5
- Helm image tag updated to 6a0d9d5.
- Argo CD automatically synchronized the Git change.
- Argo CD Status: Synced
- Argo CD Health: Healthy
- Argo CD deployment history shows the sync was initiated by the automated sync policy.
- Kubernetes deployment status: 2/2 replicas ready
- 2 application pods running.
- Deployed image: 196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker:6a0d9d5

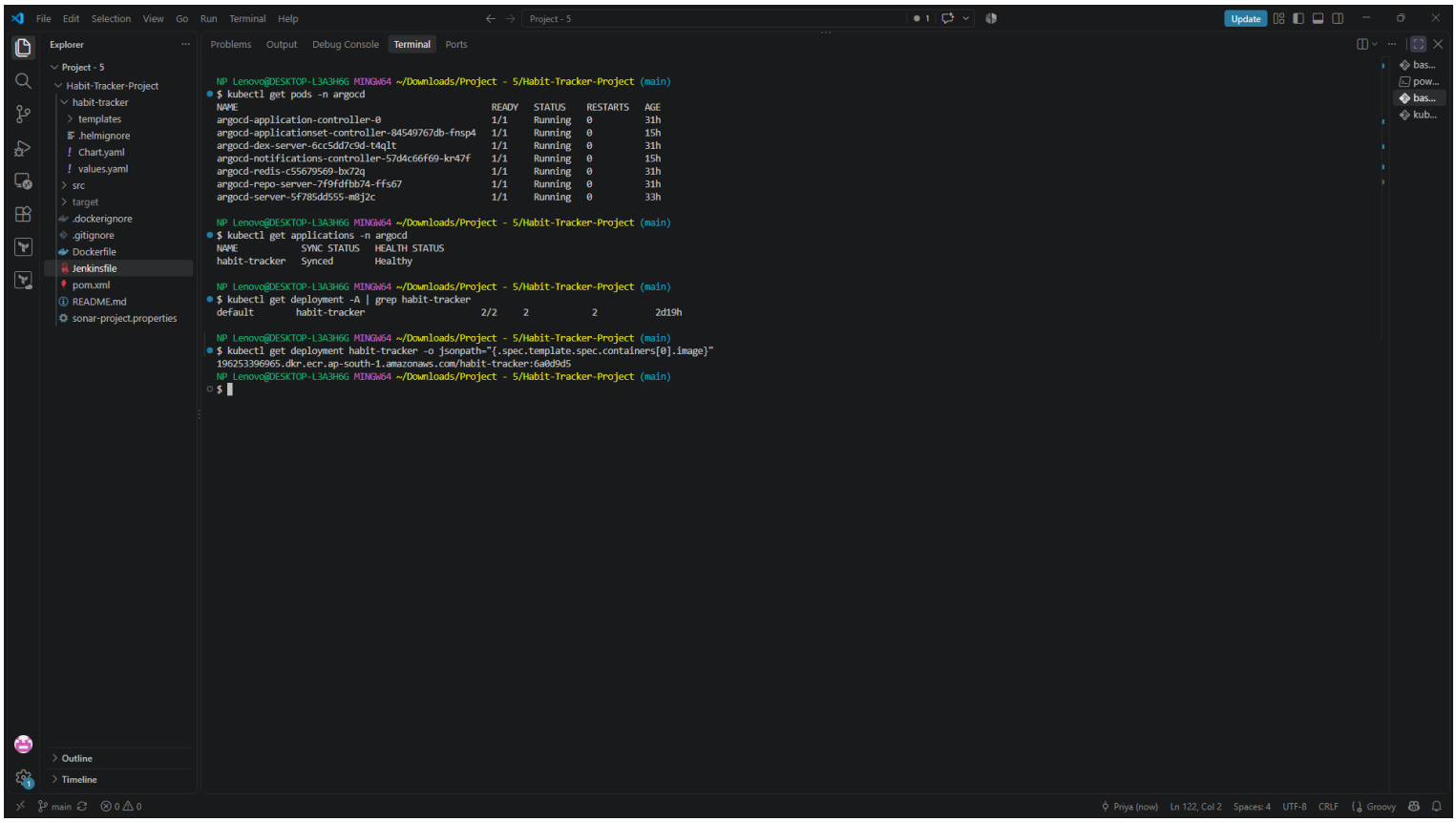
Result

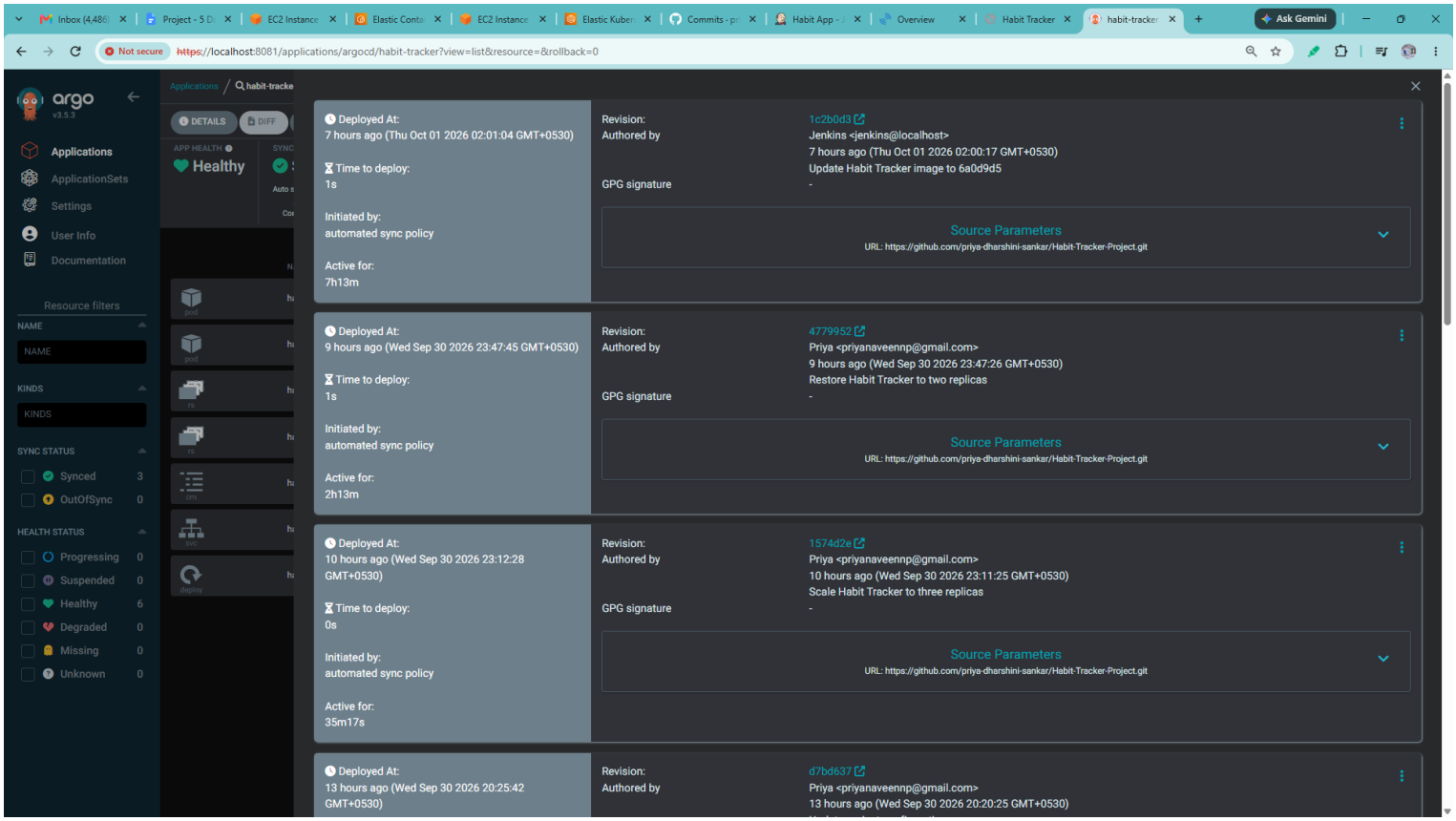
Jenkins handles CI, while Argo CD automatically performs CD to Kubernetes through GitOps workflow.

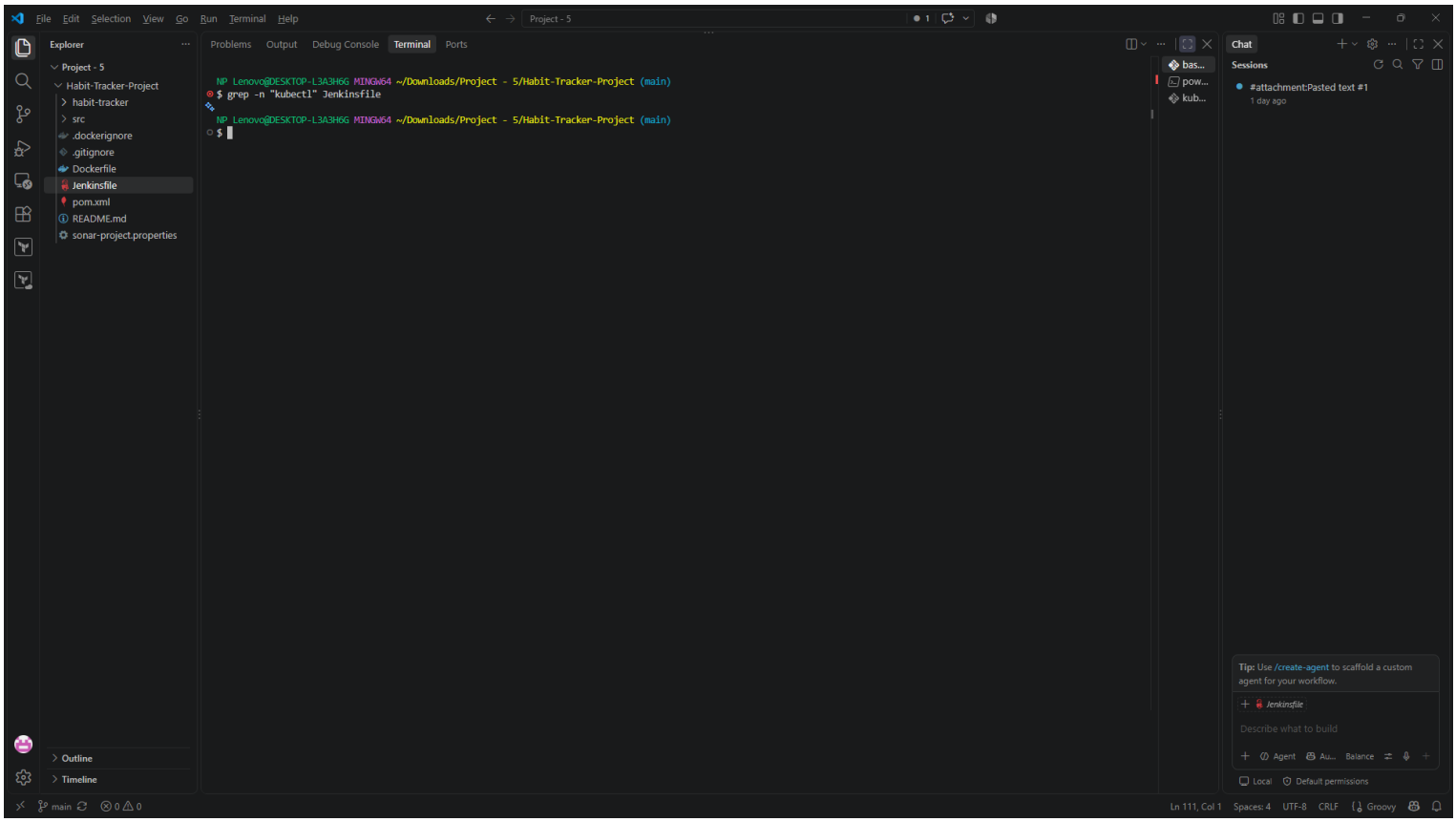


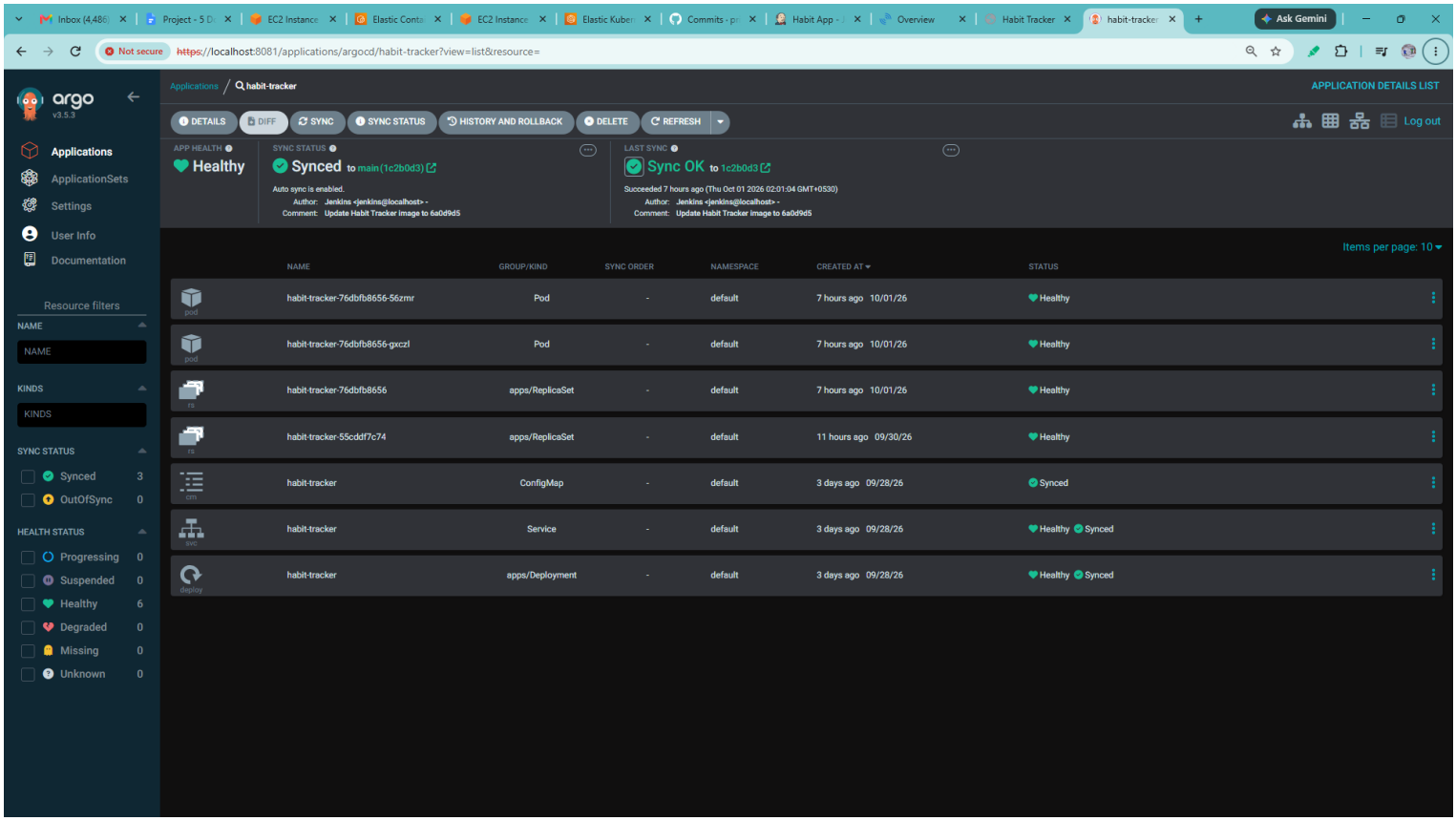
Jenkins 2.568.3




File Edit Selection View Go Run Terminal Help
Project - 5
Update
Explorer
Project - 5
Habit-Tracker-Project
habit-tracker
templates
.helmignore
Chart.yaml
values.yaml
src
target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get pods -n argocd
NAME READY STATUS RESTARTS AGE
argocd-application-controller-0 1/1 Running 0 31h
argocd-applicationset-controller-84549767db-fnsp4 1/1 Running 0 15h
argocd-dex-server-6cc5dd7c9d-t4qlt 1/1 Running 0 31h
argocd-notifications-controller-57d4c66f69-kr47f 1/1 Running 0 15h
argocd-redis-c55679569-bx72q 1/1 Running 0 31h
argocd-repo-server-7f9fdfbb74-ffs67 1/1 Running 0 31h
argocd-server-5f785dd555-m8j2c 1/1 Running 0 33h
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get applications -n argocd
NAME SYNC STATUS HEALTH STATUS
habit-tracker Synced Healthy
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get deployment -A | grep habit-tracker
default habit-tracker 2/2 2 2 2d19h
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get deployment habit-tracker -o jsonpath="{.spec.template.spec.containers[0].image}"
196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker:6a0d9d5
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Outline
Timeline
main
Priya (now) Ln 122, Col 2 Spaces: 4 UTF-8 CRLF Groovy


Not secure https://localhost:8081/applications/argocd/habit-tracker?view=list&resource=&rollback=0
Ask Gemini
argo
v3.5.3
Applications
ApplicationSets
Settings
User Info
Documentation
Resource filters
NAME
KINDS
SYNC STATUS
Synced 3
OutOfSync 0
HEALTH STATUS
Progressing 0
Suspended 0
Healthy 6
Degraded 0
Missing 0
Unknown 0
DETAILS DIFF
APP HEALTH
Healthy
SYNC
Deployed At:
7 hours ago (Thu Oct 01 2026 02:01:04 GMT+0530)
Time to deploy:
1s
Initiated by:
automated sync policy
Active for:
7h13m
Revision: 1c2b0d3
Authored by Jenkins <jenkins@localhost>
7 hours ago (Thu Oct 01 2026 02:00:17 GMT+0530)
Update Habit Tracker image to 6a0d9d5
GPG signature -
Source Parameters
URL: https://github.com/priya-dharshini-sankar/Habit-Tracker-Project.git
Deployed At:
9 hours ago (Wed Sep 30 2026 23:47:45 GMT+0530)
Time to deploy:
1s
Initiated by:
automated sync policy
Active for:
2h13m
Revision: 4779952
Authored by Priya <priyanaveennp@gmail.com>
9 hours ago (Wed Sep 30 2026 23:47:26 GMT+0530)
Restore Habit Tracker to two replicas
GPG signature -
Source Parameters
URL: https://github.com/priya-dharshini-sankar/Habit-Tracker-Project.git
Deployed At:
10 hours ago (Wed Sep 30 2026 23:12:28 GMT+0530)
Time to deploy:
0s
Initiated by:
automated sync policy
Active for:
35m17s
Revision: 1574d2e
Authored by Priya <priyanaveennp@gmail.com>
10 hours ago (Wed Sep 30 2026 23:11:25 GMT+0530)
Scale Habit Tracker to three replicas
GPG signature -
Source Parameters
URL: https://github.com/priya-dharshini-sankar/Habit-Tracker-Project.git
Deployed At:
13 hours ago (Wed Sep 30 2026 20:25:42 GMT+0530)
Revision: d7bd637
Authored by Priya <priyanaveennp@gmail.com>
13 hours ago (Wed Sep 30 2026 20:20:25 GMT+0530)


File Edit Selection View Go Run Terminal Help
Project - 5
Explorer
Project - 5
Habit-Tracker-Project
habit-tracker
src
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
MP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ grep -n "kubectl" Jenkinsfile
MP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Chat
Sessions
#attachment:Pasted text #1
1 day ago
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Jenkinsfile
Describe what to build
Agent
Default permissions
Outline
Timeline
main
Ln 111, Col 1
Spaces: 4
UTF-8
CRLF
Groovy


Not secure https://localhost:8081/applications/argocd/habit-tracker?view=list&resource=
argo
v3.5.3
Applications
ApplicationSets
Settings
User Info
Documentation
Resource filters
NAME
KINDS
SYNC STATUS
Synced 3
OutOfSync 0
HEALTH STATUS
Progressing 0
Suspended 0
Healthy 6
Degraded 0
Missing 0
Unknown 0
Applications / habit-tracker
APPLICATION DETAILS LIST
DETAILS DIFF SYNC SYNC STATUS HISTORY AND ROLLBACK DELETE REFRESH
Log out
APP HEALTH
Healthy
SYNC STATUS
Synced to main (1c2b0d3)
Auto sync is enabled.
Author: Jenkins <jenkins@localhost> -
Comment: Update Habit Tracker image to 6a0d9d5
LAST SYNC
Sync OK to 1c2b0d3
Succeeded 7 hours ago (Thu Oct 01 2026 02:01:04 GMT+0530)
Author: Jenkins <jenkins@localhost> -
Comment: Update Habit Tracker image to 6a0d9d5
Items per page: 10
NAME GROUP/KIND SYNC ORDER NAMESPACE CREATED AT STATUS
habit-tracker-76dbfb8656-56zmr Pod - default 7 hours ago 10/01/26 Healthy
habit-tracker-76dbfb8656-gxczl Pod - default 7 hours ago 10/01/26 Healthy
habit-tracker-76dbfb8656 apps/ReplicaSet - default 7 hours ago 10/01/26 Healthy
habit-tracker-55cddf7c74 apps/ReplicaSet - default 11 hours ago 09/30/26 Healthy
habit-tracker ConfigMap - default 3 days ago 09/28/26 Synced
habit-tracker Service - default 3 days ago 09/28/26 Healthy Synced
habit-tracker apps/Deployment - default 3 days ago 09/28/26 Healthy Synced

Task 11 : Secrets & Security

Overview

Implemented basic security practices for the Habit Tracker application by securely managing Jenkins credentials, Kubernetes Secrets, ServiceAccount access, and container execution.

Execution

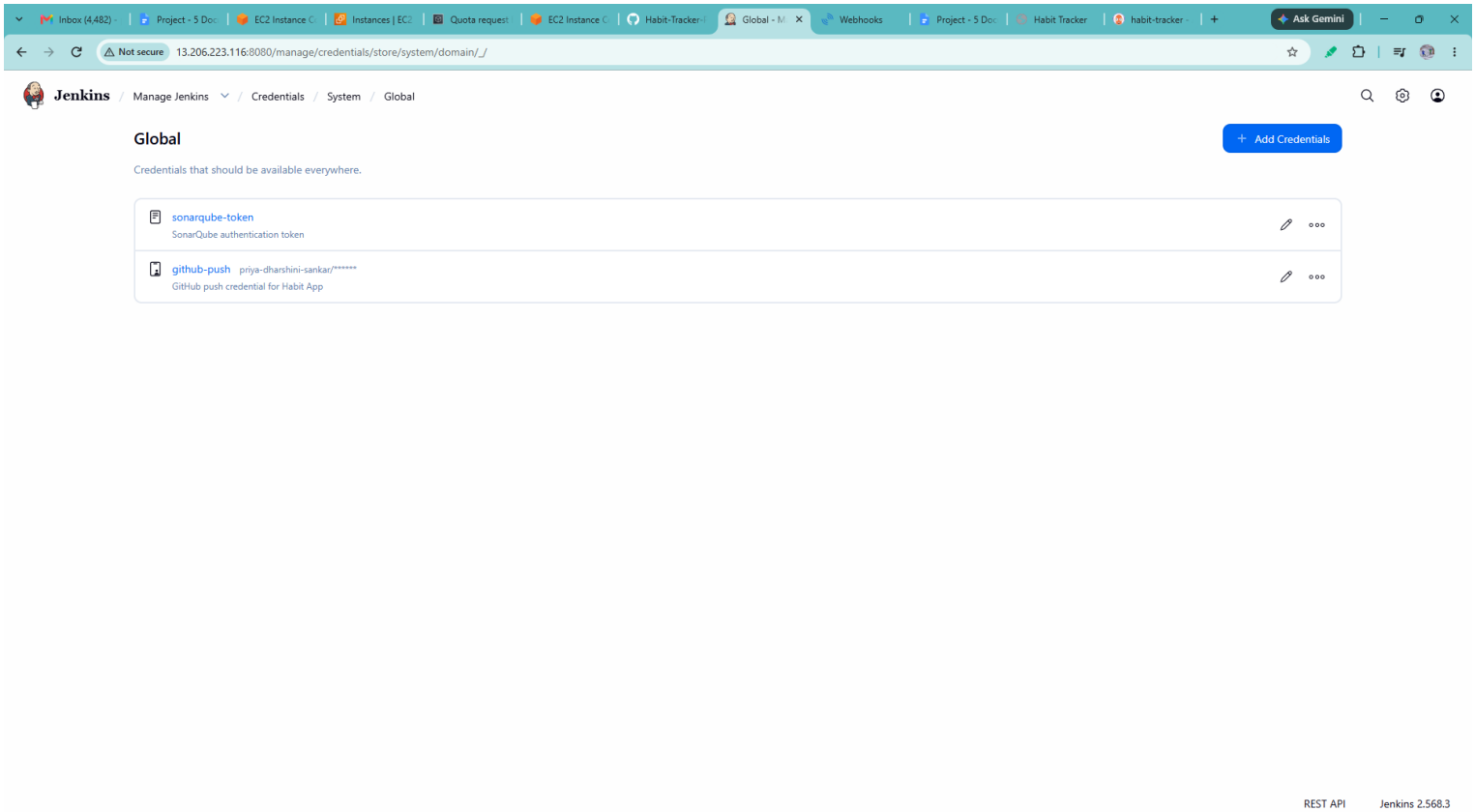
- Configured required credentials securely in Jenkins Credentials Manager.
- Used credential IDs in the Jenkins pipeline instead of hardcoding sensitive values.
- Configured the Docker container to run as a non-root user.
- Created the Kubernetes Secret habit-tracker-secret.
- Connected the application deployment to the Kubernetes Secret.
- Disabled automatic ServiceAccount token mounting.
- Restricted unnecessary ServiceAccount permissions.
- Verified that sensitive credentials were not exposed in the project files.

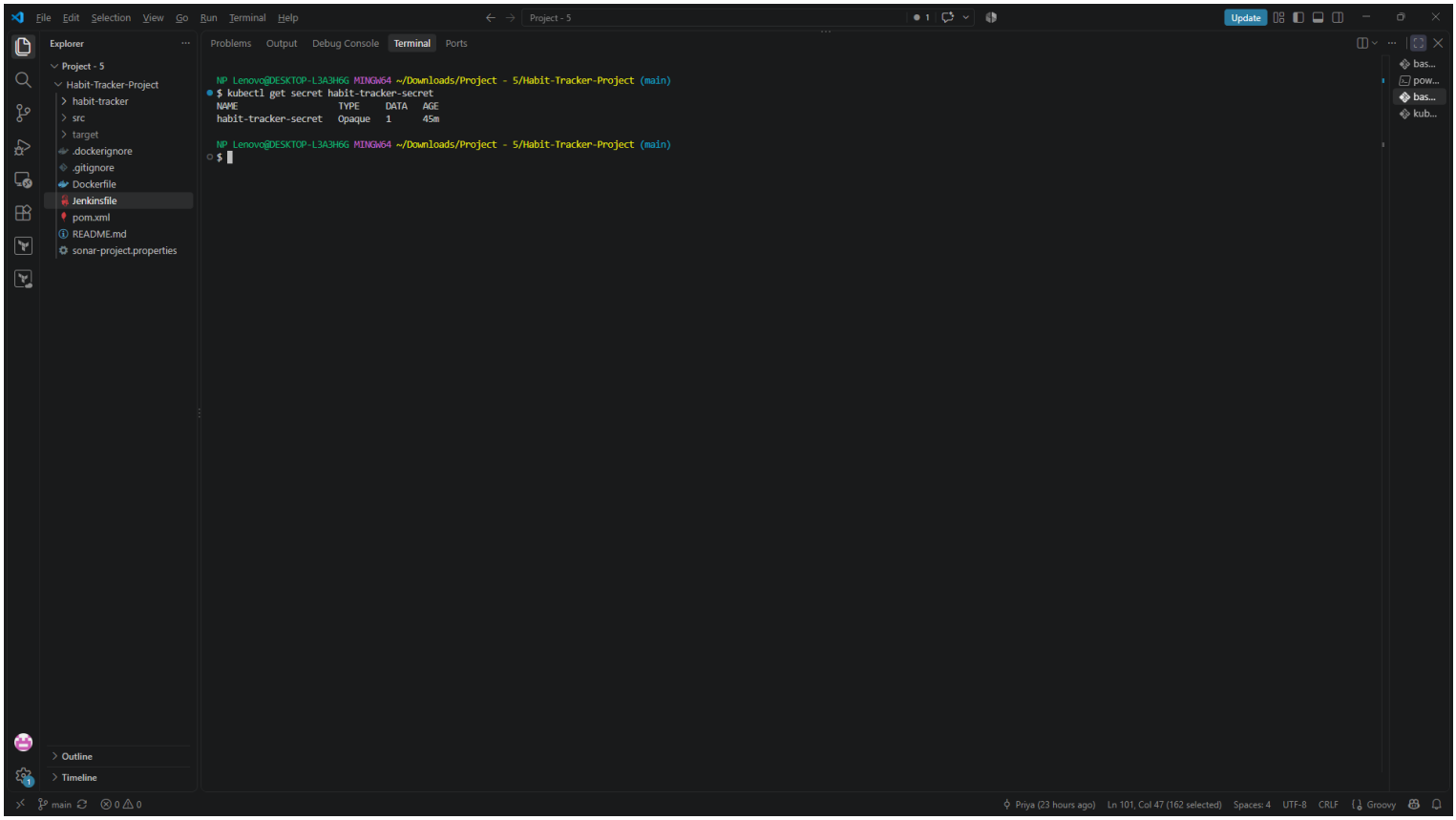
Verification

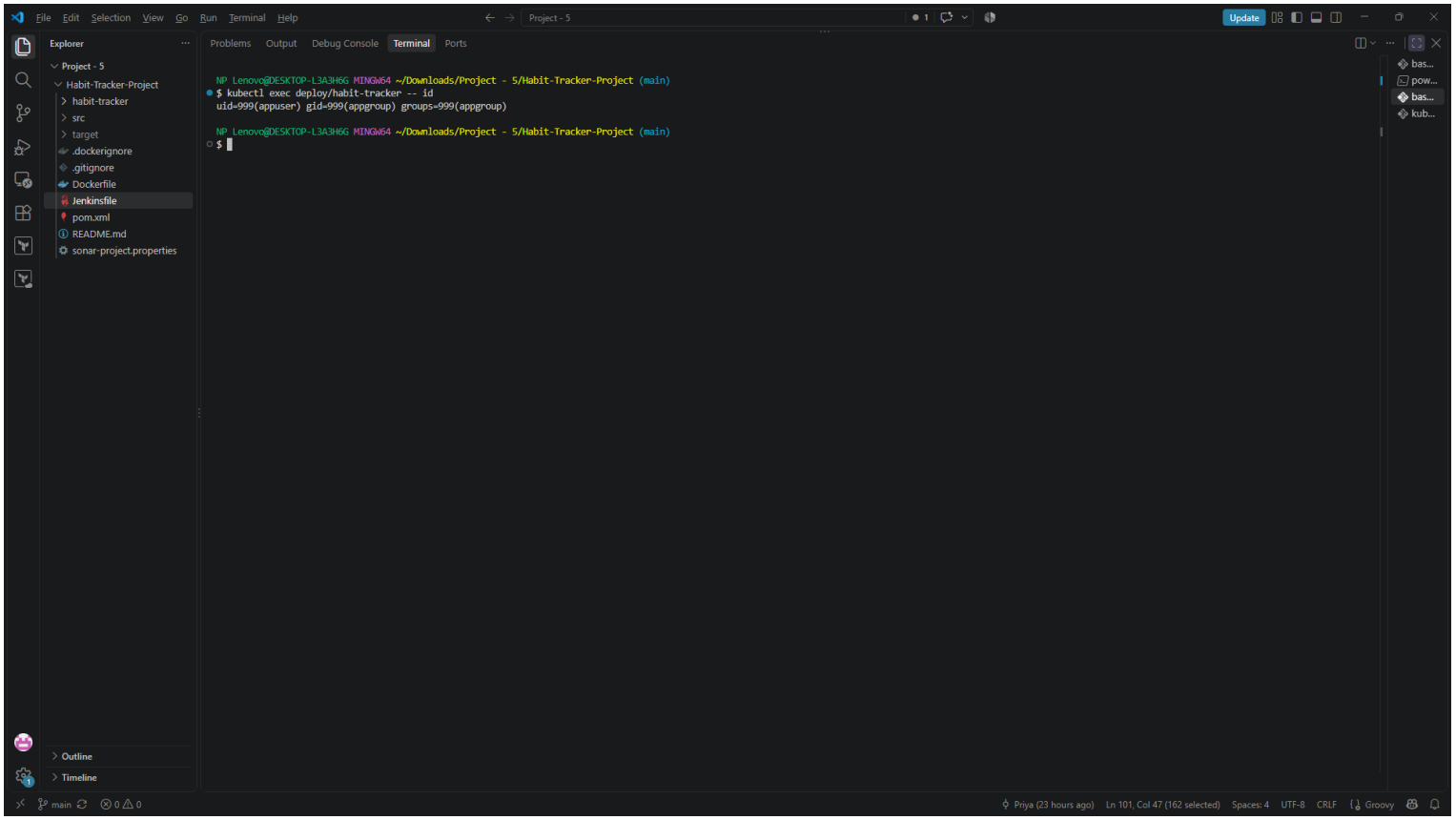
- Jenkins credentials were available in the Global Credentials store.
- habit-tracker-secret was present in Kubernetes.
- The application container was configured to run as appuser.
- ServiceAccount permissions were checked and unnecessary access was not granted.
- The Deployment correctly referenced the Kubernetes Secret.
- Project files were checked for exposed passwords, tokens, or credentials.

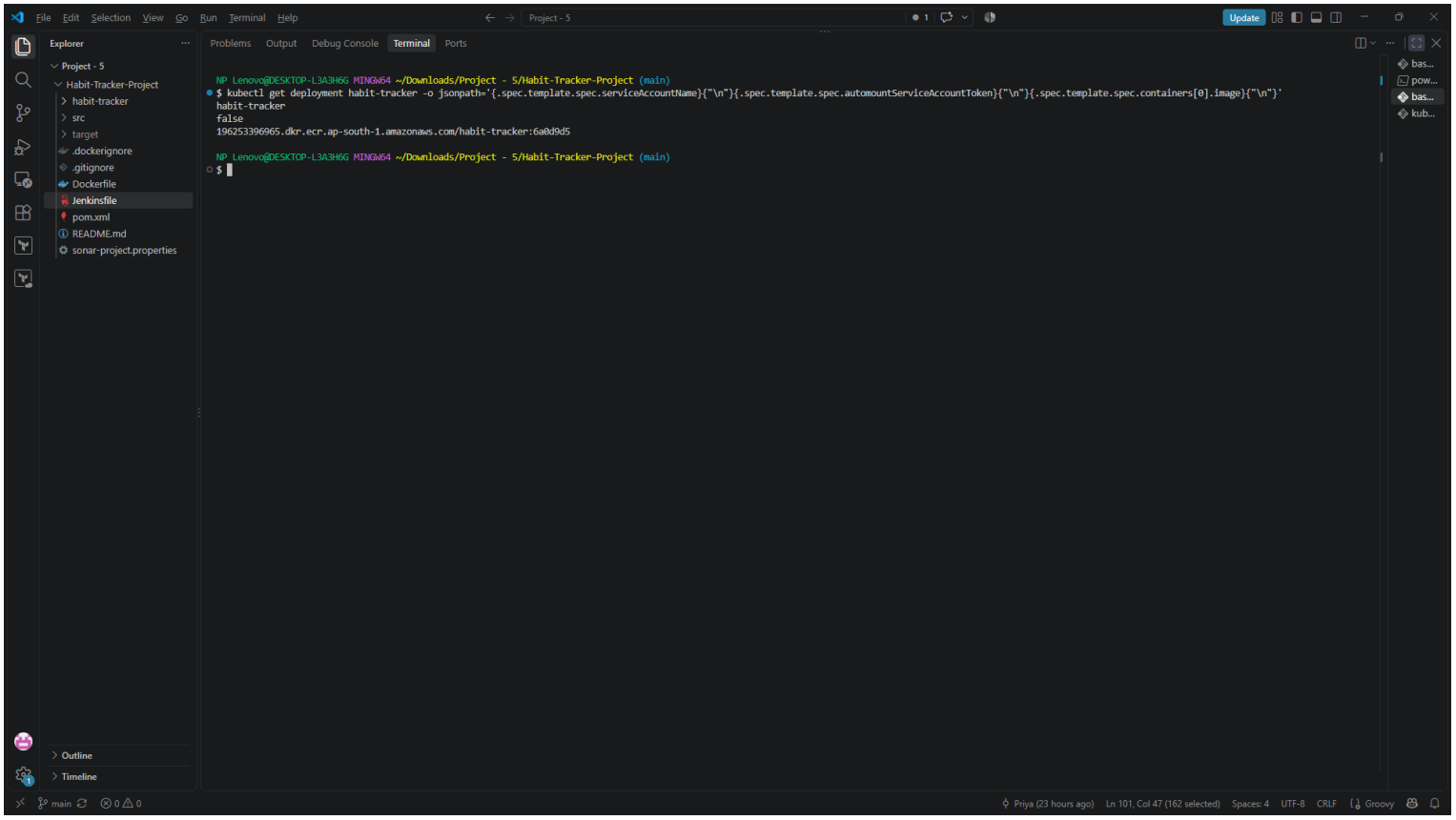
Result

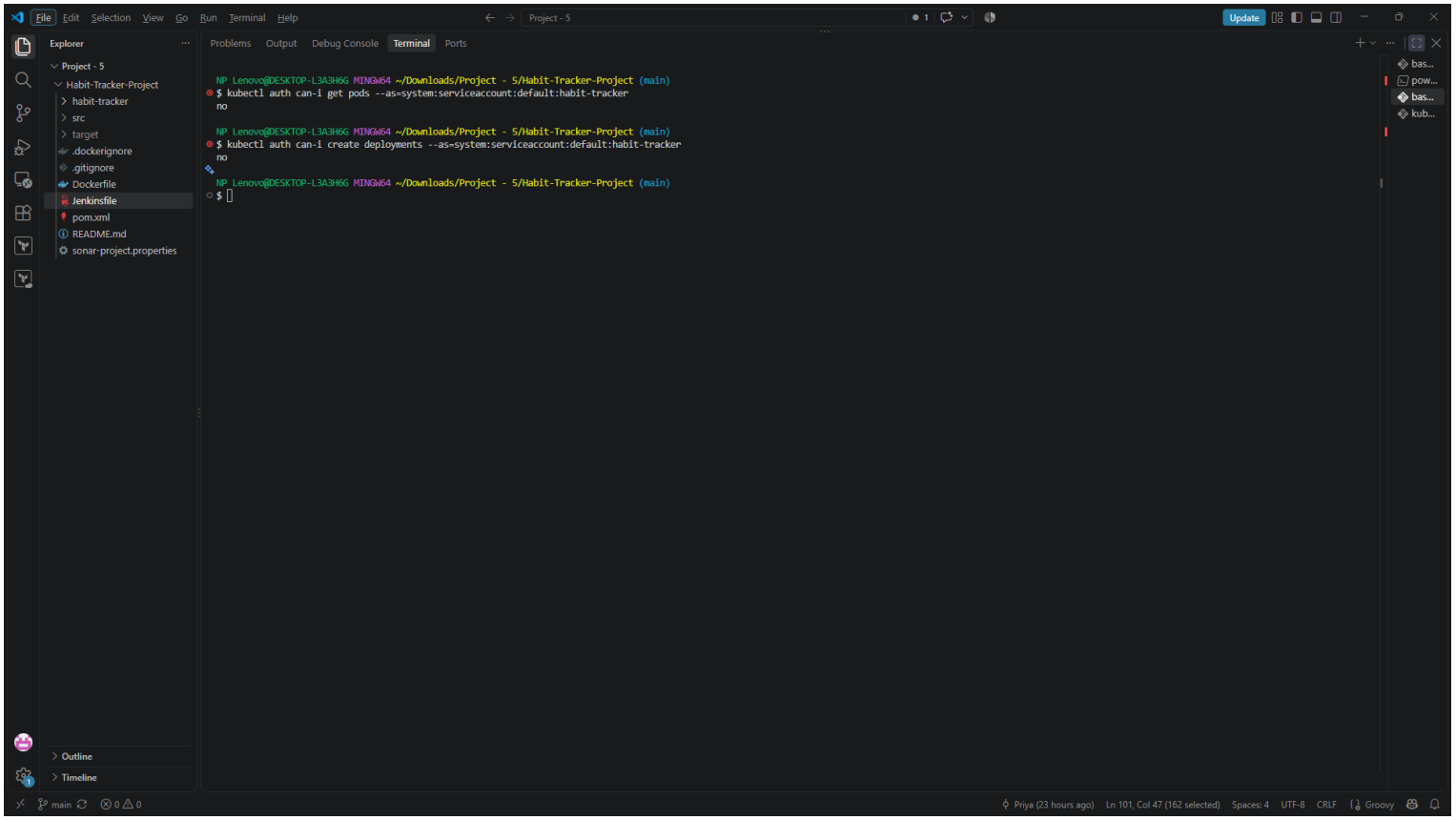
The Habit Tracker application follows the required basic security practices using Jenkins Credentials, Kubernetes Secrets, non-root container execution, and restricted ServiceAccount access.

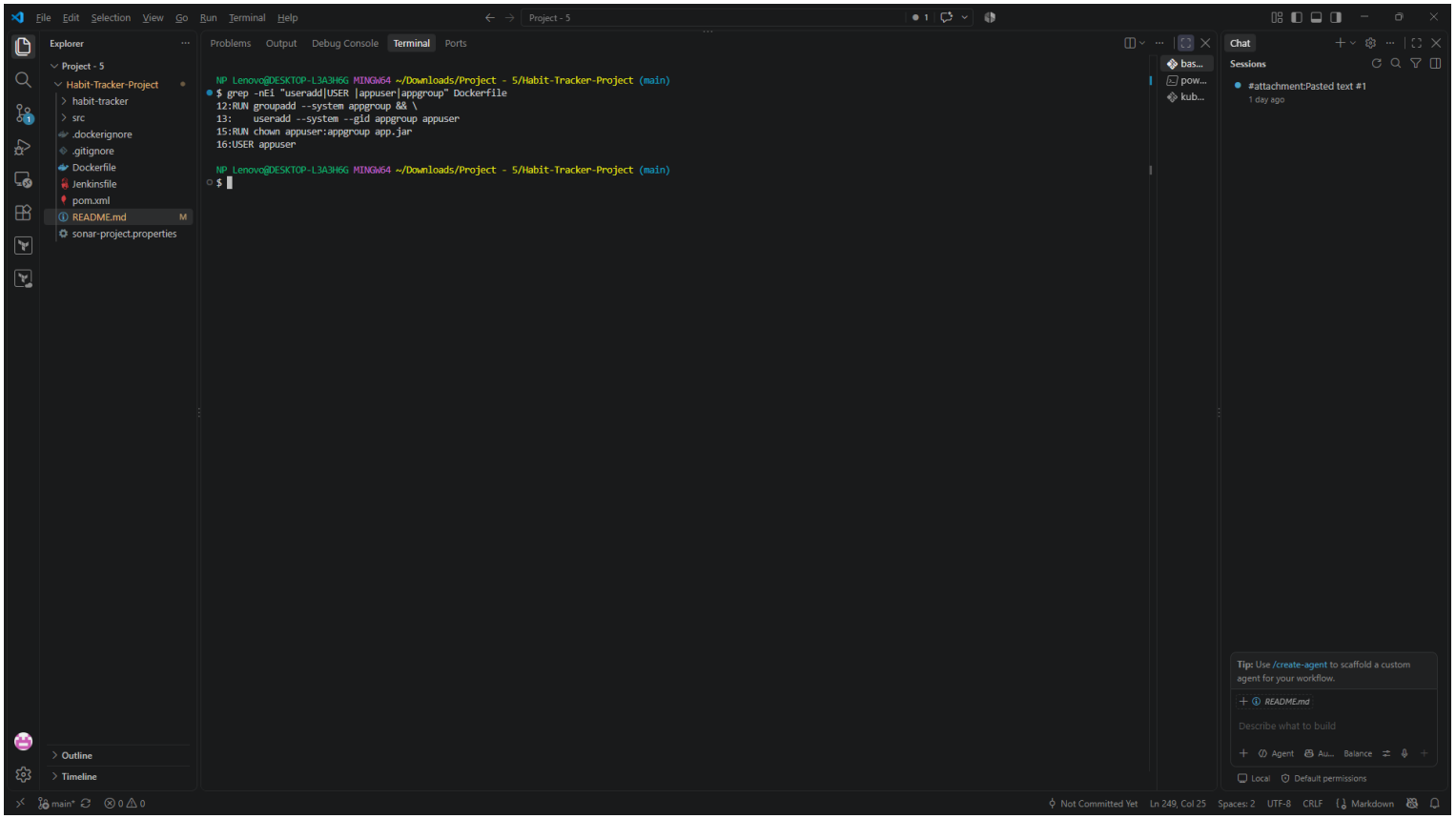

Jenkins / Manage Jenkins / Credentials / System / Global
Global
Credentials that should be available everywhere.
Add Credentials
sonarqube-token
SonarQube authentication token
github-push priya-dharshini-sankar/******
GitHub push credential for Habit App
REST API
Jenkins 2.568.3

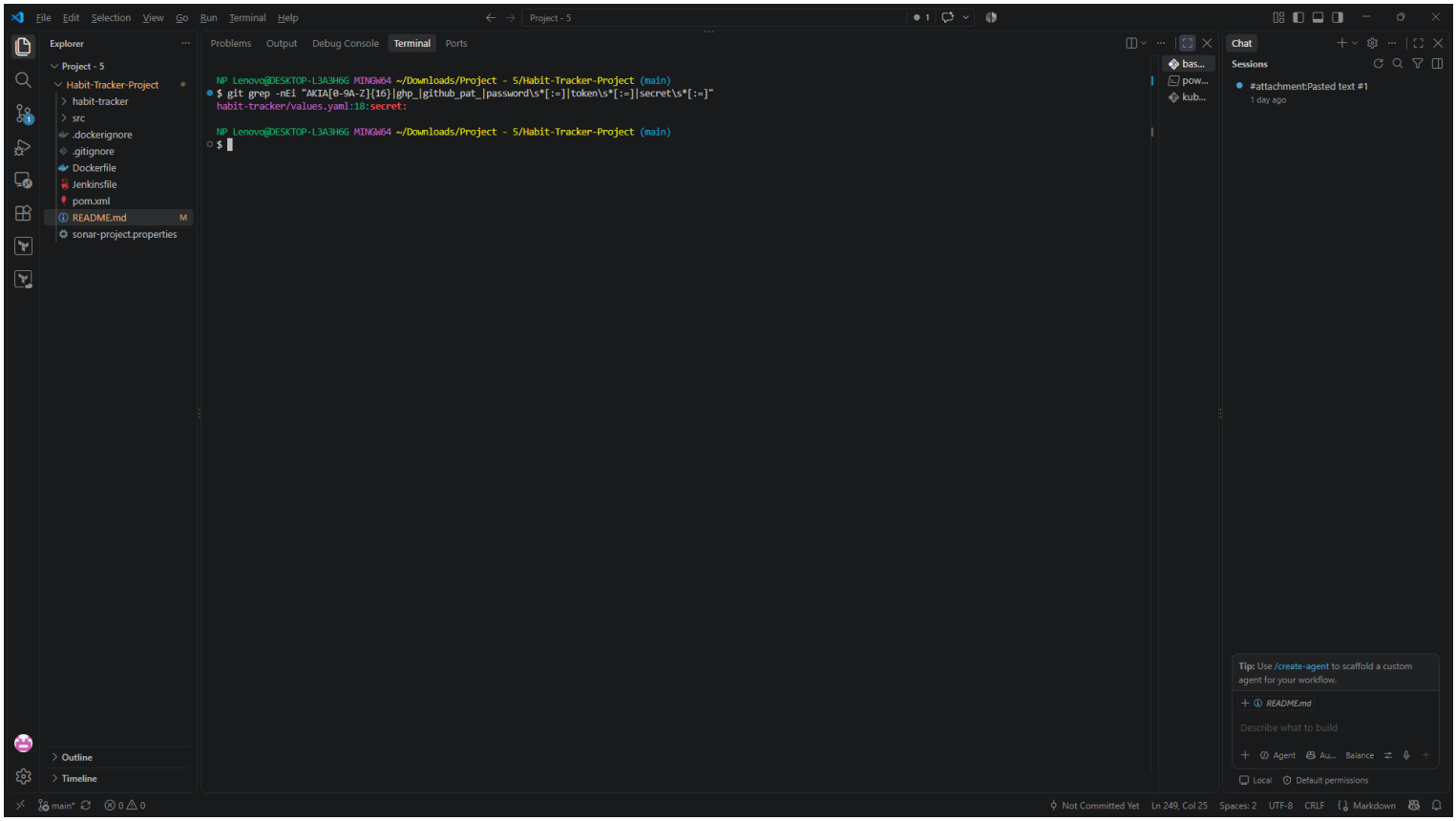

Explorer
Project - 5
Habit-Tracker-Project
habit-tracker
src
target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get secret habit-tracker-secret
NAME TYPE DATA AGE
habit-tracker-secret Opaque 1 45m
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Outline
Timeline

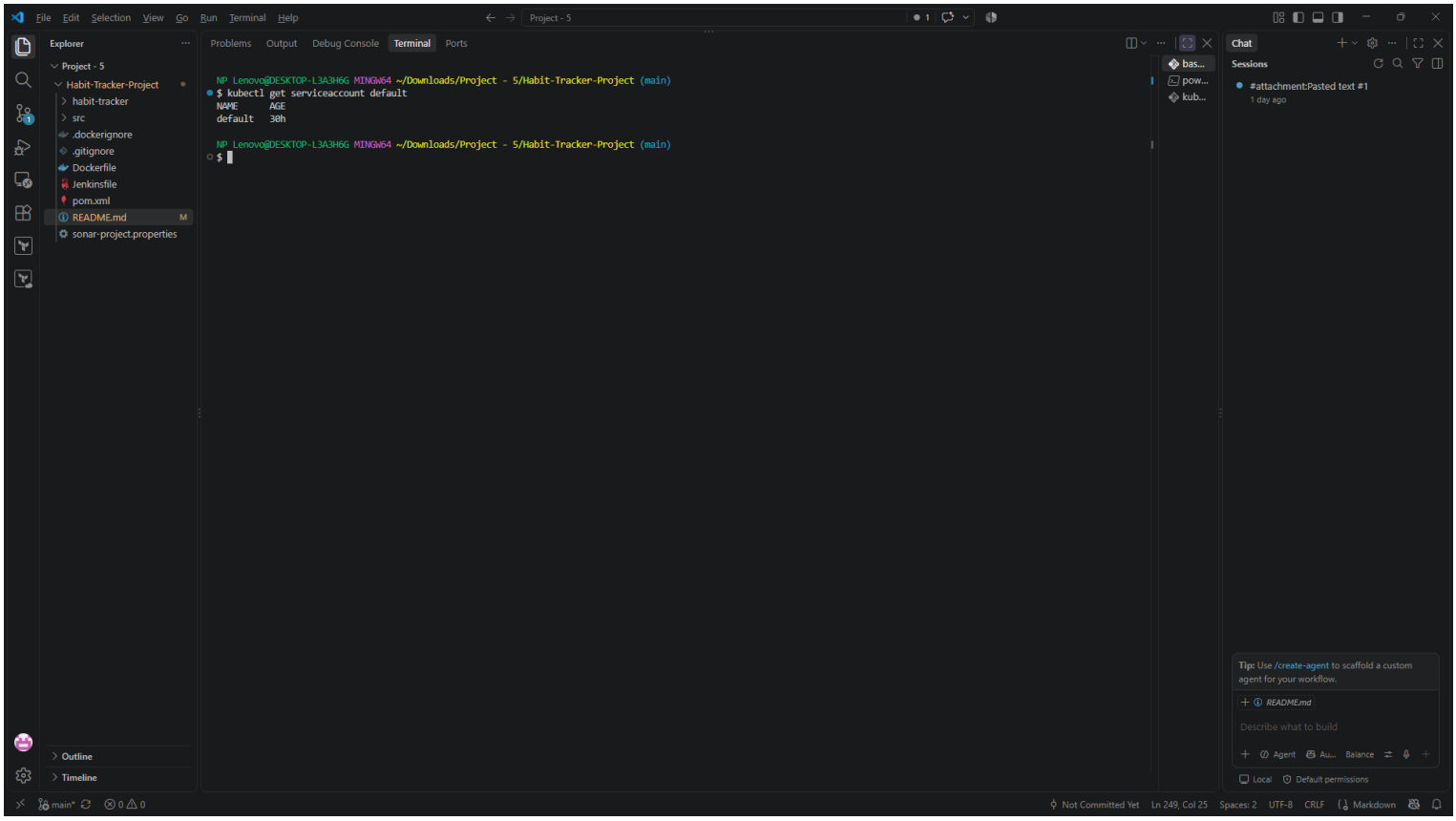

File Edit Selection View Go Run Terminal Help
Project - 5
Update
Explorer
Project - 5
Habit-Tracker-Project
habit-tracker
src
target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl exec deploy/habit-tracker -- id
uid=999(appuser) gid=999(appgroup) groups=999(appgroup)
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Outline
Timeline
main
Priya (23 hours ago) Ln 101, Col 47 (162 selected) Spaces: 4 UTF-8 CRLF Groovy

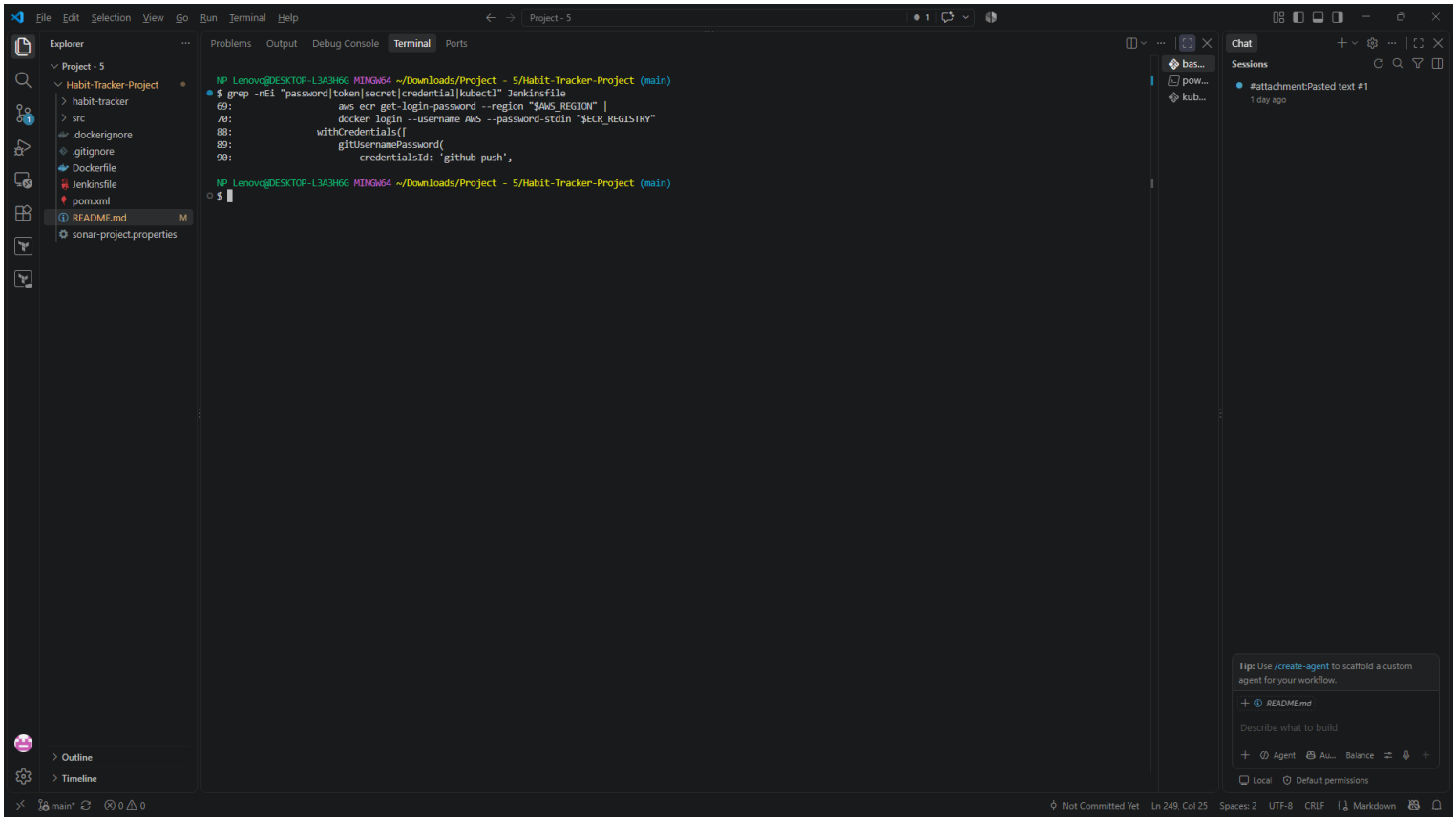

File Edit Selection View Go Run Terminal Help
Project - 5
Update
Explorer
Project - 5
Habit-Tracker-Project
habit-tracker
src
target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get deployment habit-tracker -o jsonpath='{.spec.template.spec.serviceAccountName}{"\n"}{.spec.template.spec.automountServiceAccountToken}{"\n"}{.spec.template.spec.containers[0].image}{"\n"}'
habit-tracker
false
196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker:6a0d9d5
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Outline
Timeline
main
Priya (23 hours ago) Ln 101, Col 47 (162 selected) Spaces: 4 UTF-8 CRLF Groovy

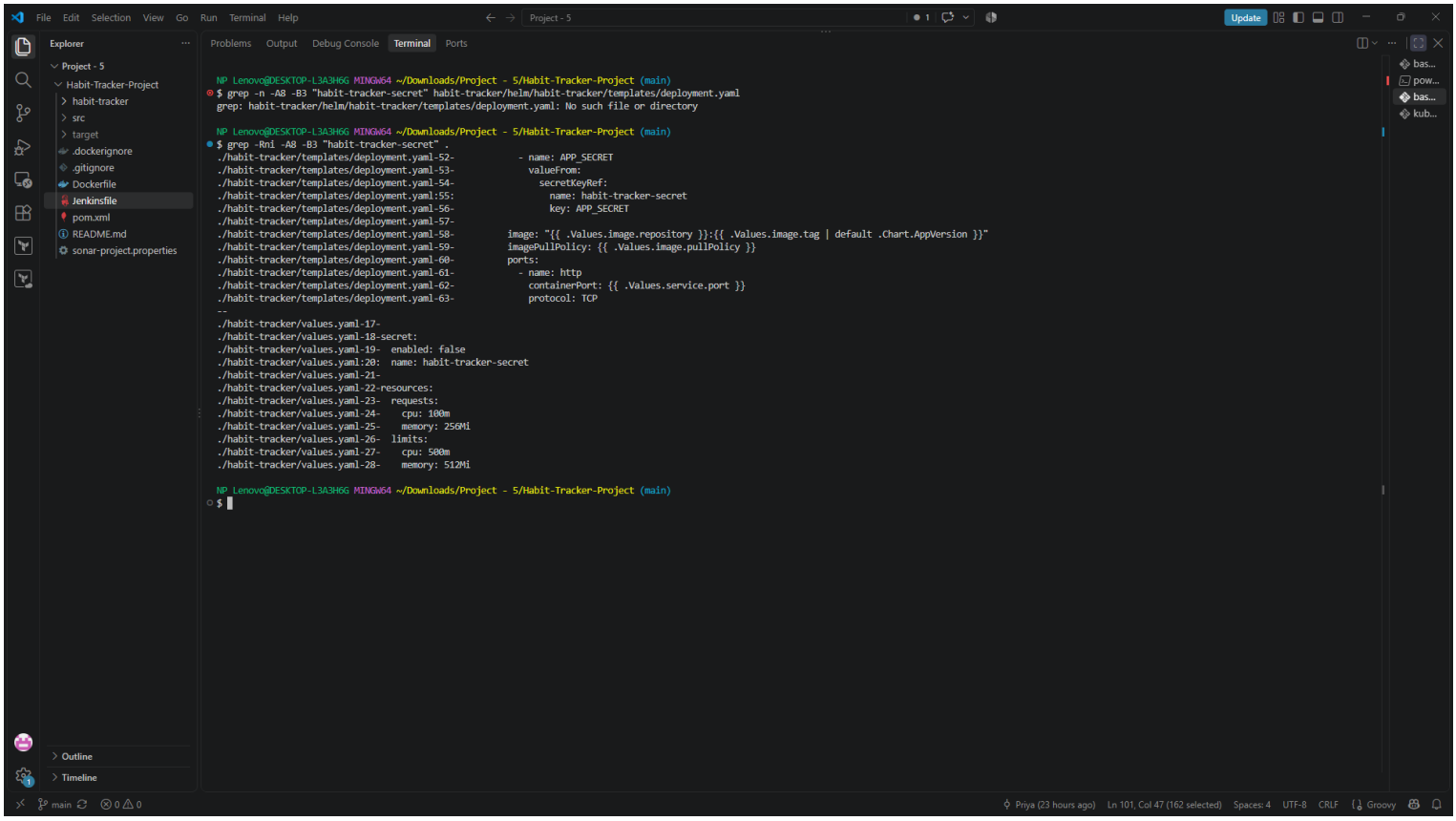

File Edit Selection View Go Run Terminal Help
Project - 5
Update
Explorer
Project - 5
Habit-Tracker-Project
habit-tracker
src
target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl auth can-i get pods --as=system:serviceaccount:default:habit-tracker
no
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl auth can-i create deployments --as=system:serviceaccount:default:habit-tracker
no
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Outline
Timeline
main
Priya (23 hours ago) Ln 101, Col 47 (162 selected) Spaces: 4 UTF-8 CRLF Groovy


File Edit Selection View Go Run Terminal Help
Project - 5
Explorer
Project - 5
Habit-Tracker-Project
habit-tracker
src
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md M
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ grep -nEi "useradd|USER |appuser|appgroup" Dockerfile
12:RUN groupadd --system appgroup && \
13: useradd --system --gid appgroup appuser
15:RUN chown appuser:appgroup app.jar
16:USER appuser
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Chat
Sessions
#attachment:Pasted text #1
1 day ago
Tip: Use /create-agent to scaffold a custom agent for your workflow.
README.md
Describe what to build
Agent
Local
Default permissions
Outline
Timeline
main*
Not Committed Yet Ln 249, Col 25 Spaces: 2 UTF-8 CRLF Markdown


File Edit Selection View Go Run Terminal Help
Project - 5
Explorer
Project - 5
Habit-Tracker-Project
habit-tracker
src
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md M
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ git grep -nEi "AKIA[0-9A-Z]{16}|ghp_|github_pat_|password\s*[:=]|token\s*[:=]|secret\s*[:=]"
habit-tracker/values.yaml:18:secret:
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Chat
Sessions
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UTF-8
CRLF
Markdown


File Edit Selection View Go Run Terminal Help
Project - 5
Explorer
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Habit-Tracker-Project
habit-tracker
src
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md M
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get serviceaccount default
NAME AGE
default 30h
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Chat
Sessions
#attachment:Pasted text #1
1 day ago
Tip: Use /create-agent to scaffold a custom agent for your workflow.
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Project - 5
Explorer
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habit-tracker
src
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ grep -nEi "password|token|secret|credential|kubectl" Jenkinsfile
69: aws ecr get-login-password --region "$AWS_REGION" |
70: docker login --username AWS --password-stdin "$ECR_REGISTRY"
88: withCredentials([
89: gitUsernamePassword(
90: credentialsId: 'github-push',
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Chat
Sessions
#attachment:Pasted text #1
1 day ago
Tip: Use /create-agent to scaffold a custom agent for your workflow.
README.md
Describe what to build
Agent
Balance
Local
Default permissions
Outline
Timeline
main*
Not Committed Yet
Ln 249, Col 25
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Project - 5
Update
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Project - 5
Habit-Tracker-Project
habit-tracker
src
target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ grep -n -A8 -B3 "habit-tracker-secret" habit-tracker/helm/habit-tracker/templates/deployment.yaml
grep: habit-tracker/helm/habit-tracker/templates/deployment.yaml: No such file or directory
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ grep -Rni -A8 -B3 "habit-tracker-secret" .
./habit-tracker/templates/deployment.yaml-52- - name: APP_SECRET
./habit-tracker/templates/deployment.yaml-53- valueFrom:
./habit-tracker/templates/deployment.yaml-54- secretKeyRef:
./habit-tracker/templates/deployment.yaml:55: name: habit-tracker-secret
./habit-tracker/templates/deployment.yaml-56- key: APP_SECRET
./habit-tracker/templates/deployment.yaml-57-
./habit-tracker/templates/deployment.yaml-58- image: "{{ .Values.image.repository }}:{{ .Values.image.tag | default .Chart.AppVersion }}"
./habit-tracker/templates/deployment.yaml-59- imagePullPolicy: {{ .Values.image.pullPolicy }}
./habit-tracker/templates/deployment.yaml-60- ports:
./habit-tracker/templates/deployment.yaml-61- - name: http
./habit-tracker/templates/deployment.yaml-62- containerPort: {{ .Values.service.port }}
./habit-tracker/templates/deployment.yaml-63- protocol: TCP
--
./habit-tracker/values.yaml-17-
./habit-tracker/values.yaml-18-secret:
./habit-tracker/values.yaml-19- enabled: false
./habit-tracker/values.yaml:20: name: habit-tracker-secret
./habit-tracker/values.yaml-21-
./habit-tracker/values.yaml-22-resources:
./habit-tracker/values.yaml-23- requests:
./habit-tracker/values.yaml-24- cpu: 100m
./habit-tracker/values.yaml-25- memory: 256Mi
./habit-tracker/values.yaml-26- limits:
./habit-tracker/values.yaml-27- cpu: 500m
./habit-tracker/values.yaml-28- memory: 512Mi
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Outline
Timeline
main
Priya (23 hours ago)
Ln 101, Col 47 (162 selected)
Spaces: 4
UTF-8
CRLF
Groovy

Task 12 : Deployment & Rollback

Overview

Implemented and verified rolling deployment and rollback for the Habit Tracker application using Kubernetes, Jenkins, Helm, and Argo CD.

Execution

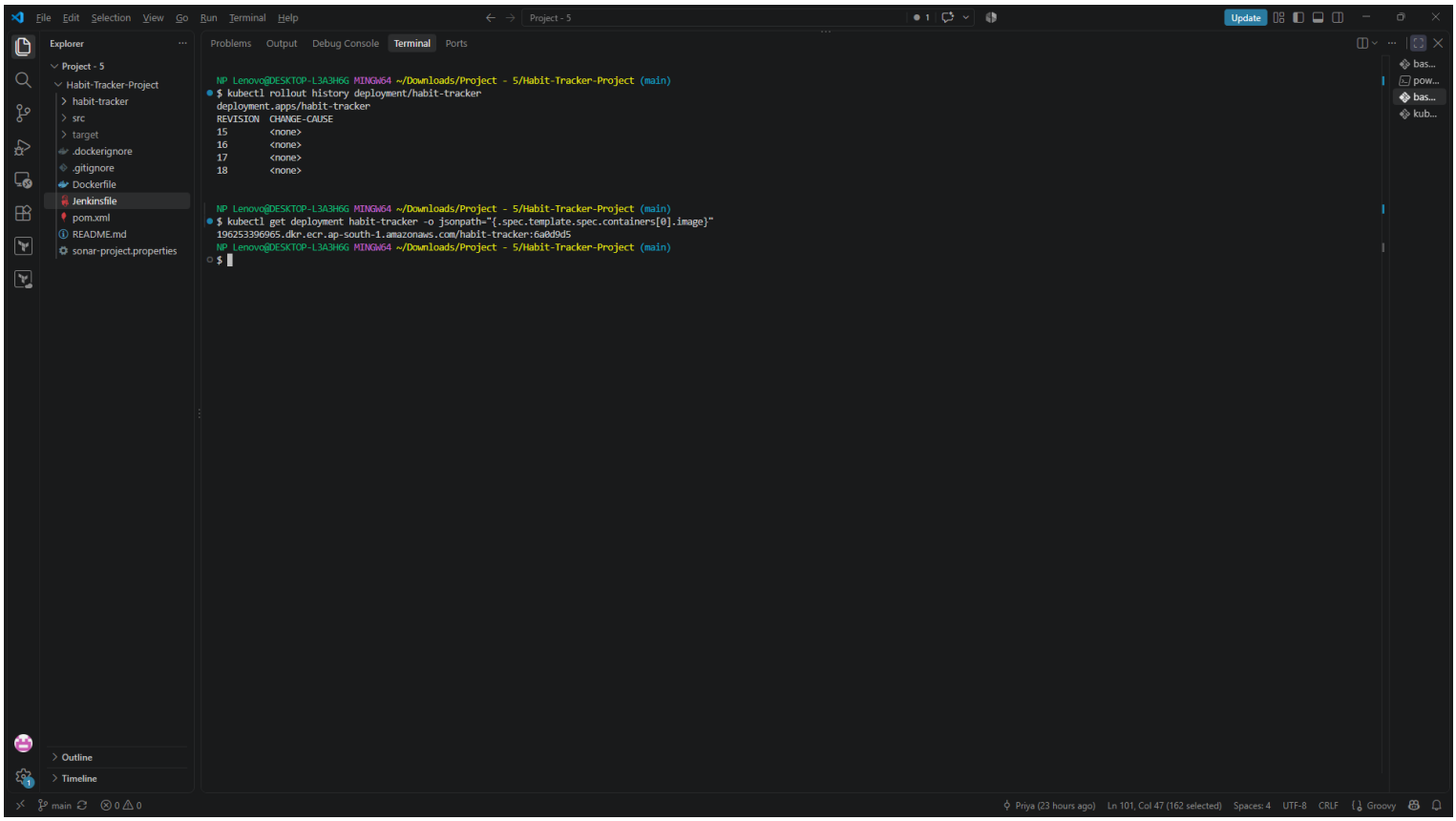
- Created a new Habit Tracker application version with Git commit d7cc5bb.
- Jenkins built and tested the application, created the Docker image, pushed it to Amazon ECR, and updated the Helm image tag.
- Argo CD synchronized the new application version to the Kubernetes cluster.
- Verified the rolling deployment using Kubernetes rollout status.
- Verified that the deployment completed successfully with 2/2 replicas available.
- Performed Kubernetes rollback using kubectl rollout undo deployment/habit-tracker
- Restored the previous working image 6a0d9d5.
- Created a Git rollback commit : 5deb4fe Rollback Habit Tracker image to 6a0d9d5
- Argo CD reflected the rollback state from Git and the final application was Synced and Healthy.

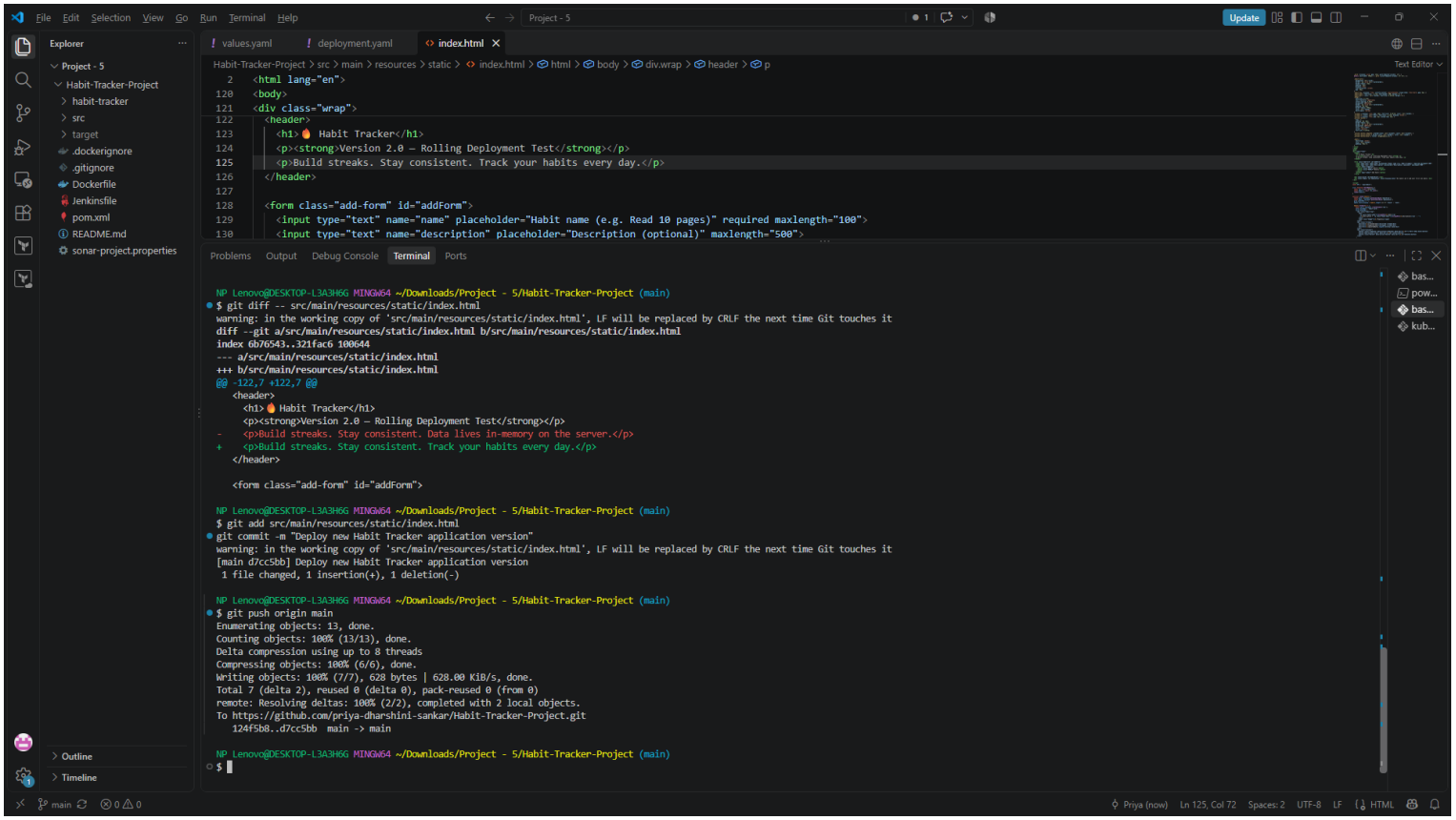
Verification

- New application image d7cc5bb was deployed successfully.
- Rolling deployment completed successfully.
- Kubernetes Deployment showed 2/2 replicas available.
- All application pods were running successfully.
- Rollback restored image 6a0d9d5.
- Rollback rollout completed successfully.
- Final Deployment status was 2/2 READY, 2/2 AVAILABLE.
- Final application pods were in Running state with 0 restarts.
- Argo CD final status was Synced and Healthy.
- Git history clearly records the deployment and rollback:
    - d7cc5bb – Deploy new Habit Tracker application version
    - 32552c8 – Update Habit Tracker image to d7cc5bb
    - 5deb4fe – Rollback Habit Tracker image to 6a0d9d5

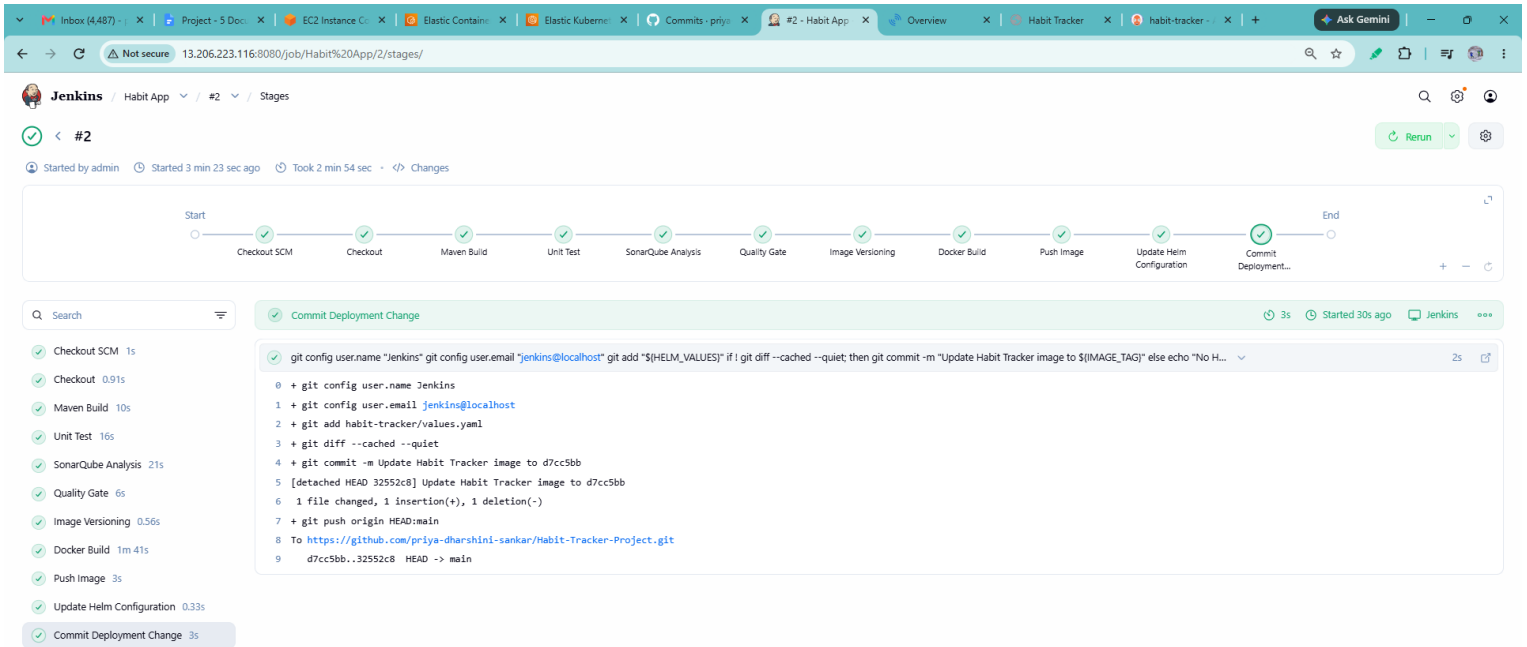
Result

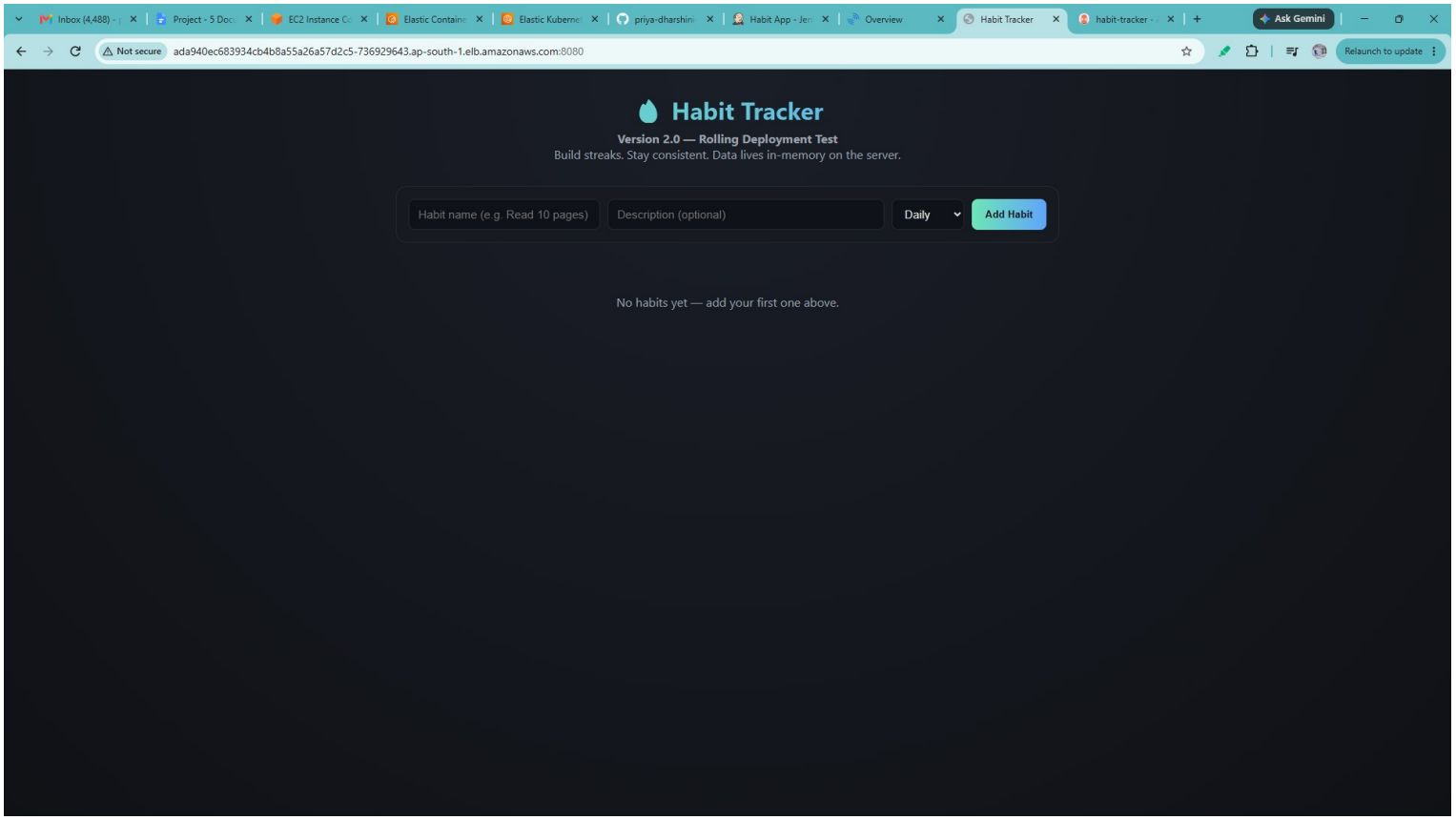
The Habit Tracker application successfully supports rolling deployment and rollback. The new application version was deployed and verified, and the previous working version was restored successfully through Kubernetes and GitOps.

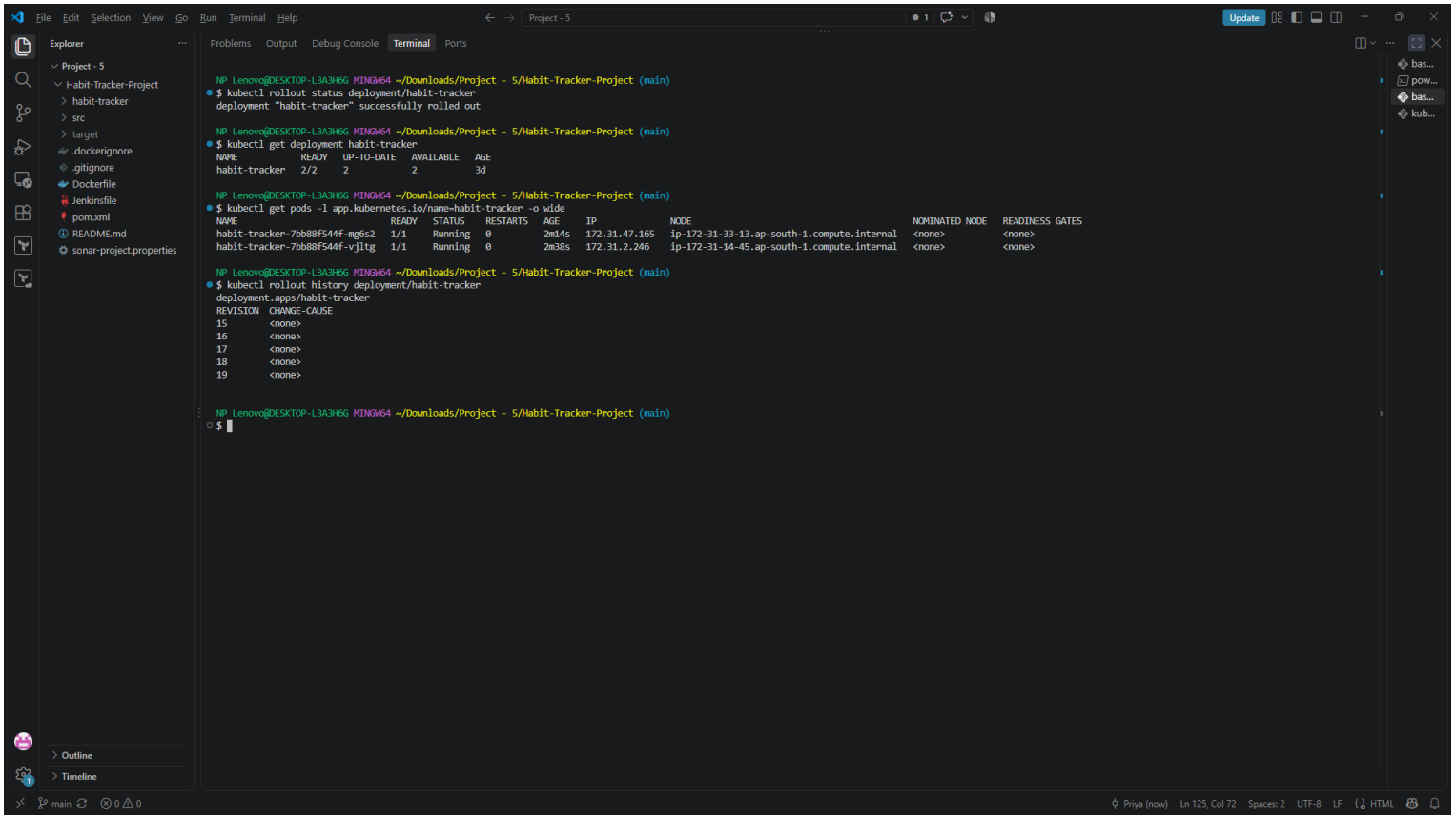


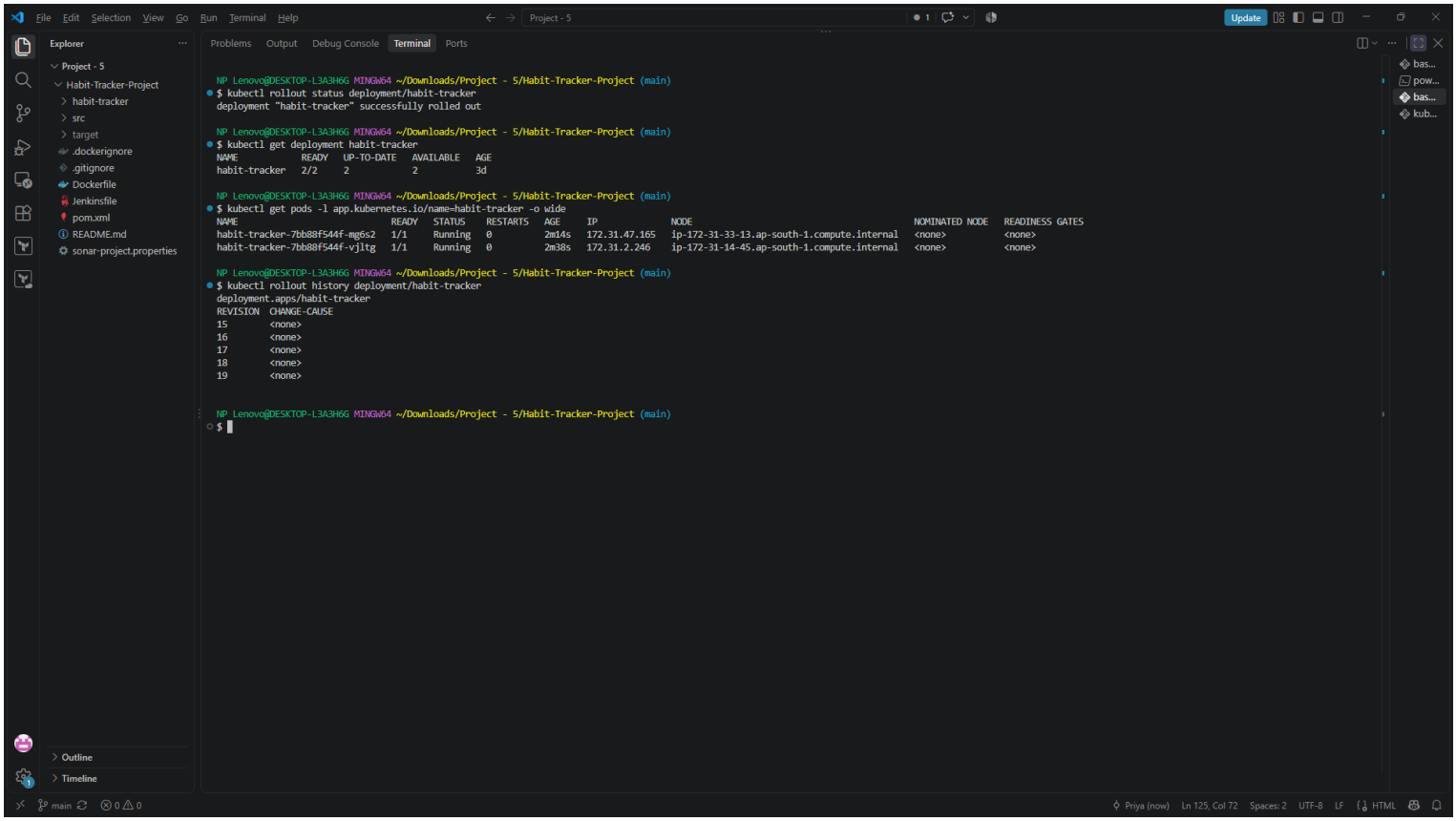

File Edit Selection View Go Run Terminal Help
Project - 5
Update
Explorer
Project - 5
Habit-Tracker-Project
habit-tracker
src
target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
values.yaml
deployment.yaml
index.html
Habit-Tracker-Project > src > main > resources > static > index.html > html > body > div.wrap > header > p
<html lang="en">
<body>

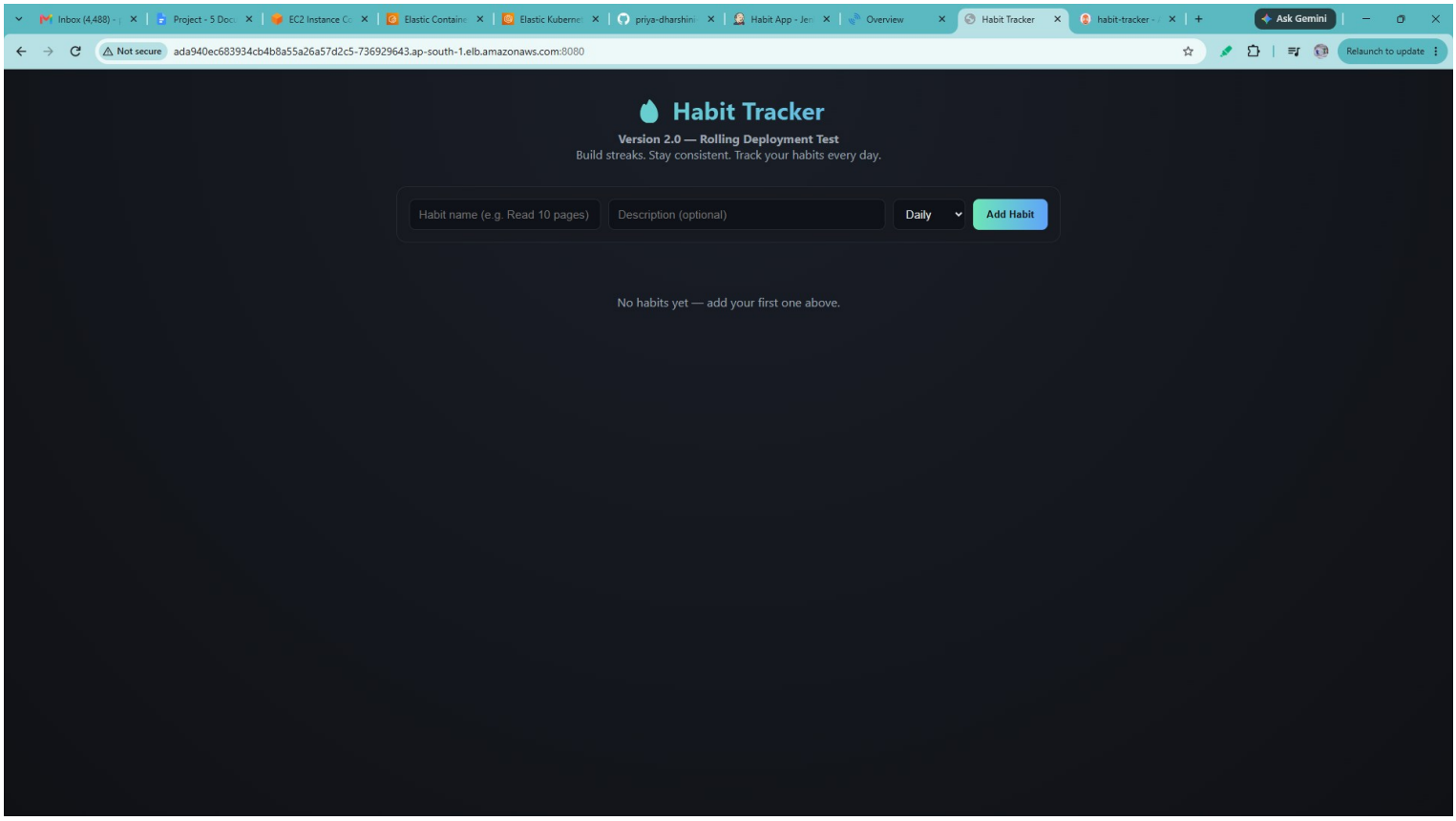
<header>
<h1>🔥 Habit Tracker</h1>
Version 2.0 — Rolling Deployment Test
Build streaks. Stay consistent. Track your habits every day.
</header>
<form class="add-form" id="addForm">
<input type="text" name="name" placeholder="Habit name (e.g. Read 10 pages)" required maxlength="100">
<input type="text" name="description" placeholder="Description (optional)" maxlength="500">
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ git diff -- src/main/resources/static/index.html
warning: in the working copy of 'src/main/resources/static/index.html', LF will be replaced by CRLF the next time Git touches it
diff --git a/src/main/resources/static/index.html b/src/main/resources/static/index.html
index 6b76543..321fac6 100644
--- a/src/main/resources/static/index.html
+++ b/src/main/resources/static/index.html
@@ -122,7 +122,7 @@
<header>
<h1>🔥 Habit Tracker</h1>
Version 2.0 — Rolling Deployment Test
- Build streaks. Stay consistent. Data lives in-memory on the server.
+ Build streaks. Stay consistent. Track your habits every day.
</header>
<form class="add-form" id="addForm">
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ git add src/main/resources/static/index.html
git commit -m "Deploy new Habit Tracker application version"
warning: in the working copy of 'src/main/resources/static/index.html', LF will be replaced by CRLF the next time Git touches it
[main d7cc5bb] Deploy new Habit Tracker application version
1 file changed, 1 insertion(+), 1 deletion(-)
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ git push origin main
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 628 bytes | 628.00 KiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/priya-dharshini-sankar/Habit-Tracker-Project.git
124f5b8..d7cc5bb main -> main
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
Outline
Timeline
main
Priya (now) Ln 125, Col 72 Spaces: 2 UTF-8 LF HTML

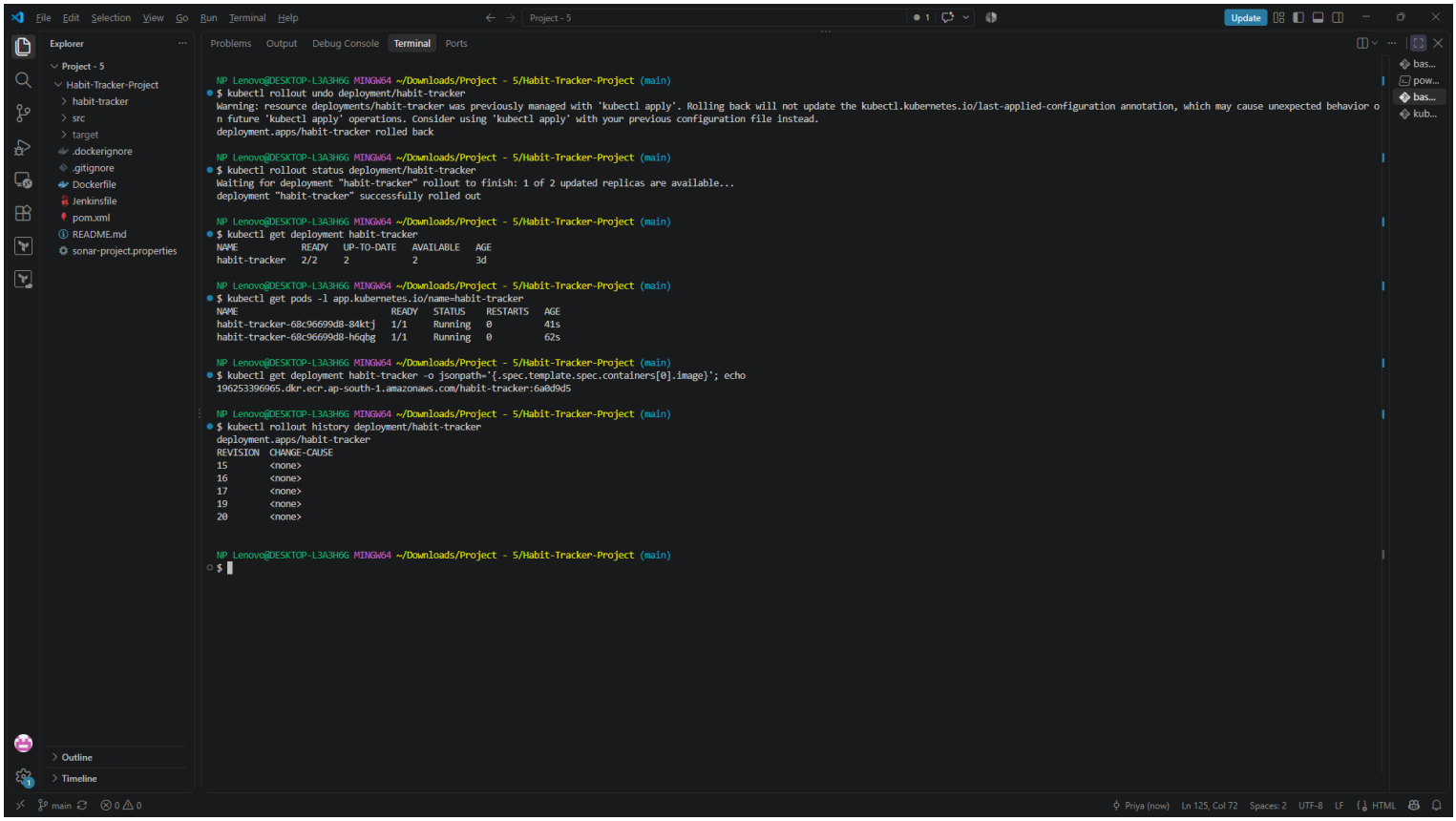

Not secure 13.206.223.116:8080/job/Habit%20App/2/stages/
Jenkins / Habit App / #2 / Stages
#2
Rerun
Started by admin Started 3 min 23 sec ago Took 2 min 54 sec Changes
Start
Checkout SCM
Checkout
Maven Build
Unit Test
SonarQube Analysis
Quality Gate
Image Versioning
Docker Build
Push Image
Update Helm Configuration
Commit Deployment...
End
Search
Checkout SCM 1s
Checkout 0.91s
Maven Build 10s
Unit Test 16s
SonarQube Analysis 21s
Quality Gate 6s
Image Versioning 0.56s
Docker Build 1m 41s
Push Image 3s
Update Helm Configuration 0.33s
Commit Deployment Change 3s
Commit Deployment Change
3s Started 30s ago Jenkins
git config user.name "Jenkins" git config user.email "jenkins@localhost" git add "${HELM_VALUES}" if ! git diff --cached --quiet; then git commit -m "Update Habit Tracker image to ${IMAGE_TAG}" else echo "No H... 2s
+ git config user.name Jenkins
+ git config user.email jenkins@localhost
+ git add habit-tracker/values.yaml
+ git diff --cached --quiet
+ git commit -m Update Habit Tracker image to d7cc5bb
[detached HEAD 32552c8] Update Habit Tracker image to d7cc5bb
1 file changed, 1 insertion(+), 1 deletion(-)
+ git push origin HEAD:main
To https://github.com/priya-dharshini-sankar/Habit-Tracker-Project.git
d7cc5bb..32552c8 HEAD -> main
Jenkins 2.568.3

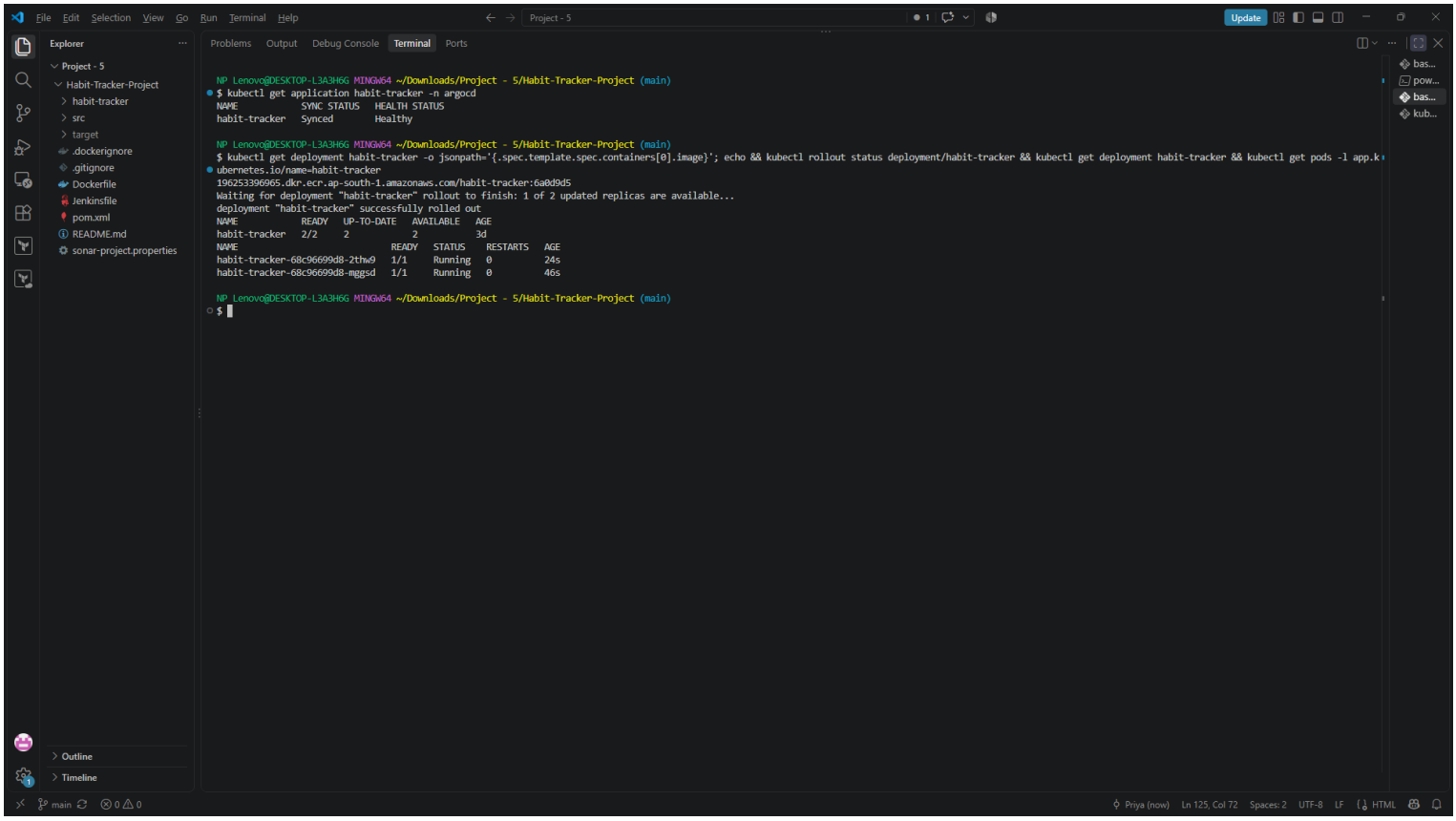

Habit Tracker
Version 2.0 — Rolling Deployment Test
Build streaks. Stay consistent. Data lives in-memory on the server.
Habit name (e.g. Read 10 pages)
Description (optional)
Daily
Add Habit
No habits yet — add your first one above.

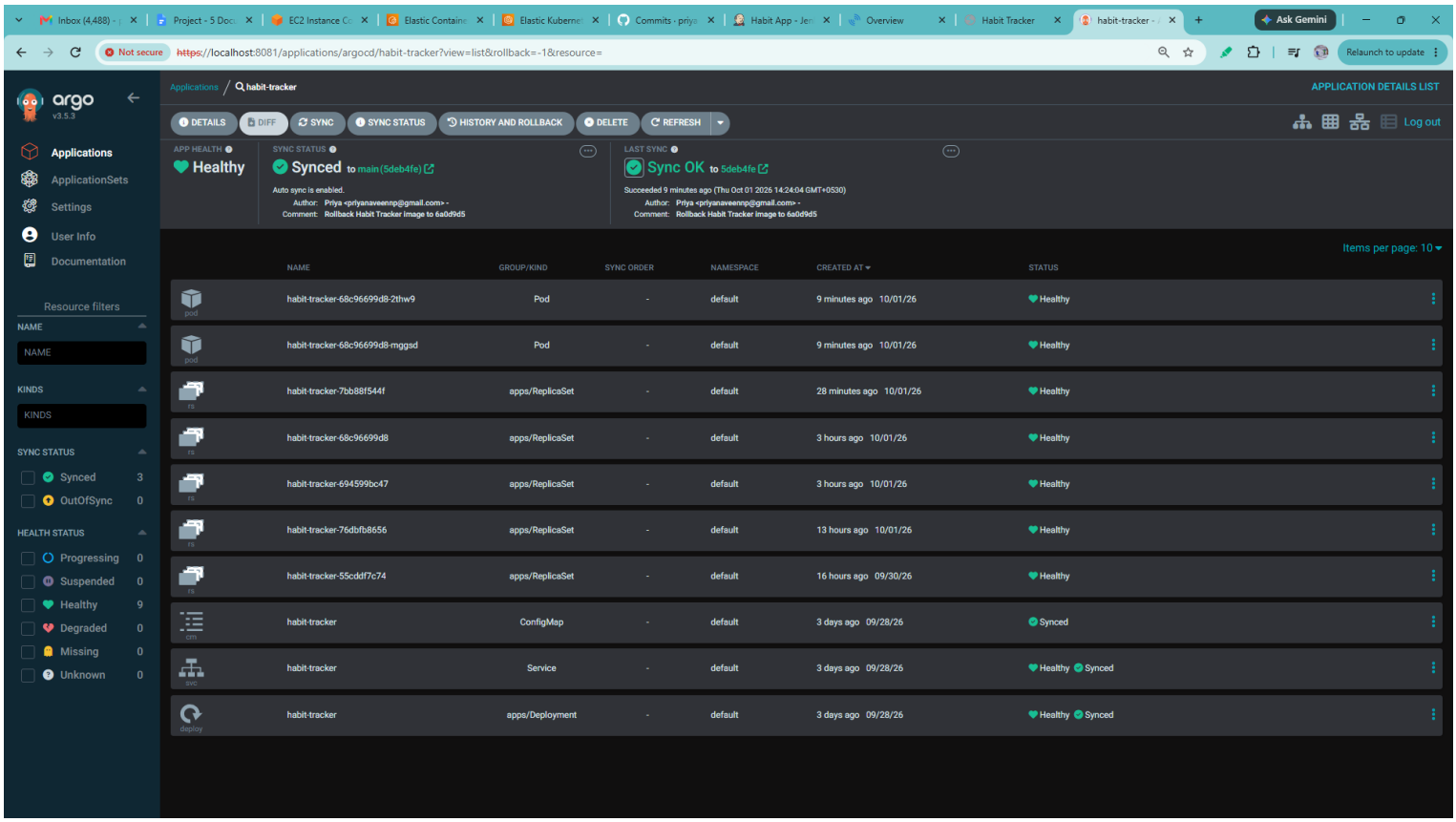

NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl rollout status deployment/habit-tracker
deployment "habit-tracker" successfully rolled out
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get deployment habit-tracker
NAME READY UP-TO-DATE AVAILABLE AGE
habit-tracker 2/2 2 2 3d
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get pods -l app.kubernetes.io/name=habit-tracker -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
habit-tracker-7bb88f544f-mg6s2 1/1 Running 0 2m14s 172.31.47.165 ip-172-31-33-13.ap-south-1.compute.internal <none> <none>
habit-tracker-7bb88f544f-vjltg 1/1 Running 0 2m38s 172.31.2.246 ip-172-31-14-45.ap-south-1.compute.internal <none> <none>
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl rollout history deployment/habit-tracker
deployment.apps/habit-tracker
REVISION CHANGE-CAUSE
15 <none>
16 <none>
17 <none>
18 <none>
19 <none>
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$

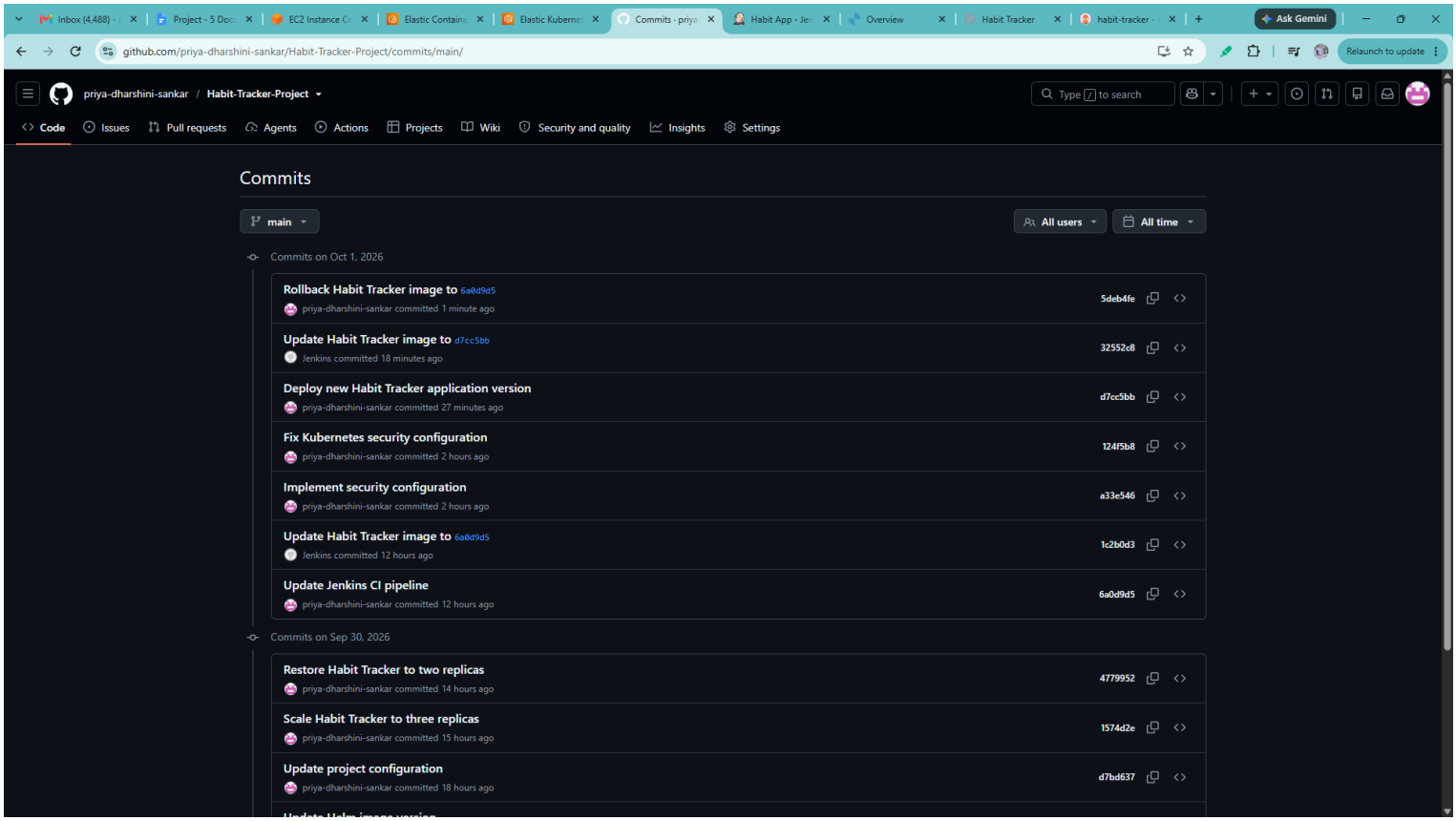

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src
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.dockerignore
.gitignore
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pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl rollout status deployment/habit-tracker
deployment "habit-tracker" successfully rolled out
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get deployment habit-tracker
NAME READY UP-TO-DATE AVAILABLE AGE
habit-tracker 2/2 2 2 3d
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get pods -l app.kubernetes.io/name=habit-tracker -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
habit-tracker-7bb88f544f-mg6s2 1/1 Running 0 2m14s 172.31.47.165 ip-172-31-33-13.ap-south-1.compute.internal <none> <none>
habit-tracker-7bb88f544f-vjltg 1/1 Running 0 2m38s 172.31.2.246 ip-172-31-14-45.ap-south-1.compute.internal <none> <none>
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl rollout history deployment/habit-tracker
deployment.apps/habit-tracker
REVISION CHANGE-CAUSE
15 <none>
16 <none>
17 <none>
18 <none>
19 <none>
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
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Ln 125, Col 72 Spaces: 2 UTF-8 LF HTML


Habit Tracker
Not secure
ada940ec683934cb4b8a55a26a57d2c5-736929643.ap-south-1.elb.amazonaws.com:8080
Ask Gemini
Relaunch to update
Habit Tracker
Version 2.0 — Rolling Deployment Test
Build streaks. Stay consistent. Track your habits every day.
Habit name (e.g. Read 10 pages)
Description (optional)
Daily
Add Habit
No habits yet — add your first one above.


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Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl rollout undo deployment/habit-tracker
Warning: resource deployments/habit-tracker was previously managed with 'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last-applied-configuration annotation, which may cause unexpected behavior o
n future 'kubectl apply' operations. Consider using 'kubectl apply' with your previous configuration file instead.
deployment.apps/habit-tracker rolled back
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl rollout status deployment/habit-tracker
Waiting for deployment "habit-tracker" rollout to finish: 1 of 2 updated replicas are available...
deployment "habit-tracker" successfully rolled out
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get deployment habit-tracker
NAME READY UP-TO-DATE AVAILABLE AGE
habit-tracker 2/2 2 2 3d
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get pods -l app.kubernetes.io/name=habit-tracker
NAME READY STATUS RESTARTS AGE
habit-tracker-68c96699d8-84ktj 1/1 Running 0 41s
habit-tracker-68c96699d8-h6qbg 1/1 Running 0 62s
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get deployment habit-tracker -o jsonpath='{.spec.template.spec.containers[0].image}'; echo
196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker:6a0d9d5
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl rollout history deployment/habit-tracker
deployment.apps/habit-tracker
REVISION CHANGE-CAUSE
15 <none>
16 <none>
17 <none>
19 <none>
20 <none>
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
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Priya (now) Ln 125, Col 72 Spaces: 2 UTF-8 LF HTML


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Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get application habit-tracker -n argocd
NAME SYNC STATUS HEALTH STATUS
habit-tracker Synced Healthy
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get deployment habit-tracker -o jsonpath='{.spec.template.spec.containers[0].image}'; echo && kubectl rollout status deployment/habit-tracker && kubectl get deployment habit-tracker && kubectl get pods -l app.k
ubernetes.io/name=habit-tracker
196253396965.dkr.ecr.ap-south-1.amazonaws.com/habit-tracker:6a0d9d5
Waiting for deployment "habit-tracker" rollout to finish: 1 of 2 updated replicas are available...
deployment "habit-tracker" successfully rolled out
NAME READY UP-TO-DATE AVAILABLE AGE
habit-tracker 2/2 2 2 3d
NAME READY STATUS RESTARTS AGE
habit-tracker-68c96699d8-2thw9 1/1 Running 0 24s
habit-tracker-68c96699d8-mggsd 1/1 Running 0 46s
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
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argo
v3.5.3
Applications
ApplicationSets
Settings
User Info
Documentation
Resource filters
NAME
KINDS
SYNC STATUS
Synced 3
OutOfSync 0
HEALTH STATUS
Progressing 0
Suspended 0
Healthy 9
Degraded 0
Missing 0
Unknown 0
Applications / habit-tracker
APPLICATION DETAILS LIST
DETAILS
DIFF
SYNC
SYNC STATUS
HISTORY AND ROLLBACK
DELETE
REFRESH
Log out
APP HEALTH
Healthy
SYNC STATUS
Synced to main (5deb4fe)
Auto sync is enabled.
Author: Priya <priyanaveennp@gmail.com> -
Comment: Rollback Habit Tracker image to 6a0d9d5
LAST SYNC
Sync OK to 5deb4fe
Succeeded 9 minutes ago (Thu Oct 01 2026 14:24:04 GMT+0530)
Author: Priya <priyanaveennp@gmail.com> -
Comment: Rollback Habit Tracker image to 6a0d9d5
Items per page: 10
NAME
GROUP/KIND
SYNC ORDER
NAMESPACE
CREATED AT
STATUS
habit-tracker-68c96699d8-2thw9 Pod default 9 minutes ago 10/01/26 Healthy
habit-tracker-68c96699d8-mggsd Pod default 9 minutes ago 10/01/26 Healthy
habit-tracker-7bb88f544f apps/ReplicaSet default 28 minutes ago 10/01/26 Healthy
habit-tracker-68c96699d8 apps/ReplicaSet default 3 hours ago 10/01/26 Healthy
habit-tracker-694599bc47 apps/ReplicaSet default 3 hours ago 10/01/26 Healthy
habit-tracker-76dbfb8656 apps/ReplicaSet default 13 hours ago 10/01/26 Healthy
habit-tracker-55cddf7c74 apps/ReplicaSet default 16 hours ago 09/30/26 Healthy
habit-tracker ConfigMap default 3 days ago 09/28/26 Synced
habit-tracker Service default 3 days ago 09/28/26 Healthy Synced
habit-tracker apps/Deployment default 3 days ago 09/28/26 Healthy Synced


github.com/priya-dharshini-sankar/Habit-Tracker-Project/commits/main/
priya-dharshini-sankar / Habit-Tracker-Project
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Commits
main
All users
All time
Commits on Oct 1, 2026
Rollback Habit Tracker image to 6a0d9d5
priya-dharshini-sankar committed 1 minute ago
5deb4fe
Update Habit Tracker image to d7cc5bb
Jenkins committed 18 minutes ago
32552c8
Deploy new Habit Tracker application version
priya-dharshini-sankar committed 27 minutes ago
d7cc5bb
Fix Kubernetes security configuration
priya-dharshini-sankar committed 2 hours ago
124f5b8
Implement security configuration
priya-dharshini-sankar committed 2 hours ago
a33e546
Update Habit Tracker image to 6a0d9d5
Jenkins committed 12 hours ago
1c2b0d3
Update Jenkins CI pipeline
priya-dharshini-sankar committed 12 hours ago
6a0d9d5
Commits on Sep 30, 2026
Restore Habit Tracker to two replicas
priya-dharshini-sankar committed 14 hours ago
4779952
Scale Habit Tracker to three replicas
priya-dharshini-sankar committed 15 hours ago
1574d2e
Update project configuration
priya-dharshini-sankar committed 18 hours ago
d7bd637

Task 13 : Monitoring & Logging

Overview

Implemented and verified basic monitoring and logging for the Habit Tracker Kubernetes deployment using kubectl and Argo CD.

Execution

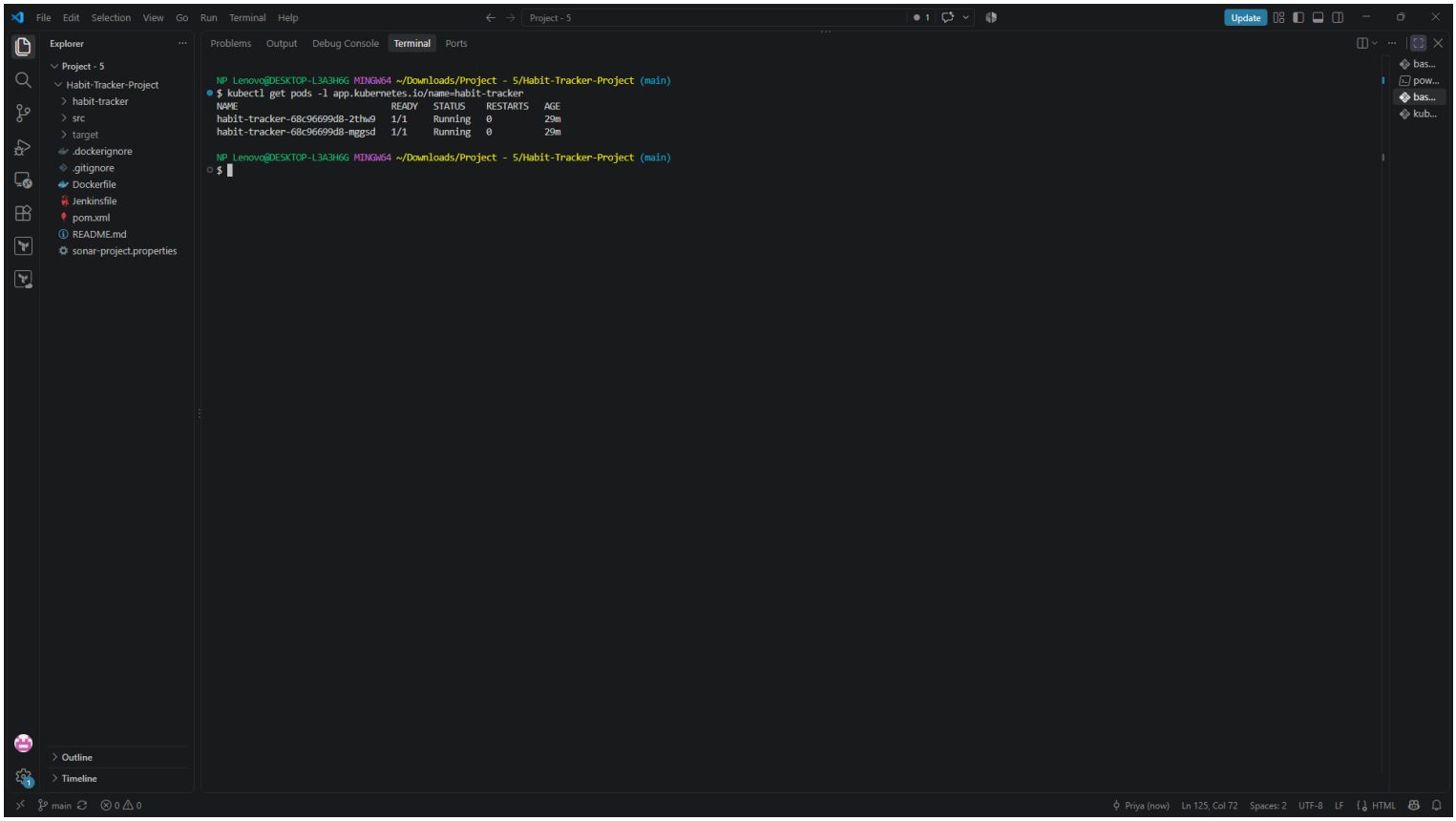
- Monitored CPU and memory usage of the Habit Tracker pods using `kubectl top pods`.
- Verified pod status and restart count using `kubectl get pods`.
- Verified Kubernetes Deployment availability using `kubectl get deployment`.
- Inspected application logs using `kubectl logs`.
- Verified Spring Boot application startup and runtime logs.
- Checked Argo CD application health and synchronization status.
- Verified the Kubernetes resources managed by Argo CD.

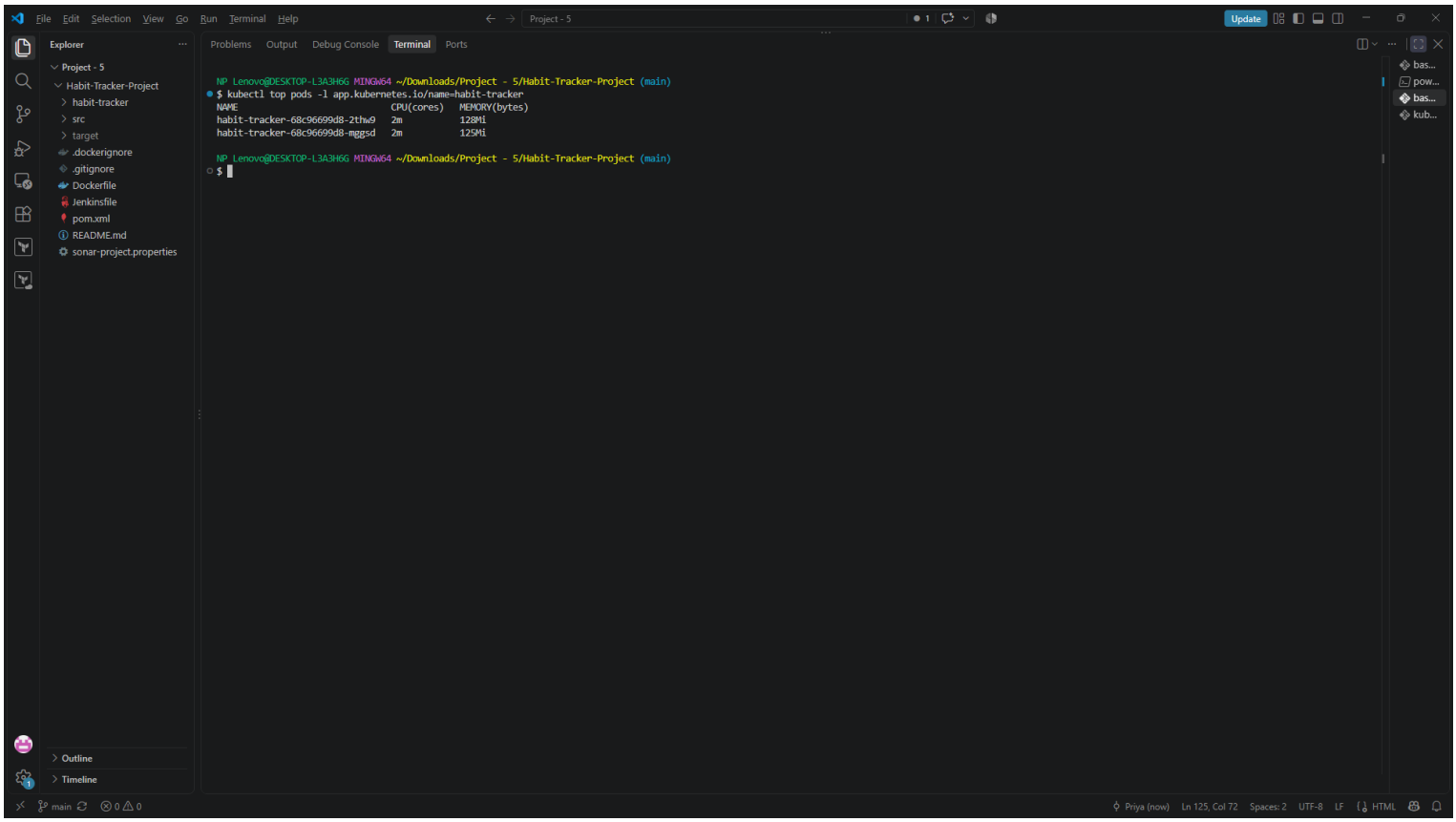
Verification

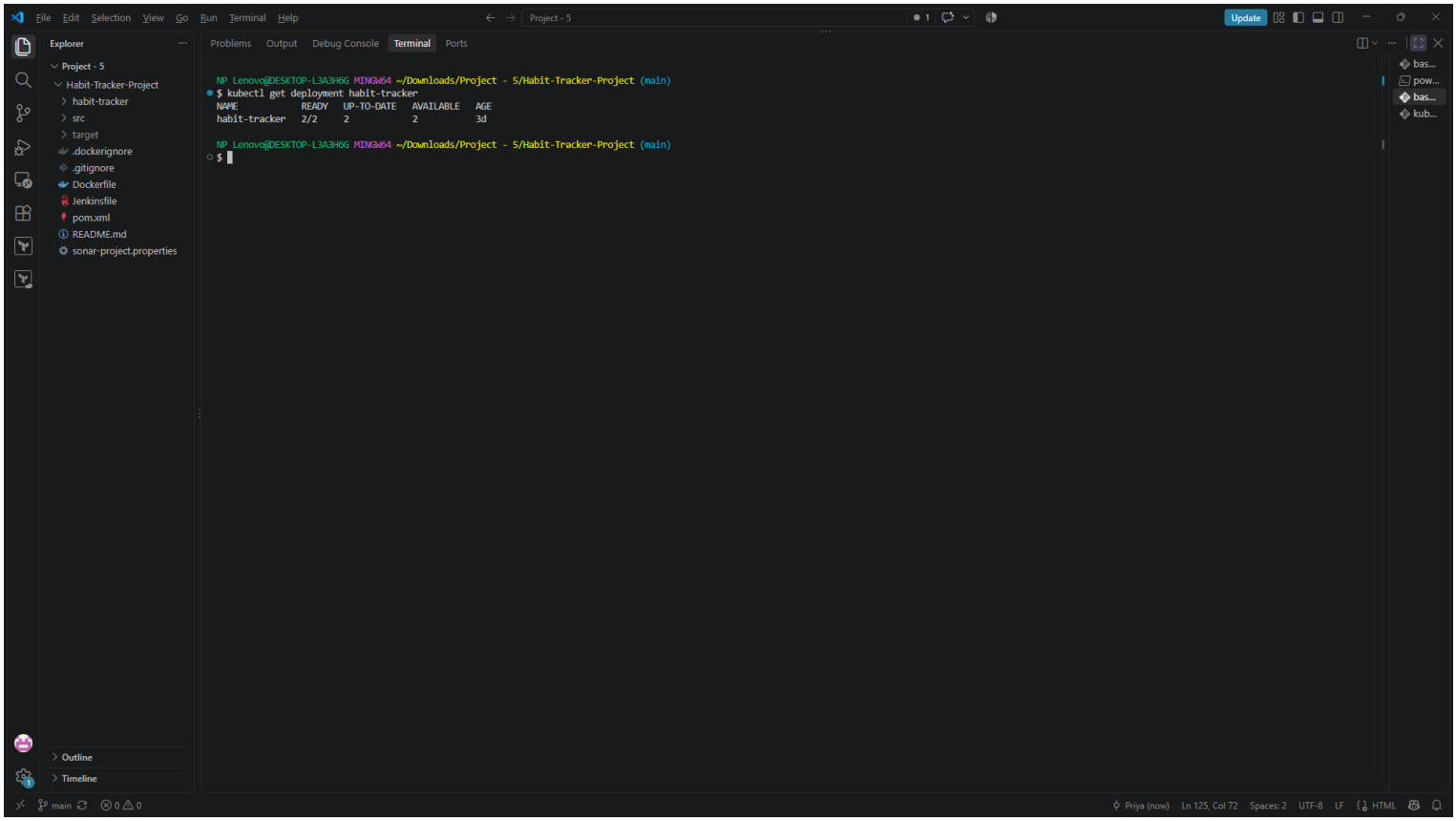
- CPU and memory usage was successfully retrieved for both running application pods.
- Both application pods were in `Running` state with `0` restarts.
- Kubernetes Deployment showed `2/2` replicas ready and available.
- Application logs were successfully retrieved.
- Spring Boot startup logs confirmed successful application startup.
- Argo CD application status was `Synced` and `Healthy`.
- Kubernetes resources were visible through Argo CD.

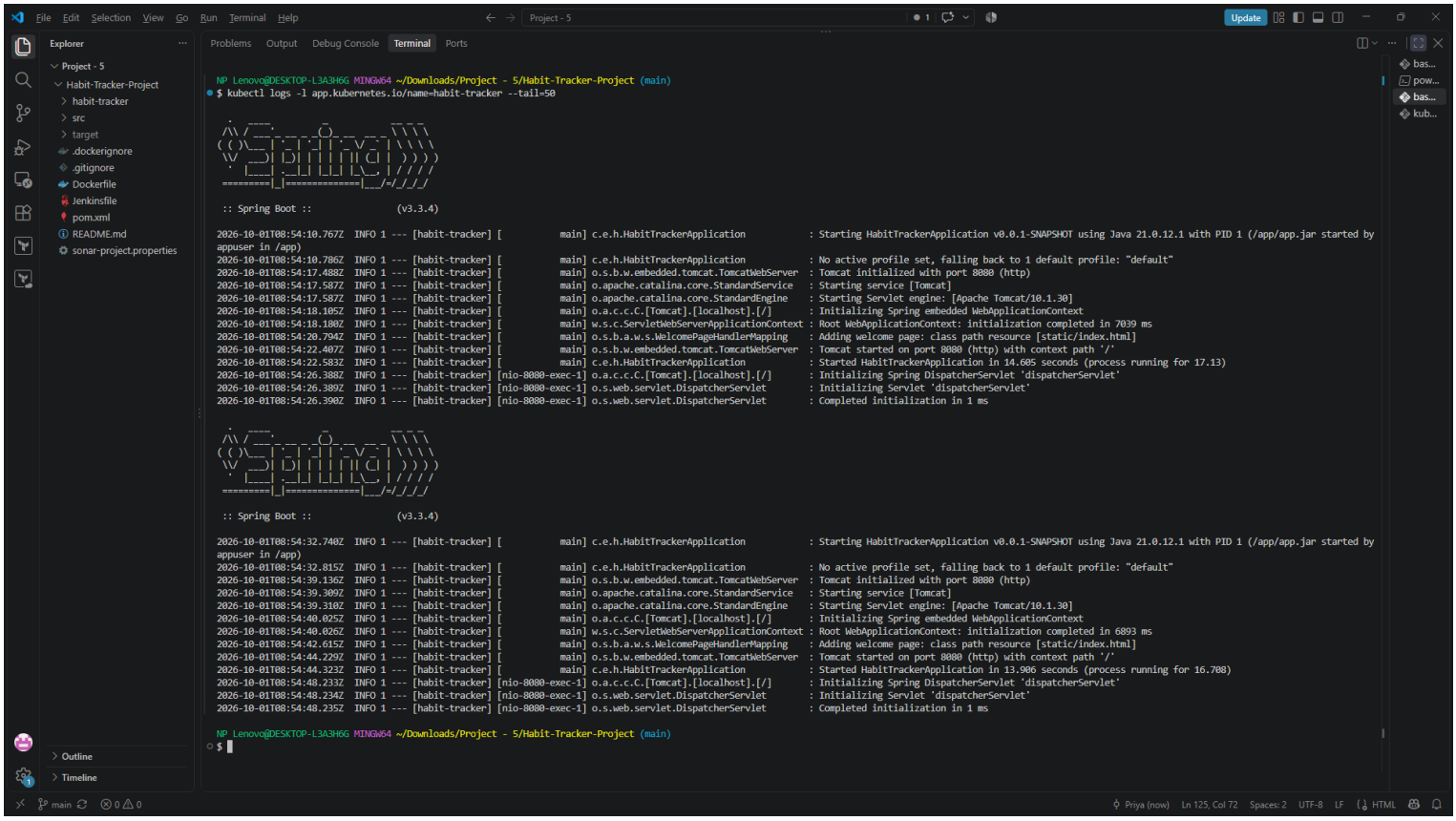
Result

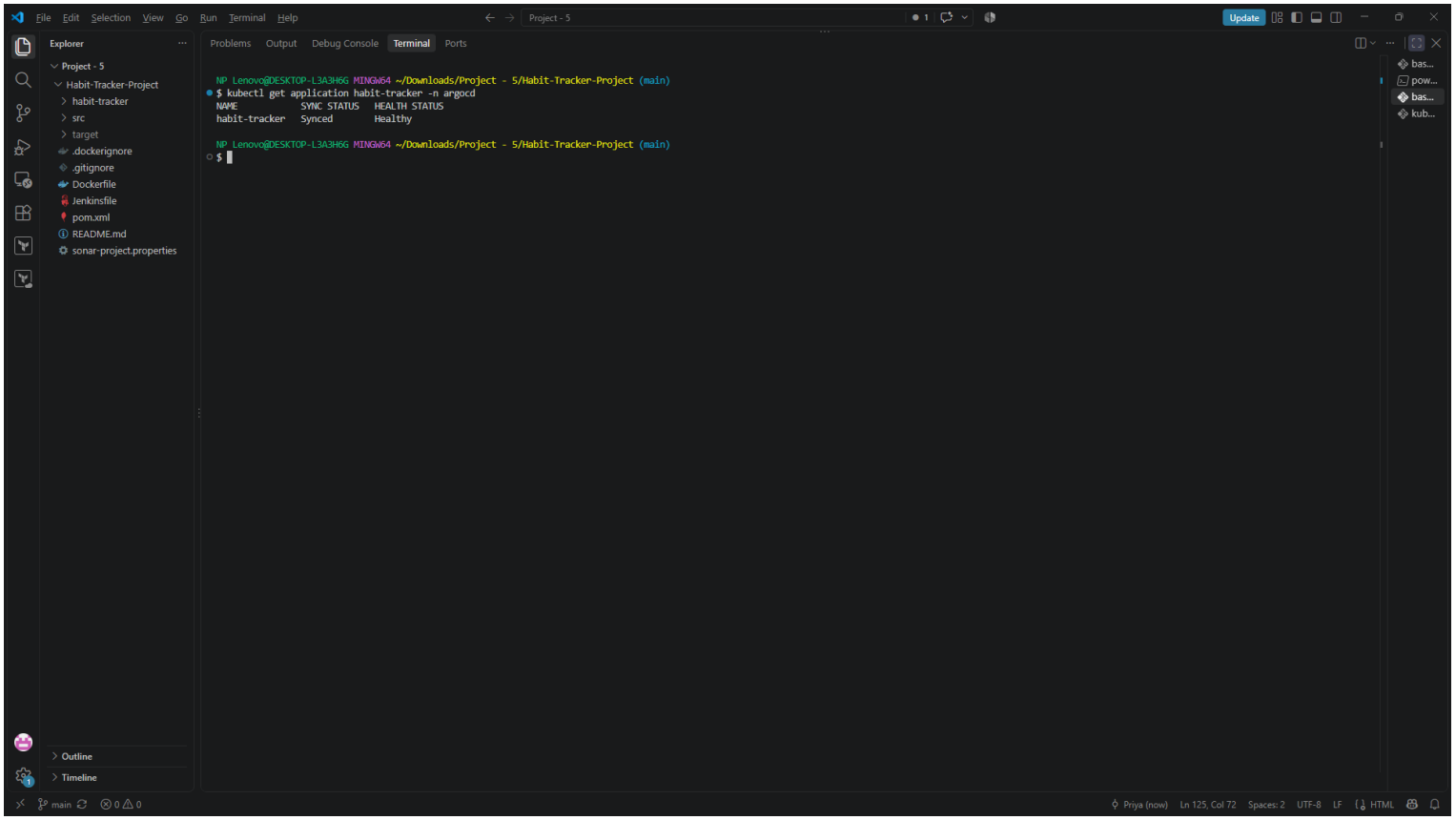
Basic monitoring and logging for the Habit Tracker application were successfully implemented and verified using Kubernetes and Argo CD.

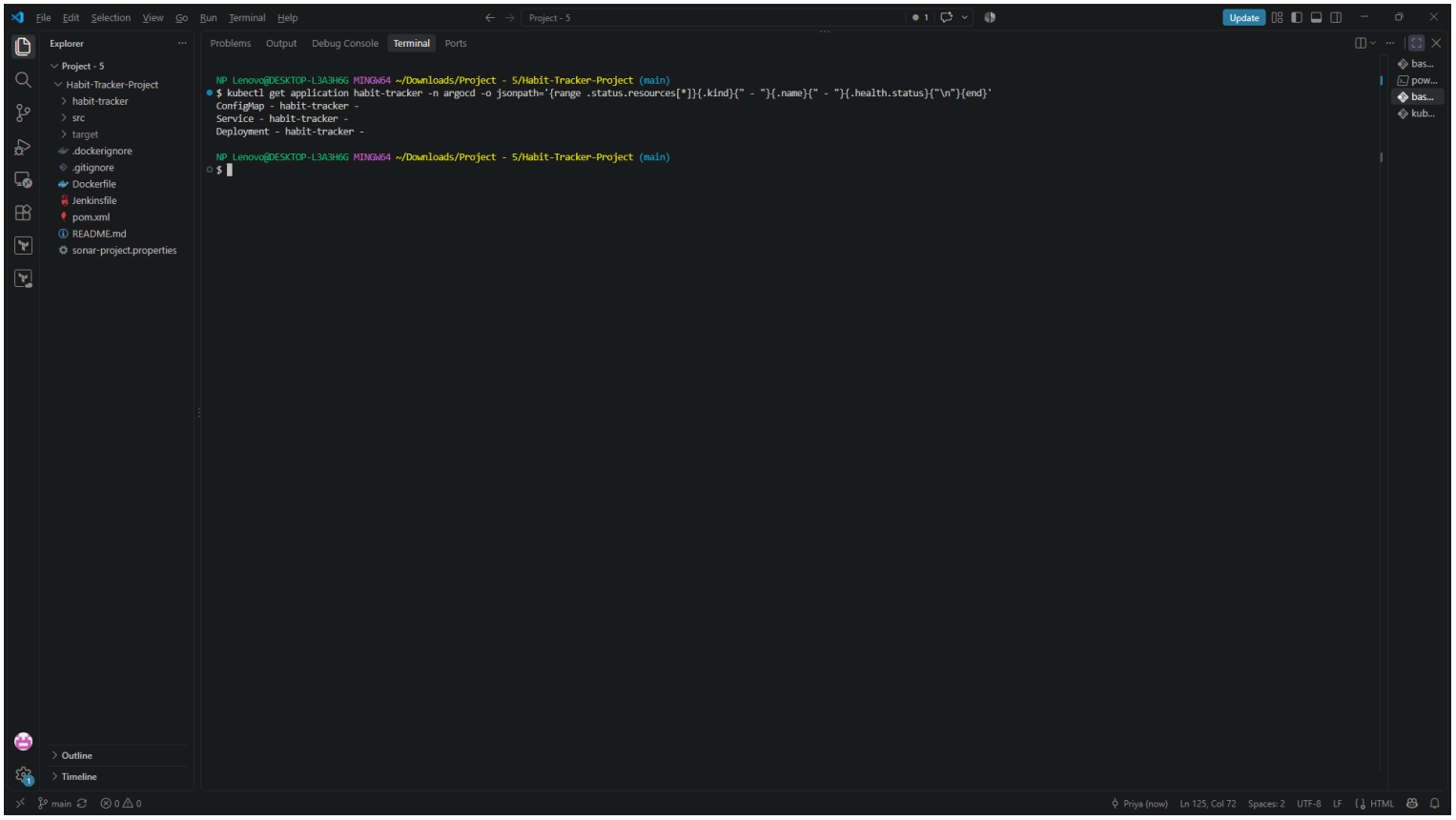

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target
.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get pods -l app.kubernetes.io/name=habit-tracker
NAME READY STATUS RESTARTS AGE
habit-tracker-68c96699d8-2thw9 1/1 Running 0 29m
habit-tracker-68c96699d8-mggsd 1/1 Running 0 29m
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
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main
Priya (now) Ln 125, Col 72 Spaces: 2 UTF-8 LF HTML

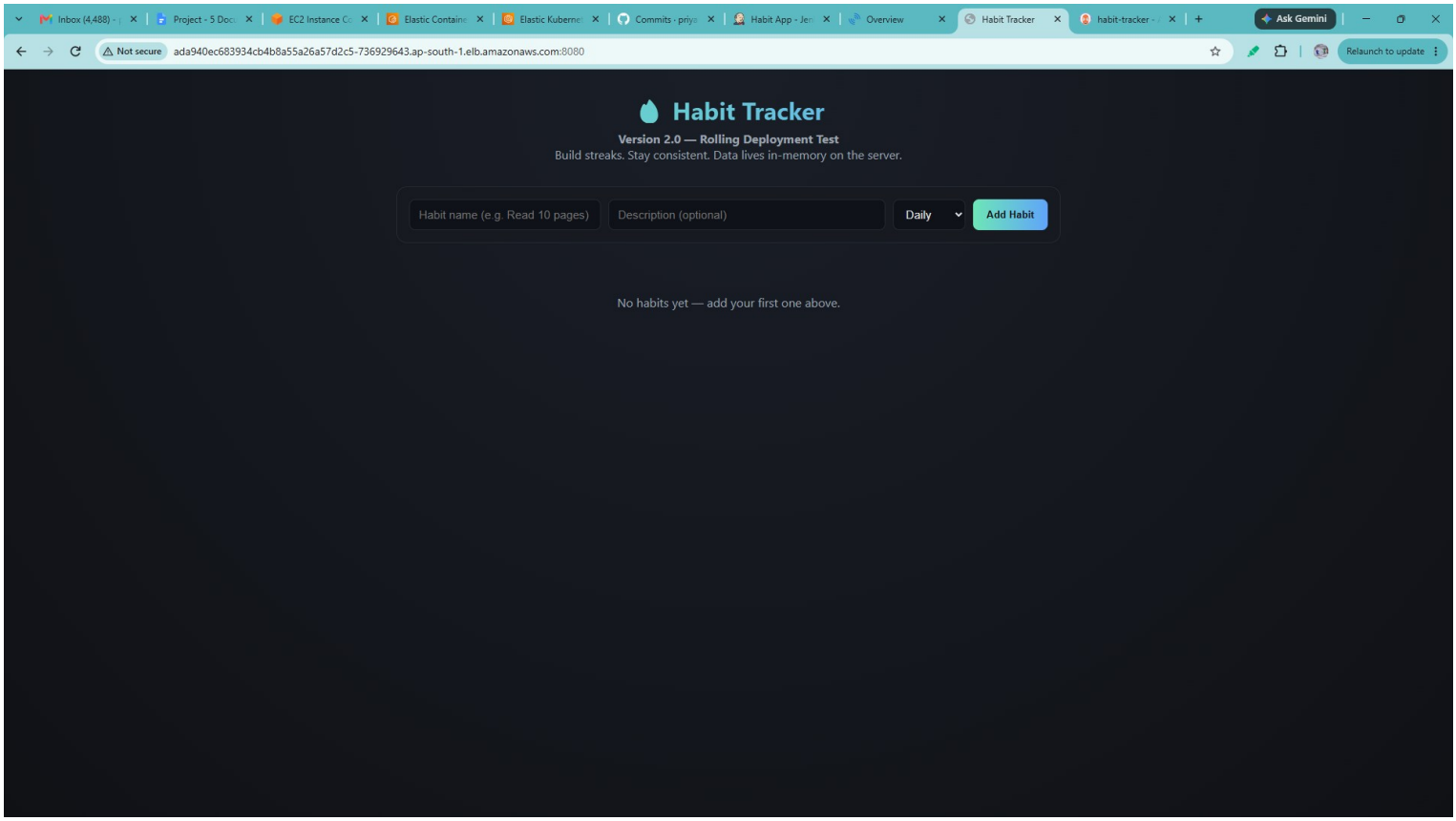

File Edit Selection View Go Run Terminal Help
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.dockerignore
.gitignore
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sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl top pods -l app.kubernetes.io/name=habit-tracker
NAME CPU(cores) MEMORY(bytes)
habit-tracker-68c96699d8-2thw9 2m 128Mi
habit-tracker-68c96699d8-mggsd 2m 125Mi
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
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pom.xml
README.md
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Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get deployment habit-tracker
NAME READY UP-TO-DATE AVAILABLE AGE
habit-tracker 2/2 2 2 3d
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
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.dockerignore
.gitignore
Dockerfile
Jenkinsfile
pom.xml
README.md
sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl logs -l app.kubernetes.io/name=habit-tracker --tail=50
:: Spring Boot :: (v3.3.4)
2026-10-01T08:54:10.767Z INFO 1 --- [habit-tracker] [main] c.e.h.HabitTrackerApplication : Starting HabitTrackerApplication v0.0.1-SNAPSHOT using Java 21.0.12.1 with PID 1 (/app/app.jar started by appuser in /app)
2026-10-01T08:54:10.786Z INFO 1 --- [habit-tracker] [main] c.e.h.HabitTrackerApplication : No active profile set, falling back to 1 default profile: "default"
2026-10-01T08:54:17.488Z INFO 1 --- [habit-tracker] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-10-01T08:54:17.587Z INFO 1 --- [habit-tracker] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-10-01T08:54:17.587Z INFO 1 --- [habit-tracker] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.30]
2026-10-01T08:54:18.105Z INFO 1 --- [habit-tracker] [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-10-01T08:54:18.180Z INFO 1 --- [habit-tracker] [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 7039 ms
2026-10-01T08:54:20.794Z INFO 1 --- [habit-tracker] [main] o.s.b.a.w.s.WelcomePageHandlerMapping : Adding welcome page: class path resource [static/index.html]
2026-10-01T08:54:22.407Z INFO 1 --- [habit-tracker] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-10-01T08:54:22.583Z INFO 1 --- [habit-tracker] [main] c.e.h.HabitTrackerApplication : Started HabitTrackerApplication in 14.605 seconds (process running for 17.13)
2026-10-01T08:54:26.388Z INFO 1 --- [habit-tracker] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-10-01T08:54:26.389Z INFO 1 --- [habit-tracker] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-10-01T08:54:26.390Z INFO 1 --- [habit-tracker] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
:: Spring Boot :: (v3.3.4)
2026-10-01T08:54:32.740Z INFO 1 --- [habit-tracker] [main] c.e.h.HabitTrackerApplication : Starting HabitTrackerApplication v0.0.1-SNAPSHOT using Java 21.0.12.1 with PID 1 (/app/app.jar started by appuser in /app)
2026-10-01T08:54:32.815Z INFO 1 --- [habit-tracker] [main] c.e.h.HabitTrackerApplication : No active profile set, falling back to 1 default profile: "default"
2026-10-01T08:54:39.136Z INFO 1 --- [habit-tracker] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-10-01T08:54:39.309Z INFO 1 --- [habit-tracker] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-10-01T08:54:39.310Z INFO 1 --- [habit-tracker] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.30]
2026-10-01T08:54:40.025Z INFO 1 --- [habit-tracker] [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-10-01T08:54:40.026Z INFO 1 --- [habit-tracker] [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 6893 ms
2026-10-01T08:54:42.615Z INFO 1 --- [habit-tracker] [main] o.s.b.a.w.s.WelcomePageHandlerMapping : Adding welcome page: class path resource [static/index.html]
2026-10-01T08:54:44.229Z INFO 1 --- [habit-tracker] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-10-01T08:54:44.323Z INFO 1 --- [habit-tracker] [main] c.e.h.HabitTrackerApplication : Started HabitTrackerApplication in 13.906 seconds (process running for 16.708)
2026-10-01T08:54:48.233Z INFO 1 --- [habit-tracker] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-10-01T08:54:48.234Z INFO 1 --- [habit-tracker] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-10-01T08:54:48.235Z INFO 1 --- [habit-tracker] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
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Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get application habit-tracker -n argocd
NAME SYNC STATUS HEALTH STATUS
habit-tracker Synced Healthy
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
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sonar-project.properties
Problems Output Debug Console Terminal Ports
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$ kubectl get application habit-tracker -n argocd -o jsonpath='{range .status.resources[*]}{.kind}{" - "}{.name}{" - "}{.health.status}{"\n"}{end}'
ConfigMap - habit-tracker -
Service - habit-tracker -
Deployment - habit-tracker -
NP Lenovo@DESKTOP-L3A3H6G MINGW64 ~/Downloads/Project - 5/Habit-Tracker-Project (main)
$
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Habit Tracker
Version 2.0 — Rolling Deployment Test
Build streaks. Stay consistent. Data lives in-memory on the server.
Habit name (e.g. Read 10 pages)
Description (optional)
Daily
Add Habit
No habits yet — add your first one above.
Not secure
ada940ec683934cb4b8a55a26a57d2c5-736929643.ap-south-1.elb.amazonaws.com:8080
Ask Gemini
Relaunch to update

Application URL

The deployed Habit Tracker Application is externally accessible through the AWS LoadBalancer:

URL : http://ada940ec683934cb4b8a55a26a57d2c5-736929643.ap-south-1.elb.amazonaws.com:8080